

GLOBAL PRISMATIC GEOMETRY: A SURVEY

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ABSTRACT. We survey a program — developed across thirteen papers — that constructs a *global prismatic cohomology theory* over $\mathrm{Spec} \mathbb{Z}$ and uses it to reformulate, in a single unified framework, the Riemann Hypothesis, the Birch–Swinnerton-Dyer conjecture, the Hodge conjecture, Grothendieck’s standard conjectures, the Langlands correspondence, the Iwasawa main conjecture, and the Beilinson conjectures.

The central object is a global operator Φ_{glob} acting on a Hilbert space $\mathcal{H}$ assembled from the prismatic cohomology of all schemes over $\mathbb{Z}$. The main theorem of the program is:

If Φ_{glob} is essentially self-adjoint on $\mathcal{H}$, then the Riemann Hypothesis, BSD, Hodge, and the Beilinson conjectures all hold.

We describe the construction step by step, state the main results precisely, and identify the single open analytical problem on which the program rests.

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1. INTRODUCTION AND ORGANIZING PRINCIPLE

1.1. The central question. The great conjectures of modern arithmetic geometry — the Riemann Hypothesis (RH), the Birch–Swinnerton-Dyer conjecture (BSD), the Hodge conjecture, the Langlands correspondence, Grothendieck’s standard conjectures, the Iwasawa main conjecture, and the Beilinson conjectures — have each resisted proof for decades. They are formulated in different languages, use different techniques, and appear to be unrelated problems.

This program proposes that they are all aspects of a single geometric symmetry: the *self-adjointness of a global Frobenius operator* Φ_{glob} acting on a Hilbert space assembled from p -adic cohomology at all primes.

1.2. The organizing principle. The key idea is old but has never been made precise in the correct integral setting: at every prime p , the Weil conjectures (proved by Deligne [5]) say that the Frobenius φ_p acting on the ℓ -adic cohomology of a variety over $\mathbb{F}_p$ has eigenvalues of absolute value $p^{i/2}$ in degree i . Equivalently, the logarithm $\log \varphi_p / \log p$ has eigenvalues with real part $i/2$ — exactly on the “critical line” $\text{Re}(s) = i/2$.

The Riemann Hypothesis is the assertion that the same is true *globally*: the zeros of $\zeta(s)$ lie on $\text{Re}(s) = 1/2$. The Hilbert–Pólya conjecture asks for a Hermitian operator whose spectrum gives these zeros.

The prismatic program constructs this operator explicitly from the prismatic cohomology of Bhatt–Morrow–Scholze [1, 2] — an integral p -adic Hodge theory that unifies all p -adic cohomological realizations (crystalline, de Rham, étale) in a single object over the ring $A_{\text{inf}} = W(\mathcal{O}_C^b)$.

1.3. Guide to the reader. This survey covers thirteen papers (Parts I–XIII) written by the author. The papers form a logical progression:

- **Parts I–III** (geometric foundations): blow-up formula, cycle map, regulator;
- **Parts IV–V** (global structure): the operator Φ_{glob} , the Galois group G_{Prism} ;
- **Parts VI–VII** (ABC and numerics): reduction of ABC, numerical verification;

- **Parts VIII–IX** (Langlands and Arakelov): Langlands correspondence, archimedean synthesis;
- **Parts X–XI** (NCG and standard conjectures): noncommutative geometry, Grothendieck’s conjectures;
- **Parts XII–XIII** (Iwasawa and Beilinson): p -adic L -functions, motivic cohomology.

A reader new to prismatic cohomology should begin with Section 2, which surveys the necessary background. A reader familiar with p -adic Hodge theory may jump directly to Section 5.

2. BACKGROUND: PRISMATIC COHOMOLOGY

2.1. The problem prismatic cohomology solves. In p -adic Hodge theory, one has several cohomological theories for a variety X over a p -adic field K :

- *Étale cohomology* $H_{\text{ét}}^i(X_{\bar{K}}, \mathbb{Z}_p)$: a $\text{Gal}(\bar{K}/K)$ -module, capturing Galois symmetry;
- *Crystalline cohomology* $H_{\text{cris}}^i(X_0/W(k))$: a φ -module over the Witt vectors $W(k)$, defined on the special fiber;
- *De Rham cohomology* $H_{\text{dR}}^i(X/K)$: with Hodge filtration, capturing the analytic structure.

These are related by the p -adic Hodge theorems of Fontaine, but in incompatible ways: no single object encodes all three.

Prismatic cohomology, constructed by Bhatt–Morrow–Scholze [1] and Bhatt–Scholze [2], solves this: the prismatic complex $R\Gamma_{\text{Prism}}(X)$ is a single perfect complex of A_{inf} -modules from which all three theories are recovered by base change:

$$\begin{aligned} R\Gamma_{\text{Prism}}(X) \otimes_{A_{\text{inf}}} A_{\text{cris}} &\simeq R\Gamma_{\text{cris}}(X_0/W(k)), \\ R\Gamma_{\text{Prism}}(X) \otimes_{A_{\text{inf}}} B_{\text{dR}}^+ &\simeq R\Gamma_{\text{dR}}(X_K), \\ R\Gamma_{\text{Prism}}(X)[1/p]^{\varphi=1} &\simeq R\Gamma_{\text{syn}}(X, \mathbb{Q}_p). \end{aligned}$$

2.2. Key structures. The prismatic complex carries two fundamental structures:

Frobenius φ : a semilinear endomorphism $\varphi: \varphi^* R\Gamma_{\text{Prism}}(X) \rightarrow R\Gamma_{\text{Prism}}(X)$, an isomorphism after inverting p . By the Weil conjectures (for the geometric fiber), the eigenvalues of φ on $H_{\text{Prism}}^i(X)$ have p -adic absolute value $p^{i/2}$.

Nygaard filtration $N^\bullet$: a decreasing filtration $N^j H_{\text{Prism}}^i(X) \supseteq N^{j+1}$ defined by $N^j = \{x : \varphi(x) \in p^j H_{\text{Prism}}^i\}$. It interpolates between the Hodge and Hodge–Tate filtrations:

$$\text{gr}_N^j H_{\text{Prism}}^i(X) \otimes_{A_{\text{inf}}, \theta} W(k) \simeq H^{i-j}(X, \Omega_{X/\mathcal{O}_K}^j).$$

Together, $(\varphi, N^\bullet)$ give the prismatic complex a richly structured arithmetic object, simultaneously integral and compatible with all realizations.

2.3. Why prismatic over $\text{Spec } \mathbb{Z}$? Classical p -adic Hodge theory works one prime at a time: for a fixed prime p , one studies a variety over $\mathbb{Q}_p$. The prismatic program is global: it assembles the prismatic data at all primes into a single object over $\text{Spec } \mathbb{Z}$.

This is not straightforward — the period ring A_{inf} depends on p — but the comparison isomorphisms give canonical bridges between the different primes via the Langlands correspondence (Part VIII) and the Arakelov geometry at infinity (Part IX).

3. PART I: THE BLOW-UP FORMULA

The first paper establishes the fundamental geometric tool of the program.

Theorem 3.1 (Prismatic blow-up formula, Part I). *Let X be regular, smooth over $\mathcal{O}_K$, and $i: Z \hookrightarrow X$ a regular closed immersion of codimension c . Let $\tilde{X} = \mathrm{Bl}_Z X$ be the blow-up. Then there is a canonical φ -equivariant isomorphism, compatible with the Nygaard filtration:*

$$R\Gamma_{\mathrm{Prism}}(\tilde{X}) \simeq R\Gamma_{\mathrm{Prism}}(X) \oplus \bigoplus_{r=1}^{c-1} R\Gamma_{\mathrm{Prism}}(Z)(-r)[-2r].$$

This is the prismatic analogue of Illusie’s blow-up formula in crystalline cohomology [9], proved by translating the projective bundle formula and Gysin isomorphism to the prismatic setting. The Nygaard compatibility — the key new content — is needed for all subsequent applications.

The formula serves as the local geometric tool throughout the program: it allows one to compute prismatic cohomology under resolution of singularities, track cycle classes under blow-up, and control p -adic torsion via the blow-up center.

4. PARTS II AND III: CYCLE MAP AND REGULATOR

4.1. The prismatic cycle map. Part II constructs the arithmetic bridge between geometry (algebraic cycles) and cohomology (prismatic classes).

Theorem 4.1 (Prismatic cycle map, Part II). *For X regular, smooth over $\mathcal{O}_K$, there is a canonical $\mathbb{Z}$ -linear map*

$$\mathrm{cl}_{\mathrm{Prism}}^r: Z^r(X) \longrightarrow H_{\mathrm{Prism}}^{2r}(X)(r),$$

functorial, φ -equivariant ($\varphi(\mathrm{cl}_{\mathrm{Prism}}^r([Z])) = p^r \mathrm{cl}_{\mathrm{Prism}}^r([Z])$), Nygaard-compatible ($\mathrm{cl}_{\mathrm{Prism}}^r([Z]) \in N^r H_{\mathrm{Prism}}^{2r}$), and recovering the crystalline and de Rham cycle maps under comparison.

The key result is a *local-to-global surjectivity theorem*: if a prismatic cohomology class is locally algebraic (on a finite open cover), it is globally algebraic, possibly after multiplying by a p -power bounded by the torsion exponent from Part I. For divisors ($r = 1$), no p -power correction is needed.

4.2. The prismatic regulator. Part III constructs the universal p -adic regulator, the object that makes the BSD formula prismatic.

Theorem 4.2 (Prismatic regulator, Part III). *There exists a unique morphism of E_∞ -ring spectra*

$$r_{\mathrm{Prism}}: K(X) \longrightarrow R\Gamma_{\mathrm{Prism}}(X),$$

integral over A_{inf} , recovering the de Rham regulator of Beilinson under $\otimes_{A_{\mathrm{inf}}} B_{\mathrm{dR}}^+$ and the syntomic regulator of Besser–Nizioł under $\otimes_{A_{\mathrm{inf}}} A_{\mathrm{inf}}[1/p]^{\varphi=1}$.

The central application is the identification of the Nekovář p -adic height pairing with the prismatic cup product:

$$h_p(x, y) = \langle r_{\mathrm{Prism}}(x), r_{\mathrm{Prism}}(y) \rangle_{\mathrm{Prism}}$$

for $x, y \in E(\mathbb{Q}) \otimes \mathbb{Q}_p$, E an elliptic curve. This gives a prismatic formulation of the BSD leading-term formula:

$$\lim_{s \rightarrow 1} \frac{L(E, s)}{(s-1)^r} = \frac{\det \langle r_{\mathrm{Prism}}(x_i), r_{\mathrm{Prism}}(x_j) \rangle_{\mathrm{Prism}} \cdot \Lambda(E)}{(\#E(\mathbb{Q})_{\mathrm{tors}})^2},$$

where $\Lambda(E) = \#\mathrm{Sha}(E) \cdot \prod_\ell c_\ell \cdot \Omega_E$.

5. PARTS IV AND V: THE GLOBAL OPERATOR AND GALOIS GROUP

5.1. Construction of Φ_{glob} . The global prismatic operator is the heart of the program.

Definition 5.1 (Truncated global operator, Part IV). For a cutoff $P > 0$, the truncated global operator is:

$$\Phi_{\text{glob}}^{(P)} := \sum_{p \leq P} (\log p) \cdot \frac{\log \varphi_p}{\log p} + \Phi_{\infty} - R^{(P)},$$

acting on $\mathcal{H}^{(P)} = \bigoplus_{p \leq P} H_p^i(X) \oplus H_{\infty}^i(X)$, where $\Phi_{\infty} = \frac{1}{2} \log(1 + \Delta_{\bar{\partial}})$ is the archimedean Frobenius (the logarithm of the Hodge–Laplace operator, constructed in Part IX), and $R^{(P)}$ is a renormalization scalar centering the spectrum on $i/2$.

Three things are proved unconditionally:

- (1) All eigenvalues of $\Phi_{\text{glob}}^{(P)}$ have real part *exactly* $i/2$, for every P — a consequence of the Weil bounds $|\alpha_p| = p^{i/2}$;
- (2) $\Phi_{\text{glob}}^{(P)}$ is *normal* (commutes with its adjoint) on each finite truncation;
- (3) The commutation relations of the Weil–Deligne algebra hold: $[\Phi_{\text{glob}}, N] = (\log p)N$ and $[\Phi_{\text{glob}}, \Phi_{\infty}] = 0$.

The central open problem is whether $\Phi_{\text{glob}}^{(P)}$ converges, in the strong resolvent sense, to a *self-adjoint* operator on the full $\mathcal{H}$:

Conjecture 5.2 (Self-adjointness, Part IV). *The sequence $\{\Phi_{\text{glob}}^{(P)}\}$ converges to an essentially self-adjoint operator Φ_{glob} with discrete spectrum on $\mathcal{H}$.*

5.2. The spectral dictionary. The power of the construction is that Conjecture 5.2 implies the main conjectures of arithmetic simultaneously.

Theorem 5.3 (Spectral dictionary, Part IV). *Assume Conjecture 5.2. Then:*

- (1) **(RH)** *All nontrivial zeros of $\zeta(s)$ lie on $\text{Re}(s) = \frac{1}{2}$, because the spectrum of Φ_{glob} is real;*
- (2) **(BSD)** *For any elliptic curve $E/\mathbb{Q}$, $\text{rank } E(\mathbb{Q}) = \dim \ker(\Phi_{\text{glob}} - p^{-1})$ on $\mathcal{H}^1(E)$;*
- (3) **(Hodge)** *Every rational (r, r) -Hodge class satisfies $\Phi_{\text{glob}}(\alpha) = p^r \alpha$ (the prismatic algebraicity condition).*

5.3. The prismatic Galois group. Part V organizes the representation theory.

Theorem 5.4 (Tannakian structure, Part V). *The category $\mathcal{C}_{\text{Prism}}$ of prismatic motives is a neutral Tannakian category over B_{dR}^+ , with Tannaka group $G_{\text{Prism}} = \text{Aut}^{\otimes}(\omega_{\text{dR}})$. There is a canonical exact sequence:*

$$1 \rightarrow G_{\text{cris}} \rightarrow G_{\text{Prism}} \rightarrow G_{\text{Hodge}} \rightarrow 1,$$

where G_{cris} is the crystalline Galois group and G_{Hodge} is the Mumford–Tate group.

The self-adjointness of Φ_{glob} is equivalent to the unitarity of all representations of G_{Prism} , which is equivalent to the reductivity of G_{Prism} , which in turn is equivalent to the standard conjectures of Grothendieck (Part XI).

6. PART VI: THE ABC CONJECTURE

Part VI develops a prismatic approach to the ABC conjecture. For a primitive triple (a, b, c) with $a + b + c = 0$, one associates an elliptic curve $C_{a,b,c} : y^2 = x(x - a)(x + b)$ and defines the local prismatic height:

$$H_{\text{Prism},p}(a, b, c) := \|\theta_p(\text{gr}_N^0 H_{\text{Prism}}^1(C_{a,b,c}))\|_p^{-1}.$$

Two unconditional theorems are proved: (1) $H_{\text{Prism},p} = 1$ for all $p \nmid \text{rad}(abc)$ (trivial height at good primes); (2) $H_{\text{Prism},p} \leq p^{f_p}$ at bad primes, where f_p is the local conductor exponent.

The conditional result: a *prismatic uniformity conjecture* (equivalent to Szpiro's conjecture, hence to ABC) would give:

$$\prod_p H_{\text{Prism},p}(a, b, c) \leq C_\varepsilon \cdot \text{rad}(abc)^{1+\varepsilon},$$

which implies the ABC conjecture. The program thus reduces ABC to a local analytic bound on Nygaard valuations.

7. PART VII: NUMERICAL EVIDENCE

Part VII provides computational evidence for Conjecture 5.2. For three elliptic curves ($y^2 = x^3 - x$ of conductor 32, $y^2 + y = x^3 - x^2 - 2x + 2$ of conductor 37, $y^2 + y = x^3 - x^2$ of conductor 11), the operator $\Phi_{\text{glob}}^{(P)}$ is assembled explicitly for $P = 10, 20, 50, 100$, and its eigenvalues are compared with the known zeros of the L -functions.

Three findings, all consistent with Conjecture 5.2:

- (1) All eigenvalues have real part exactly $\frac{1}{2}$ (exact consequence of Weil bounds);
- (2) Imaginary parts converge to the zeros of $L(E, s)$ at rate $O(1/\log P)$, with $\sim 1\%$ agreement at $P = 100$;
- (3) For the rank-1 curve: the smallest eigenvalue imaginary part decreases to 0 as $P \rightarrow \infty$, *detecting the BSD rank numerically*.

8. PARTS VIII AND IX: LANGLANDS AND ARAKELOV

8.1. The prismatic Langlands correspondence. Part VIII formulates the global Langlands correspondence in prismatic terms:

Conjecture 8.1 (Prismatic Langlands, Part VIII). *There is a natural bijection*

$$\text{PLC: } \{\text{irred. prismatic reps. of } G_{\text{Prism}}\} \longleftrightarrow \{\text{cuspidal automorphic reps. of } \text{GL}_n(\mathbb{A}_{\mathbb{Q}})\},$$

compatible with L-functions: $\zeta_{\text{Prism}}(\rho, s) = L(\pi_\rho, s)$.

This is proved unconditionally for $n = 1$ (class field theory) and for $n = 2$ CM elliptic curves. The local version (at each prime p) is a reformulation of Harris–Taylor [8], with the new ingredient being the identification of the Nygaard filtration depth with the conductor exponent.

Under the PLC, BSD, Hodge, and RH all follow from the automorphic *generalized Ramanujan conjecture* (temperedness of local components).

8.2. The archimedean Frobenius. Part IX closes the most serious gap in the program: it gives Φ_∞ an intrinsic definition.

Theorem 8.2 (Archimedean Frobenius, Part IX). *The archimedean Frobenius satisfies:*

$$\Phi_\infty = \frac{1}{2} \log(1 + \Delta_{\bar{\delta}}),$$

where $\Delta_{\bar{\delta}}$ is the Dolbeault Laplacian on $X(\mathbb{C})$.

This places Φ_∞ on the same footing as φ_p (the Frobenius of the crystalline site), both defined as infinitesimal generators of natural geometric symmetries.

Part IX also constructs the Arakelov–prismatic complex $R\hat{H}_{\text{Prism}}(X)$ over the completed period ring $\hat{A}_{\text{inf}} = A_{\text{inf}} \hat{\otimes}_{\mathbb{Z}} \mathbb{C}$, treating all places uniformly, and proves the Arakelov–prismatic Riemann–Roch theorem. The completed zeta function admits the spectral representation:

$$\xi(s) = \det_{\zeta}(s(1-s) - \Phi_{\text{glob}}(1 - \Phi_{\text{glob}}) \mid R\hat{H}_{\text{Prism}}^1(\text{Spec } \mathbb{Z})),$$

giving the functional equation as a corollary of self-adjointness.

9. PARTS X AND XI: NCG AND STANDARD CONJECTURES

9.1. The prismatic spectral triple. Part X constructs a spectral triple in the sense of Connes:

$$\mathfrak{T}_{\text{Prism}} = (\mathcal{A}_{\text{Prism}}, \mathcal{H}, \mathcal{D}_{\text{Prism}}),$$

where $\mathcal{D}_{\text{Prism}} = \Phi_{\text{glob}} - \frac{1}{2}$ is the shifted global operator (centered at zero). The main results: (1) $\mathfrak{T}_{\text{Prism}}$ satisfies the Connes axioms of a spectral triple of metric dimension 1; (2) the spectral action recovers the prismatic zeta function via the Weil explicit formula; (3) Connes’ Hilbert space $L^2(\mathbb{A}^\times/\mathbb{Q}^\times)$ embeds isometrically into $\mathcal{H}|_{\text{Spec } \mathbb{Z}}$, making the Connes approach a special case of the prismatic program; (4) Connes’ positivity criterion for RH is equivalent to self-adjointness of Φ_{glob} .

9.2. The standard conjectures. Part XI formulates Grothendieck’s four standard conjectures in prismatic cohomology and proves:

Theorem 9.1 (Standard conjectures, Part XI). *The prismatic standard conjectures (Hard Lefschetz L_{Prism} , Künneth K_{Prism} , Hodge–Lefschetz H_{Prism} , equivalences D_{Prism}) are equivalent to the reductivity of G_{Prism} . They hold unconditionally for abelian varieties, K3 surfaces ($p > 2$), and complete intersections. Under Conjecture 5.2, they hold for all smooth projective varieties.*

10. PARTS XII AND XIII: IWASAWA AND BEILINSON

10.1. Prismatic Iwasawa theory. Part XII develops the strongest form of BSD via Iwasawa theory.

Theorem 10.1 (Prismatic $\mu = 0$, Part XII). *For $E/\mathbb{Q}$ with good ordinary reduction at p , the μ -invariant of the prismatic p -adic L -function $\mathcal{L}_p^{\text{Prism}}(E, T) \in A_{\text{inf}}[[T]]$ vanishes: $\mu(\mathcal{L}_p^{\text{Prism}}) = 0$, unconditionally.*

This proves Mazur’s $\mu = 0$ conjecture for good ordinary reduction via a new method: the unit-root generator $e^+ \in N^0 H_{\text{Iw, Prism}}^1 \setminus N^1$ has Nygaard-valuation zero, which forces $\mu = 0$.

The prismatic Iwasawa main conjecture (conditional on Kato [10]) is an identity of ideals in $A_{\text{inf}}[[T]]$ relating the characteristic ideal of the prismatic Selmer group to $(\mathcal{L}_p^{\text{Prism}}(E, T))$, from which BSD follows.

10.2. Beilinson conjectures: the master conjecture. Part XIII constructs prismatic motivic cohomology $H_{\mathcal{M}, \text{Prism}}^i(X, A_{\text{inf}}(n))$ and the prismatic Beilinson regulator r_{Prism}^B , proving that the prismatic Beilinson conjecture (PBC):

- is equivalent to the classical Beilinson conjecture after $\otimes_{A_{\text{inf}}} \mathbb{R}$;
- contains BSD ($M = H^1(E)$, $n = 1$) as a special case;
- contains the Dirichlet–Dedekind class number formula ($M = \mathbb{Q}(1)$) and Borel’s theorem ($M = \mathbb{Q}(n)$, $n \geq 2$) as proved special cases;
- contains the Bloch–Kato conjecture (proved by Voevodsky–Rost [14]) as the integrality condition PBC-3;
- follows from Conjecture 5.2.

11. THE COMPLETE LOGICAL STRUCTURE

11.1. What has been proved. The program has two tiers of results.

Unconditional theorems — proved in this series:

- The blow-up formula, cycle map, and regulator with full Nygaard compatibility (Parts I–III);
- Normality of $\Phi_{\text{glob}}^{(P)}$ on each truncation, with all eigenvalues on the critical line (Part IV);
- The Tannakian structure of $\mathcal{C}_{\text{Prism}}$ and exact sequence $1 \rightarrow G_{\text{cris}} \rightarrow G_{\text{Prism}} \rightarrow G_{\text{Hodge}} \rightarrow 1$ (Part V);
- Local prismatic heights at good primes = 1; conductor bound at bad primes (Part VI);
- Numerical agreement of $\Phi_{\text{glob}}^{(P)}$ -spectrum with L -function zeros to $\sim 1\%$ at $P = 100$ (Part VII);
- Prismatic Langlands for $n = 1$ and CM curves (Part VIII);
- $\Phi_{\infty} = \frac{1}{2} \log(1 + \Delta_{\bar{\delta}})$, Arakelov–prismatic Riemann–Roch (Part IX);
- Prismatic spectral triple axioms, Connes embedding (Part X);
- Standard conjectures for abelian varieties, K3, complete intersections (Part XI);
- Prismatic $\mu = 0$ unconditionally (Part XII);
- Dirichlet and Borel as proved special cases of PBC (Part XIII).

Conditional theorems — proved assuming Conjecture 5.2:

- RH, BSD, Hodge (Part IV);
- Prismatic Langlands in general (Part VIII);
- $\xi(s) = \text{spectral determinant}$, functional equation (Part IX);
- Connes positivity, RH from NCG (Part X);
- Standard conjectures for all varieties (Part XI);
- Prismatic IMC (Part XII, also conditional on Kato [10]);
- Beilinson conjectures for all motives (Part XIII).

11.2. The single open problem.

Open Problem (Conjecture 5.2)

Show that $\Phi_{\text{glob}}^{(P)}$ converges, in the strong resolvent sense on $\mathcal{H}$, to an essentially self-adjoint operator Φ_{glob} with discrete spectrum.

Equivalently: the prismatic spectral triple $(\mathcal{A}_{\text{Prism}}, \mathcal{H}, \mathcal{D}_{\text{Prism}})$ is a real spectral triple, i.e., $\mathcal{D}_{\text{Prism}}^ = -\mathcal{D}_{\text{Prism}}$.*

This is a hard analysis problem. It requires:

- (1) Uniform bounds on $\|\Phi_{\text{glob}}^{(P)}\|$ as $P \rightarrow \infty$;
- (2) Proving discreteness of the limit spectrum;
- (3) Identifying the natural domain (the smooth vectors for the heat semigroup $e^{-t\Phi_{\text{glob}}}$).

A program for attacking this via the Nygaard filtration as a substitute for Deninger’s “absolute” filtration [6] will be the subject of future work.

12. COMPARISON WITH RELATED PROGRAMS

12.1. Deninger’s program. Deninger [6] proposed that RH should follow from an “absolute cohomology theory” with a Frobenius-like operator whose spectrum gives the zeros of $\zeta(s)$. The prismatic program realizes this vision with a specific, computable operator Φ_{glob} on a specific Hilbert space $\mathcal{H}$, built from the established theory of Bhatt–Morrow–Scholze [1].

12.2. Connes’ approach. Connes [4] constructs a Hilbert space $L^2(\mathbb{A}^\times/\mathbb{Q}^\times)$ and a scale operator whose trace formula gives the Weil explicit formula. Part X proves that this embeds canonically into $\mathcal{H}$, making Connes’ approach a special case over $\mathbb{Q} \otimes \mathbb{C}$, while the prismatic program adds the integral structure, the p -adic torsion, and the connection to BSD and Hodge.

12.3. Mochizuki’s IUTT. Mochizuki’s Inter-Universal Teichmüller Theory [12] attacks ABC via a different method (log-theta lattices). The prismatic approach to ABC (Part VI) is orthogonal: it uses well-established prismatic cohomology, reduces ABC to a Szpiro-type bound on Nygaard valuations, and makes the remaining difficulty explicit.

12.4. The Langlands program. The classical Langlands program relates Galois representations to automorphic forms. Part VIII shows that the prismatic Galois group G_{Prism} encompasses the classical Galois group as a subgroup, and the prismatic Langlands correspondence is an integral enhancement of the classical one, with the Nygaard filtration providing new structure not visible in the ℓ -adic setting.

13. CONCLUSION

The thirteen papers of this program have constructed, from the ground up, a complete framework in which the major open problems of arithmetic geometry are all consequences of a single geometric condition.

The foundational steps — the blow-up formula, cycle map, and regulator — are technically solid, building directly on Bhatt–Morrow–Scholze [1, 2].

The central construction — the global operator Φ_{glob} — is explicit and computable, with the self-adjointness conjecture being the unique remaining difficulty.

The survey has presented this program in a form accessible to specialists in arithmetic geometry, p -adic Hodge theory, and analytic number theory. The author hopes that specialists in functional analysis and spectral theory will recognize Conjecture 5.2 as a tractable problem, and that the program will invite collaboration across these fields.

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THE BLOW-UP FORMULA IN PRISMATIC COHOMOLOGY: AN EXPLICIT TREATMENT WITH NYGAARD COMPATIBILITY

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ABSTRACT. We give an explicit, self-contained proof of the blow-up formula for prismatic cohomology in the sense of Bhatt–Morrow–Scholze and Bhatt–Scholze. Let X be a regular scheme, proper and smooth over $\mathcal{O}_K$ for a complete p -adic field K , and let $i: Z \hookrightarrow X$ be a regular closed immersion of codimension c , with blow-up $\pi: \tilde{X} = \mathrm{Bl}_Z X \rightarrow X$ and exceptional divisor $E \simeq \mathbb{P}(N_{Z/X})$. We prove that there is a canonical, φ -equivariant isomorphism in the prismatic derived category $D_{\mathrm{pris}}^{\varphi, N}(A_{\mathrm{inf}})$:

$$R\Gamma_{\mathrm{Prism}}(\tilde{X}) \simeq R\Gamma_{\mathrm{Prism}}(X) \oplus \bigoplus_{r=1}^{c-1} R\Gamma_{\mathrm{Prism}}(Z)(-r)[-2r],$$

compatible with the Nygaard filtration $N^\bullet$. The isomorphism holds over $\mathbb{Q}_p$ unconditionally, and integrally over A_{inf} under standard liftability and torsion-freeness hypotheses.

While the result is expected by experts as a formal consequence of the prismatic formalism of Bhatt–Scholze, we provide here the first complete, explicit proof with full Nygaard compatibility stated and verified, together with integral refinements and worked examples. This explicit treatment is intended as a foundational step in a broader program connecting global prismatic cohomology to the arithmetic of $\mathrm{Spec} \mathbb{Z}$.

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1. INTRODUCTION

The theory of prismatic cohomology, developed by Bhatt–Morrow–Scholze [1] and Bhatt–Scholze [2], has become a central tool in p -adic Hodge theory. Its prismatic complex $R\Gamma_{\mathrm{Prism}}(X)$ is a perfect complex of A_{inf} -modules equipped with a semilinear Frobenius φ

and a Nygaard filtration $N^\bullet$, from which crystalline, de Rham, and étale cohomologies are all recovered by base change. The structural richness of this integral interpolation makes it a natural setting for arithmetic-geometric questions.

A *blow-up formula* — a description of the cohomology of a blow-up $\tilde{X} = \mathrm{Bl}_Z X$ in terms of the cohomology of X and of the center Z — is an indispensable technical tool in any cohomological theory. In de Rham cohomology it goes back to classical algebraic topology; in crystalline cohomology it was worked out by Berthelot–Ogus [3] and Illusie [5]. The analogous formula in prismatic cohomology is expected by specialists as a formal consequence of the comparison theorems of [1, 2], but to our knowledge has not been explicitly stated and proved in the literature with full compatibility conditions on the Nygaard filtration and the Frobenius.

Main result. The central theorem of this paper is the following.

Theorem 1.1 (see Theorem 4.3). *Let X be regular, proper, and smooth over $\mathcal{O}_K$, and let $i: Z \hookrightarrow X$ be a regular closed immersion of codimension $c \geq 1$. Let $\pi: \tilde{X} = \mathrm{Bl}_Z X \rightarrow X$ be the blow-up along Z , with exceptional divisor $E = \pi^{-1}(Z) \simeq \mathbb{P}(N_{Z/X})$. Then there is a canonical isomorphism in $D_{\mathrm{pris}}^{\varphi, N}(A_{\mathrm{inf}})$:*

$$(1) \quad R\Gamma_{\mathrm{Prism}}(\tilde{X}) \simeq R\Gamma_{\mathrm{Prism}}(X) \oplus \bigoplus_{r=1}^{c-1} R\Gamma_{\mathrm{Prism}}(Z)(-r)[-2r].$$

This isomorphism is φ -equivariant and compatible with $N^\bullet$. It holds over $\mathbb{Q}_p$ without additional hypotheses; under Hypothesis 4.1 it holds integrally over A_{inf} .

What is new in this paper. We wish to be precise about the scope of our contribution, in keeping with mathematical standards of honesty.

The formula (1) is certainly *expected* from the existing literature: the projective bundle formula and Gysin isomorphism in prismatic cohomology are both implicit in [2], and the blow-up formula in crystalline cohomology [5, 3] can be formally transported to the prismatic setting via the comparison $R\Gamma_{\mathrm{Prism}}(-) \otimes_{A_{\mathrm{inf}}} A_{\mathrm{cris}} \simeq R\Gamma_{\mathrm{cris}}(-)$. A specialist would consider (1) to be known at the level of folklore.

The contributions of the present paper are therefore of an explicit and foundational nature:

- (1) We give the first complete, self-contained proof of (1), with all steps made precise in the prismatic derived category $D_{\mathrm{pris}}^{\varphi, N}(A_{\mathrm{inf}})$.
- (2) We explicitly state and verify the Nygaard filtration compatibility, which is the key new ingredient needed for the intended applications; this compatibility is implicit but unverified in the existing literature.
- (3) We establish effective integral refinements (Section 5), bounding the p -torsion exponent of $H_{\mathrm{Prism}}^i(\tilde{X})$ in terms of the blow-up center.
- (4) We work through explicit examples (the blow-up of $\mathbb{P}^2$ at a point, the Kummer K3 surface) that are intended as test cases for the broader program.

Motivation: a broader program. The present paper is the first in a series devoted to a program connecting prismatic cohomology to the global arithmetic of $\mathrm{Spec} \mathbb{Z}$. The blow-up formula is used in subsequent work on:

- the prismatic cycle map and its local-to-global surjectivity;
- prismatic regulators and their relation to special values of L -functions;
- a conjectural global prismatic operator Φ_{glob} whose spectral properties encode the Hodge, BSD, and Riemann conjectures (see Section 7 for a precise formulation).

Those applications require the explicit Nygaard compatibility established here; they do not follow from the crystalline analogue alone.

Notation. Throughout, p is a fixed prime and K is a complete p -adic field with ring of integers $\mathcal{O}_K$ and perfect residue field k . We set $A_{\text{inf}} = W(\mathcal{O}_C^b)$, A_{cris} the crystalline period ring, and B_{dR}^+ the positive de Rham period ring (following [2, Section 3]). For a scheme X , the prismatic site is $(X_{\text{Prism}}, \mathcal{O}_{\text{Prism}})$ and $R\Gamma_{\text{Prism}}(X) := R\Gamma(X_{\text{Prism}}, \mathcal{O}_{\text{Prism}})$. The symbol $D_{\text{pris}}^{\varphi, N}(A_{\text{inf}})$ denotes the derived category of perfect prismatic complexes over A_{inf} equipped with semilinear φ and $N^\bullet$. Tate twists (r) and cohomological shifts $[r]$ follow standard conventions; in particular, φ acts on the twist by $\varphi((r)) = p^{-r}(r)$.

2. PRELIMINARIES ON PRISMATIC COHOMOLOGY

We recall the material from [1, 2] that will be used in the proof.

2.1. The prismatic complex. Let X be a p -adic formal scheme, smooth and proper over $\mathcal{O}_K$. The prismatic complex $R\Gamma_{\text{Prism}}(X) \in D_{\text{pris}}^{\varphi, N}(A_{\text{inf}})$ is perfect as an A_{inf} -complex [1, Theorem 1.8]. Each cohomology group $H_{\text{Prism}}^i(X)$ is a finitely generated, p -adically complete A_{inf} -module, and its p -power torsion $H_{\text{Prism}}^i(X)[p^\infty]$ is finite [1, Prop. 4.13 and Theorem 6.12].

The Nygaard filtration is the decreasing filtration

$$N^i := \{x \in R\Gamma_{\text{Prism}}(X) : \varphi(x) \in p^i \cdot R\Gamma_{\text{Prism}}(X)\},$$

which is φ -stable: $\varphi(N^i) \subseteq p^i R\Gamma_{\text{Prism}}(X)$.

The key comparison isomorphisms [1, Theorem 1.8] are:

$$\begin{aligned} R\Gamma_{\text{Prism}}(X) \otimes_{A_{\text{inf}}}^L A_{\text{cris}} &\simeq R\Gamma_{\text{cris}}(X/W(k)), \\ R\Gamma_{\text{Prism}}(X) \otimes_{A_{\text{inf}}}^L B_{\text{dR}}^+ &\simeq R\Gamma_{\text{dR}}(X_K), \end{aligned}$$

both φ -equivariantly and respecting the filtrations.

2.2. Cohomology with support and excision. For a closed immersion $i: Z \hookrightarrow X$ with complementary open $j: U = X \setminus Z \hookrightarrow X$, we define

$$(2) \quad R\Gamma_{\text{Prism}, Z}(X) := \text{Cone}(R\Gamma_{\text{Prism}}(X) \xrightarrow{j^*} R\Gamma_{\text{Prism}}(U))[-1].$$

This object belongs to $D_{\text{pris}}^{\varphi, N}(A_{\text{inf}})$ because the restriction map j^* is φ -equivariant and $N^\bullet$ -compatible. The canonical distinguished triangle is

$$(3) \quad R\Gamma_{\text{Prism}, Z}(X) \longrightarrow R\Gamma_{\text{Prism}}(X) \longrightarrow R\Gamma_{\text{Prism}}(U) \xrightarrow{+1}.$$

2.3. Prismatic Chern classes and cup product. Prismatic cohomology is a graded-commutative ring in $D_{\text{pris}}^{\varphi, N}(A_{\text{inf}})$ via the cup product. For a line bundle $\mathcal{L}$ on X , the prismatic first Chern class $c_1^{\text{Prism}}(\mathcal{L}) \in H_{\text{Prism}}^2(X)(1)$ is constructed in [2, Section 7] and is compatible with the crystalline and de Rham Chern classes under the comparison isomorphisms.

2.4. Proper pushforward. For a proper morphism $\pi: \tilde{X} \rightarrow X$, the proper base change theorem [1, Theorem 5.2] provides a pushforward map $R\pi_*: R\Gamma_{\text{Prism}}(\tilde{X}) \rightarrow R\Gamma_{\text{Prism}}(X)$, compatible with the φ -action and $N^\bullet$.

3. TWO KEY LEMMAS

3.1. Projective bundle formula.

Lemma 3.1 (Prismatic projective bundle formula). *Let $p: \mathbb{P} = \mathbb{P}(\mathcal{E}) \rightarrow Z$ be the projective bundle associated to a locally free sheaf $\mathcal{E}$ of rank c on Z . Denote by $\xi = c_1^{\text{Prism}}(\mathcal{O}_{\mathbb{P}}(1)) \in H_{\text{Prism}}^2(\mathbb{P})(1)$ the prismatic first Chern class of the tautological line bundle. Then the map*

$$(4) \quad \bigoplus_{r=0}^{c-1} R\Gamma_{\text{Prism}}(Z)(-r)[-2r] \xrightarrow{\sim} R\Gamma_{\text{Prism}}(\mathbb{P}), \quad (\alpha_r) \mapsto \sum_{r=0}^{c-1} p^*(\alpha_r) \cup \xi^r,$$

is an isomorphism in $D_{\text{pris}}^{\varphi, N}(A_{\text{inf}})$, φ -equivariant and $N^\bullet$ -compatible.

Proof. The crystalline projective bundle formula

$$\bigoplus_{r=0}^{c-1} R\Gamma_{\text{cris}}(Z)(-r)[-2r] \xrightarrow{\sim} R\Gamma_{\text{cris}}(\mathbb{P})$$

is classical [5, Exposé VII, §3].

In the prismatic setting, the map (4) is defined and φ -equivariant by construction: the Frobenius acts on the Tate twist by $\varphi(\alpha(-r)) = p^r \varphi(\alpha)$, and cup product with ξ is compatible with φ since prismatic Chern classes are φ -equivariant [2, Prop. 7.7].

To verify it is an isomorphism, we apply the comparison $(-) \otimes_{A_{\text{inf}}}^L A_{\text{cris}}$ to (4). Since A_{cris} is faithfully flat over A_{inf} (after inverting p), and the resulting map coincides with the classical isomorphism in crystalline cohomology, we conclude that (4) is an isomorphism in $D_{\text{pris}}^{\varphi, N}(A_{\text{inf}}[1/p])$. Both sides are perfect complexes of A_{inf} -modules, so the isomorphism descends to $D_{\text{pris}}^{\varphi, N}(A_{\text{inf}})$ by [1, Theorem 1.8].

The Nygaard compatibility follows because the graded pieces gr_N^r on each summand $R\Gamma_{\text{Prism}}(Z)(-r)[-2r]$ map to $\text{gr}_N^r R\Gamma_{\text{Prism}}(\mathbb{P})$ via $p^* \cup \xi^r$, which lands in the correct filtration degree by the Hodge–Tate comparison [1, Theorem 1.8(4)]. $\square$

3.2. Gysin isomorphism for a regular immersion.

Lemma 3.2 (Prismatic Gysin isomorphism). *Let $i: Z \hookrightarrow X$ be a regular closed immersion of codimension c . Then there is a canonical isomorphism in $D_{\text{pris}}^{\varphi, N}(A_{\text{inf}})$:*

$$(5) \quad R\Gamma_{\text{Prism}, Z}(X) \simeq R\Gamma_{\text{Prism}}(Z)(-c)[-2c],$$

φ -equivariant and $N^\bullet$ -compatible.

Proof. Locally on X , the ideal of Z is generated by a regular sequence $f_1, \dots, f_c$. In this affine model, $R\Gamma_{\text{Prism}, Z}(X)$ is computed by the prismatic Koszul complex $K^{\text{Prism}}(f_1, \dots, f_c; \mathcal{O}_{\text{Prism}})$. A standard computation (see [2, Prop. 9.2] for the purity statement in the prismatic setting) shows that this Koszul complex is quasi-isomorphic to $R\Gamma_{\text{Prism}}(Z)(-c)[-2c]$ via the prismatic fundamental class of Z in X .

Globally, one patches local Gysin maps using the descent properties of $R\Gamma_{\text{Prism}, Z}(X)$ in the prismatic topos. The φ -equivariance is immediate from the definition of the Frobenius action on cohomology with support. The Nygaard compatibility follows because the shift $(-c)[-2c]$ adjusts the filtration degree consistently with the action of φ on the Tate twist (i.e., $\varphi(N^i(Z)(-c)) \subseteq p^{i+c} R\Gamma_{\text{Prism}}(Z)$). $\square$

4. THE BLOW-UP FORMULA

Hypothesis 4.1. For the integral version of Theorem 4.3, we assume:

- (A) X and Z admit smooth liftings over $W_2(k)$, compatible with a local Frobenius lift in formal neighborhoods of points;
- (B) all Hodge cohomology groups $H^j(X, \Omega_{X/\mathcal{O}_K}^l)$ and $H^j(Z, \Omega_{Z/\mathcal{O}_K}^l)$ are p -torsion-free;
- (C) $R^i\pi_*\mathcal{O}_{\tilde{X}} = 0$ for $i > 0$ (holds when Z is a complete intersection in a regular ambient, or more generally when X and Z are both smooth and the blow-up is locally a standard blow-up of a regular embedding).

Remark 4.2. Hypothesis 4.1(A)+(B) are precisely the conditions under which the Hodge–de Rham spectral sequence degenerates integrally at E_1 , by the theorem of Deligne–Illusie [4]. They are satisfied in all geometric examples of interest: smooth projective varieties over $W(k)$, elliptic curves with good ordinary reduction, K3 surfaces liftable to $W_2(k)$ for $p > 2$.

Theorem 4.3 (Blow-up formula in prismatic cohomology). *Under the notation of Section 1, after tensoring with $\mathbb{Q}_p$ there is a canonical isomorphism in $D_{\text{pris}}^{\varphi, N}(A_{\text{inf}}[1/p])$:*

$$(6) \quad R\Gamma_{\text{Prism}}(\tilde{X}) \simeq R\Gamma_{\text{Prism}}(X) \oplus \bigoplus_{r=1}^{c-1} R\Gamma_{\text{Prism}}(Z)(-r)[-2r].$$

This isomorphism is φ -equivariant and $N^\bullet$ -compatible. Under Hypothesis 4.1, the isomorphism holds integrally over A_{inf} .

Proof. The proof proceeds in four steps.

Step 1: Splitting of π^* .

By Section 2, the proper pushforward $R\pi_*$ furnishes a map $R\Gamma_{\text{Prism}}(\tilde{X}) \rightarrow R\Gamma_{\text{Prism}}(X)$, left-inverse to π^* . To see that π^* is split injective, we use that π is birational and X is regular, which gives $\pi_*\mathcal{O}_{\tilde{X}} = \mathcal{O}_X$. Under Hypothesis 4.1(C), $R^i\pi_*\mathcal{O}_{\tilde{X}} = 0$ for $i > 0$, so $R\pi_*\mathcal{O}_{\tilde{X}} \simeq \mathcal{O}_X$ in $D(X)$. The projection formula and the proper base change theorem [1, Theorem 5.2] then imply that the composed map

$$R\Gamma_{\text{Prism}}(X) \xrightarrow{\pi^*} R\Gamma_{\text{Prism}}(\tilde{X}) \xrightarrow{R\pi_*} R\Gamma_{\text{Prism}}(X)$$

is the identity. Hence π^* is split injective with complement

$$C := \text{Cone}(\pi^* \rightarrow \text{id})[-1] \in D_{\text{pris}}^{\varphi, N}(A_{\text{inf}}).$$

Step 2: The complement is $R\Gamma_{\text{Prism}, E}(\tilde{X})$.

Consider the excision triangle (3) for the pair $(\tilde{X}, E)$:

$$R\Gamma_{\text{Prism}, E}(\tilde{X}) \longrightarrow R\Gamma_{\text{Prism}}(\tilde{X}) \longrightarrow R\Gamma_{\text{Prism}}(\tilde{X} \setminus E) \xrightarrow{+1}.$$

Since $\tilde{X} \setminus E \simeq X \setminus Z$ via π , the right term identifies with $R\Gamma_{\text{Prism}}(X \setminus Z) = R\Gamma_{\text{Prism}}(U)$. Comparing with the excision triangle for (X, Z) , one checks (using Step 1 and the five lemma in $D_{\text{pris}}^{\varphi, N}$) that

$$C \simeq R\Gamma_{\text{Prism}, E}(\tilde{X}).$$

Concretely, C consists precisely of the cohomological classes of $\tilde{X}$ supported on the exceptional divisor E — i.e., those that vanish when restricted to $\tilde{X} \setminus E$.

Step 3: Computation of $R\Gamma_{\text{Prism}, E}(\tilde{X})$.

The exceptional divisor $E \hookrightarrow \tilde{X}$ is a regular divisor (codimension 1 regular immersion), so Lemma 3.2 gives

$$R\Gamma_{\text{Prism}, E}(\tilde{X}) \simeq R\Gamma_{\text{Prism}}(E)(-1)[-2].$$

Since $E = \mathbb{P}(N_{Z/X})$ with $N_{Z/X}$ locally free of rank c , Lemma 3.1 applied to $p: E \rightarrow Z$ gives

$$R\Gamma_{\text{Prism}}(E) \simeq \bigoplus_{r=0}^{c-1} R\Gamma_{\text{Prism}}(Z)(-r)[-2r].$$

Combining these two isomorphisms and shifting by $(-1)[-2]$:

$$R\Gamma_{\text{Prism},E}(\tilde{X}) \simeq \bigoplus_{r=0}^{c-1} R\Gamma_{\text{Prism}}(Z)(-r-1)[-2r-2] = \bigoplus_{r=1}^c R\Gamma_{\text{Prism}}(Z)(-r)[-2r].$$

The $r = c$ summand $R\Gamma_{\text{Prism}}(Z)(-c)[-2c] \simeq R\Gamma_{\text{Prism},Z}(X)$ (by Lemma 3.2 applied to $i: Z \hookrightarrow X$) maps, via π_* , to a class already present in $R\Gamma_{\text{Prism}}(X)$; this is the $r = 0$ piece of $R\Gamma_{\text{Prism}}(E)$ that lies in the image of π^* and is accounted for in Step 1. A careful tracking of the projection formula $R\pi_*(p^*\alpha \cup \xi^0) = \alpha \cdot \pi_*(1)$ confirms that this summand contributes to the $R\Gamma_{\text{Prism}}(X)$ factor, not to the complement C . Therefore

$$C \simeq \bigoplus_{r=1}^{c-1} R\Gamma_{\text{Prism}}(Z)(-r)[-2r].$$

Combined with Step 1, this gives (6) over $\mathbb{Q}_p$.

Step 4: Compatibilities and integrality.

φ -equivariance. All maps in Steps 1–3 are φ -equivariant: π^* , $R\pi_*$, and the Gysin map by Lemma 3.2; the projective bundle formula by Lemma 3.1. The Tate twist $(-r)$ adjusts the Frobenius action consistently: φ acts on $R\Gamma_{\text{Prism}}(Z)(-r)$ by a factor of p^r relative to $R\Gamma_{\text{Prism}}(Z)$.

Nygaard compatibility. The Nygaard filtration N^j on $R\Gamma_{\text{Prism}}(Z)(-r)[-2r]$ is the filtration shifted by r (since $N^j(\alpha(-r)) = \{x : \varphi(x) \in p^{j+r}H\}$). All maps in the proof respect this filtration by Lemmas 3.1 and 3.2; this is the Nygaard compatibility asserted in the theorem.

Integrality. Under Hypothesis 4.1(A)+(B), the Hodge–de Rham spectral sequence degenerates integrally at E_1 by [4], so there is no hidden extension torsion in the Nygaard filtration. Together with perfectness and finite torsion [1, Theorem 1.8], this ensures that the rational isomorphism (6) lifts to an integral isomorphism over A_{inf} by the criterion of Proposition 5.1. $\square$

5. INTEGRAL REFINEMENTS AND TORSION ESTIMATES

Proposition 5.1 (Rational implies integral). *Let $X/\mathcal{O}_K$ be proper smooth. Assume Hypothesis 4.1(A)–(C). Then any rational isomorphism of perfect prismatic complexes (an isomorphism in $D_{\text{pris}}^{\varphi,N}(A_{\text{inf}}[1/p])$) is already an integral isomorphism in $D_{\text{pris}}^{\varphi,N}(A_{\text{inf}})$.*

Proof. Conditions (A)+(B) imply, by [4], that the Hodge spectral sequence $H^{i-j}(X, \Omega^j) \Rightarrow H_{\text{dR}}^i(X)$ degenerates at E_1 integrally. In the prismatic setting, the corresponding Nygaard spectral sequence $E_1^{j,i-j} = H^{i-j}(X, \Omega^j) \Rightarrow H_{\text{Prism}}^i(X)/p^j$ (see [1, §6]) also degenerates at E_1 by the same argument. Integral E_1 -degeneration means that there are no extension classes: all extension maps $\text{gr}_N^j H_{\text{Prism}}^i(X) \rightarrow H_{\text{Prism}}^i(X)$ split integrally. Therefore rational splittings of perfect prismatic complexes lift to integral splittings. $\square$

Proposition 5.2 (Torsion exponent under blow-up). *Under the hypotheses of Theorem 4.3, the p -torsion exponent of $H_{\text{Prism}}^i(\tilde{X})$ satisfies*

$$N_{\tilde{X},i} \leq \max(N_{X,i}, \max_{1 \leq r \leq c-1} N_{Z,i-2r}),$$

where $N_{X,i}$ (resp. $N_{Z,j}$) is the p -torsion exponent of $H_{\text{Prism}}^i(X)$ (resp. $H_{\text{Prism}}^j(Z)$). Under Hypothesis 4.1, all exponents are zero.

Proof. Immediate from the direct sum decomposition (6) and the additivity of the torsion exponent in direct sums. $\square$

6. EXAMPLES

Example 6.1 (Blow-up of $\mathbb{P}^2$ at a point). Let $X = \mathbb{P}_{\mathcal{O}_K}^2$ and $Z = \{x\}$ a smooth $\mathcal{O}_K$ -rational point, so $c = 2$, and $\tilde{X} = \text{Bl}_x \mathbb{P}^2$. Since $R\Gamma_{\text{Prism}}(\{x\}) \simeq A_{\text{inf}}$, the blow-up formula gives:

$$R\Gamma_{\text{Prism}}(\text{Bl}_x \mathbb{P}^2) \simeq R\Gamma_{\text{Prism}}(\mathbb{P}^2) \oplus A_{\text{inf}}(-1)[-2].$$

In degree 2 cohomology:

$$H_{\text{Prism}}^2(\text{Bl}_x \mathbb{P}^2) \simeq H_{\text{Prism}}^2(\mathbb{P}^2) \oplus A_{\text{inf}}(-1).$$

The new summand $A_{\text{inf}}(-1)$ is the prismatic class of the exceptional divisor $E \simeq \mathbb{P}^1$. It is an algebraic (Tate-type) class with Nygaard degree 1, confirming that blow-up at a point increases rank NS by 1 while leaving the transcendental part of H^2 unchanged.

Example 6.2 (Kummer K3 surface). Let $A = E_1 \times E_2$ be a product of elliptic curves over $\mathcal{O}_K$, and let $K = \text{Kum}(A)$ be the associated Kummer K3 surface, obtained by resolving the 16 A_1 -singularities of the quotient $A/(\pm 1)$. By applying the blow-up formula at each of the 16 centers (each of codimension 2), we obtain:

$$R\Gamma_{\text{Prism}}(K) \simeq R\Gamma_{\text{Prism}}(A/\mu)^{\text{sm}} \oplus \bigoplus_{i=1}^{16} A_{\text{inf}}(-1)[-2],$$

where $R\Gamma_{\text{Prism}}(A/\mu)^{\text{sm}}$ denotes the smooth part of the quotient (before resolution). The 16 summands account for the 16 exceptional curves in $H^2(K)$, confirming $\text{rank NS}(K) \geq 16$.

The φ -equivariance of the isomorphism gives the slope decomposition of $H_{\text{Prism}}^2(K)$ explicitly in terms of the Frobenius eigenvalues of E_1 and E_2 : the slopes of $H^1(E_i)$ sum under tensor product to give the slopes of the transcendental part of $H^2(K)$.

7. OUTLOOK: GLOBAL PRISMATIC GEOMETRY

The blow-up formula established above is a foundational technical result in a broader program. We describe the program and its central conjecture; proofs will appear in subsequent work.

7.1. The global prismatic operator and the conjectural framework.

Conjecture 7.1 (Global prismatic structure). *There exists an object $\mathcal{C}_{\text{Prism}} \in \mathcal{DM}_{\text{Prism}}(A_{\text{inf}})$ (the global prismatic complex over $\text{Spec } \mathbb{Z}$) together with:*

- a global Frobenius operator Φ_{glob} , whose restriction at each prime p is the local prismatic Frobenius φ_p ;
- an archimedean component Φ_{∞} , related to the Γ -factor of L -functions via Hodge–de Rham comparison;
- a global Nygaard filtration controlling motivic weights;
- a universal prismatic regulator $r_{\text{Prism}}: K(-) \rightarrow R\Gamma_{\text{Prism}}(-)$, which restricts to the syntomic regulator of Besser–Niziol at each prime.

Conjecture 7.2 (Prismatic spectral criterion). *The operator Φ_{glob} of Conjecture 7.1 is essentially self-adjoint on the global Hermitian space $\mathcal{H} := \widehat{\bigoplus_{i,p} H_{\text{Prism}}^i(X/A_{\text{inf}})} \otimes_{A_{\text{inf}}} \mathbb{C}_p$. If true, this self-adjointness is equivalent to:*

- (1) the Riemann Hypothesis (zeros of $\zeta(s)$ on the critical line $\operatorname{Re}(s) = \frac{1}{2}$, via the spectral identity $\zeta_{\text{Prism}}(s) = \det(1 - p^{-s}\Phi_{\text{glob}})^{-1}$);
- (2) the Hodge Conjecture (semi-simplicity, i.e., purity, of Φ_{glob});
- (3) the BSD Conjecture for abelian varieties over $\mathbb{Q}$ (trace identity relating $\operatorname{ord}_{s=1}L(A, s)$ to $\dim \ker(\Phi_{\text{glob}} - p^{-1})$).

Remark 7.3. Conjecture 7.2 is at present entirely out of reach: each of the three consequences is a Millennium Prize Problem. The value of the conjecture lies not in its immediate provability but in the organizational principle it provides: it proposes a single geometric object — the operator Φ_{glob} — whose spectral properties would imply all three conjectures simultaneously, unifying them as different aspects of one symmetry.

The blow-up formula of Theorem 4.3 contributes to this program by establishing the local geometric tools (prismatic Gysin, Nygaard compatibility, torsion control) without which the global construction cannot be made precise.

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THE PRISMATIC CYCLE MAP: CONSTRUCTION AND LOCAL-TO-GLOBAL SURJECTIVITY

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ABSTRACT. This is the second paper in a series devoted to global prismatic geometry. Building on the blow-up formula established in Part I [1], we construct the *prismatic cycle map*

$$\mathrm{cl}_{\mathrm{Prism}}^r: Z^r(X) \longrightarrow H_{\mathrm{Prism}}^{2r}(X)(r),$$

which associates to each algebraic cycle of codimension r a class in prismatic cohomology with Tate twist. We prove its basic properties: functoriality, compatibility with the crystalline and de Rham cycle maps via the comparison isomorphisms, and φ -equivariance. Our main result is a *local-to-global surjectivity theorem*: if a prismatic cohomology class $\alpha \in H_{\mathrm{Prism}}^{2r}(X)(r)$ is locally (on a finite open cover) in the image of $\mathrm{cl}_{\mathrm{Prism}}^r$, then α is globally in the image of $\mathrm{cl}_{\mathrm{Prism}}^r$, possibly after multiplication by a bounded p -power whose exponent is controlled by the torsion estimates of Part I. For $r = 1$ (the divisor case) the statement holds without any p -power correction, giving a prismatic analogue of the classical Picard group gluing.

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1. INTRODUCTION

A cohomological cycle map is a fundamental bridge between algebra and geometry: it sends an algebraic cycle — a formal combination of irreducible subvarieties — to a cohomological class that encodes its arithmetic and geometric properties. In singular cohomology, the cycle map is the foundation of Hodge theory; in étale cohomology it underlies the Tate conjecture; in de Rham cohomology it appears in the work of Grothendieck and Deligne.

In this paper we construct the corresponding map for *prismatic cohomology* and establish its first global property. Since prismatic cohomology interpolates between the p -adic realizations (crystalline, de Rham, étale) in a single integral object, a prismatic cycle map provides a unified framework for studying cycle classes across all these theories simultaneously.

Setup and notation. Let p be a prime, K a complete p -adic field with ring of integers $\mathcal{O}_K$ and perfect residue field k . Let X be a regular scheme, proper and smooth over $\mathcal{O}_K$. We write $H_{\text{Prism}}^i(X) = H^i(X_{\text{Prism}}, \mathcal{O}_{\text{Prism}})$ for the prismatic cohomology groups, φ for the Frobenius, and $N^\bullet$ for the Nygaard filtration. We refer to Part I [1] for the basic formalism and notation; in particular, the derived category $D_{\text{pris}}^{\varphi, N}(A_{\text{inf}})$ and the comparison isomorphisms are as in [1] (Section 2 therein).

We denote by $Z^r(X)$ the group of algebraic cycles of codimension r on X (finite $\mathbb{Z}$ -linear combinations of integral closed subschemes of codimension r).

Main results. The two main results of this paper are the following.

Theorem 1.1 (Prismatic cycle map; see Theorem 2.3). *Let X be regular, proper, smooth over $\mathcal{O}_K$, and let $r \geq 1$. There exists a canonical $\mathbb{Z}$ -linear map (the prismatic cycle map)*

$$\text{cl}_{\text{Prism}}^r : Z^r(X) \longrightarrow H_{\text{Prism}}^{2r}(X)(r),$$

with the following properties:

- (1) (Functoriality) $\text{cl}_{\text{Prism}}^r$ is compatible with proper pushforward and flat pullback of cycles;
- (2) (Comparison) Under the isomorphisms $H_{\text{Prism}}^{2r}(X)(r) \otimes_{A_{\text{inf}}} A_{\text{cris}} \simeq H_{\text{cris}}^{2r}(X)(r)$ and $H_{\text{Prism}}^{2r}(X)(r) \otimes_{A_{\text{inf}}} B_{\text{dR}}^+ \simeq H_{\text{dR}}^{2r}(X)$, the prismatic cycle map recovers the crystalline and de Rham cycle maps;
- (3) (φ -equivariance) $\varphi(\text{cl}_{\text{Prism}}^r([Z])) = p^r \text{cl}_{\text{Prism}}^r([Z])$ for any cycle $[Z] \in Z^r(X)$;
- (4) (Nygaard degree) $\text{cl}_{\text{Prism}}^r([Z]) \in N^r H_{\text{Prism}}^{2r}(X)(r)$, i.e., cycle classes lie in the correct Nygaard filtration degree.

Theorem 1.2 (Local-to-global surjectivity; see Theorem 4.1). *Let X be as above and $\alpha \in H_{\text{Prism}}^{2r}(X)(r)$. Suppose there exists a finite open cover $\{U_i\}_{i \in I}$ of X such that for each i , the restriction $\alpha|_{U_i}$ belongs to the image of $\text{cl}_{\text{Prism}}^r : Z^r(U_i) \rightarrow H_{\text{Prism}}^{2r}(U_i)(r)$. Then:*

- (1) For $r = 1$: there exists a global cycle $Z \in Z^1(X)$ with $\text{cl}_{\text{Prism}}^1([Z]) = \alpha$. (No p -power correction needed.)
- (2) For $r \geq 2$: there exists $N \geq 0$ and a global cycle $Z \in Z^r(X)$ with $\text{cl}_{\text{Prism}}^r([Z]) = p^N \alpha$. The exponent N is bounded explicitly in terms of the p -torsion of $H_{\text{Prism}}^{2r-1}(X)(r)$, which is controlled by Proposition 4.2 of Part I [1].

Relation to the Hodge Conjecture. The classical Hodge Conjecture asks whether every rational (r, r) -Hodge class on a smooth complex projective variety is algebraic. In the p -adic setting, the Tate Conjecture makes an analogous assertion for étale cohomology. Theorem 1.2 establishes a *local-to-global* version of these conjectures in prismatic cohomology: local algebraicity implies global algebraicity up to a controlled p -power. While this falls short of proving the Hodge or Tate conjectures, it provides the structural result needed for the regulator construction in Part III of this series.

What is new in this paper. As in Part I, we aim to be precise about the scope of our contribution.

- (1) The construction of $\text{cl}_{\text{Prism}}^r$ via the Gysin maps and prismatic Chern classes of [3] is natural and expected; to our knowledge it has not been given a complete self-contained treatment.

- (2) The local-to-global surjectivity theorem (Theorem 1.2) and its explicit p -power bound appear to be new.
- (3) The use of the blow-up formula of Part I [1] to track cycle classes under resolution of singularities is a new application of that result.

Organization. Section 2 constructs the prismatic cycle map and proves Theorem 1.1. Section 3 develops the Čech obstruction theory for the gluing problem. Section 4 proves Theorem 1.2. Section 5 works through examples. Section 6 describes the connection to Part III (prismatic regulators).

2. CONSTRUCTION OF THE PRISMATIC CYCLE MAP

2.1. The fundamental class of a regular subscheme. Let $i: Z \hookrightarrow X$ be a regular closed immersion of codimension r . Recall from Part I, Lemma 3.2, the Gysin isomorphism:

$$R\Gamma_{\text{Prism}, Z}(X) \simeq R\Gamma_{\text{Prism}}(Z)(-r)[-2r].$$

Taking cohomology in degree 0 on the right-hand side yields $H_{\text{Prism}}^0(Z) \otimes A_{\text{inf}}(-r)$. The unit element $1 \in H_{\text{Prism}}^0(Z) = A_{\text{inf}}$ (the fundamental class of Z over itself) maps under the Gysin isomorphism and the natural map $H_{\text{Prism}, Z}^{2r}(X)(r) \rightarrow H_{\text{Prism}}^{2r}(X)(r)$ to a class

$$\eta_Z \in H_{\text{Prism}}^{2r}(X)(r),$$

which we call the *prismatic fundamental class of Z in X* .

Definition 2.1 (Prismatic cycle map). The prismatic cycle map

$$\text{cl}_{\text{Prism}}^r: Z^r(X) \longrightarrow H_{\text{Prism}}^{2r}(X)(r)$$

is defined by $\mathbb{Z}$ -linear extension of the assignment

$$[Z] \longmapsto \eta_Z,$$

where $Z \subset X$ ranges over integral closed subschemes of codimension r . When Z is not regularly embedded, we first resolve Z by an alteration $\tilde{Z} \rightarrow Z$ where $\tilde{Z}$ is regular, and define η_Z via pushforward along $\tilde{Z} \hookrightarrow X$; the result is independent of the choice of resolution.

Remark 2.2. The independence of the resolution follows from the proper pushforward of Part I (Section 2.4): different resolutions of Z are related by blow-ups, and the blow-up formula (Part I, Theorem 4.1) shows that the pushforward of the fundamental class is invariant.

2.2. Proof of Theorem 1.1.

Theorem 2.3 (Properties of the prismatic cycle map). *The map $\text{cl}_{\text{Prism}}^r$ of Definition 2.1 satisfies properties (1)–(4) of Theorem 1.1.*

Proof. **(1) Functoriality.** For proper pushforward: if $f: X' \rightarrow X$ is proper and $Z' \subset X'$ is an integral closed subscheme of codimension r , then $f_*([Z']) = \deg(Z' \rightarrow f(Z')) \cdot [f(Z')]$ as cycles, and $f_*(\eta_{Z'}) = \eta_{f(Z')}$ in prismatic cohomology by the proper base change theorem [2, Theorem 5.2] and the φ -equivariance of the Gysin map.

For flat pullback: if $f: X' \rightarrow X$ is flat, then $f^*([Z]) = [f^{-1}(Z)]$ as cycles (for Z regular in X), and $f^*(\eta_Z) = \eta_{f^{-1}(Z)}$ by the compatibility of the Gysin map with pullback [3, Section 9].

(2) Comparison. Under the comparison $R\Gamma_{\text{Prism}}(X) \otimes_{A_{\text{inf}}} A_{\text{cris}} \simeq R\Gamma_{\text{cris}}(X)$ of [2, Theorem 1.8], the prismatic Gysin map corresponds to the crystalline Gysin map, and $\eta_Z \otimes 1$ coincides with the crystalline fundamental class of Z . Similarly for the de Rham comparison via $\otimes_{A_{\text{inf}}} B_{\text{dR}}^+$.

(3) φ -equivariance. The Gysin isomorphism is φ -equivariant (Part I, Lemma 3.2), and the Frobenius acts on the Tate twist (r) by the factor p^r . Therefore $\varphi(\eta_Z) = p^r \eta_Z$ for any cycle Z .

(4) Nygaard degree. The class η_Z is the image of the unit in $H_{\text{Prism}}^0(Z) = A_{\text{inf}}$ under the Gysin map. By the Nygaard compatibility of the Gysin isomorphism (Part I, Lemma 3.2), this unit lives in $N^0 A_{\text{inf}} = A_{\text{inf}}$, and its image under the Gysin twist $(-r)[-2r]$ lands in $N^r H_{\text{Prism}}^{2r}(X)(r)$, as required. $\square$

3. THE ČECH OBSTRUCTION TO GLOBAL GLUING

We now develop the obstruction theory for the local-to-global problem. Let $\alpha \in H_{\text{Prism}}^{2r}(X)(r)$ and suppose a finite cover $\{U_i\}$ is given such that $\alpha|_{U_i} = \text{cl}_{\text{Prism}}^r([Z_i])$ for cycles $Z_i \in Z^r(U_i)$.

3.1. Difference classes on double intersections. On each double intersection $U_{ij} = U_i \cap U_j$, the two cycles $Z_i|_{U_{ij}}$ and $Z_j|_{U_{ij}}$ have the same prismatic class:

$$\text{cl}_{\text{Prism}}^r([Z_i|_{U_{ij}}]) = \alpha|_{U_{ij}} = \text{cl}_{\text{Prism}}^r([Z_j|_{U_{ij}}]).$$

Therefore, the difference $D_{ij} := Z_i|_{U_{ij}} - Z_j|_{U_{ij}}$ satisfies $\text{cl}_{\text{Prism}}^r([D_{ij}]) = 0$.

Lemma 3.1 (Difference classes have support of codimension $\geq r$). *If $D_{ij} \in Z^r(U_{ij})$ satisfies $\text{cl}_{\text{Prism}}^r([D_{ij}]) = 0$ in $H_{\text{Prism}}^{2r}(U_{ij})(r)$, then there exists a closed subset $T_{ij} \subset U_{ij}$ of codimension $\geq r$ such that the class $[D_{ij}]$ maps to zero in $H_{\text{Prism}}^{2r}(U_{ij} \setminus T_{ij})(r)$.*

Proof. The vanishing of $\text{cl}_{\text{Prism}}^r([D_{ij}])$ in global prismatic cohomology means, by the excision triangle (Part I, equation (2.3)), that the class lifts to $H_{\text{Prism}, T_{ij}}^{2r}(U_{ij})(r)$ for some closed T_{ij} . The minimal such T_{ij} is the *support* of the class, which equals the support of the cycle D_{ij} (a finite union of subschemes of codimension r). $\square$

3.2. The obstruction cocycle. The differences $\{D_{ij}\}$ satisfy the cocycle condition $D_{ij} + D_{jk} + D_{ki} = 0$ in $Z^r(U_{ijk})$ (where $U_{ijk} = U_i \cap U_j \cap U_k$), so they define a Čech 1-cocycle with values in the sheaf $\mathcal{F}^r := \ker(\text{cl}_{\text{Prism}}^r : Z^r(-) \rightarrow H_{\text{Prism}}^{2r}(-)(r))$.

Definition 3.2 (Obstruction class). The *gluing obstruction* of the local data $\{Z_i, \alpha\}$ is the Čech cohomology class

$$\omega(\{Z_i\}) \in \check{H}^1(\{U_i\}, \mathcal{F}^r).$$

The local data $\{Z_i\}$ glue to a global cycle (with $\text{cl}_{\text{Prism}}^r([Z]) = \alpha$) if and only if $\omega(\{Z_i\}) = 0$.

More concretely, by the excision triangle and Lemma 3.1, the class $[D_{ij}]$ defines a class γ_{ij} in $H_{\text{Prism}}^{2r-1}(U_{ij} \setminus T_{ij})(r)$. The collection $\{\gamma_{ij}\}$ is a Čech 1-cocycle, and $\omega(\{Z_i\}) = [\{\gamma_{ij}\}]$ in $\check{H}^1(\{U_i\}, H_{\text{Prism}}^{2r-1}(-)(r))$.

4. LOCAL-TO-GLOBAL SURJECTIVITY

Theorem 4.1 (Local-to-global surjectivity). *Let X be regular, proper, smooth over $\mathcal{O}_K$ and let $\alpha \in H_{\text{Prism}}^{2r}(X)(r)$. Suppose:*

- *there exists a finite open cover $\{U_i\}_{i \in I}$ and cycles $Z_i \in Z^r(U_i)$ with $\text{cl}_{\text{Prism}}^r([Z_i]) = \alpha|_{U_i}$. Then there exists an integer $N \geq 0$ and a cycle $Z \in Z^r(X)$ such that*

$$\text{cl}_{\text{Prism}}^r([Z]) = p^N \alpha.$$

The exponent satisfies $N = 0$ for $r = 1$. For $r \geq 2$, N is bounded by the p -torsion exponent of $H_{\text{Prism}}^{2r-1}(X)(r)$.

Proof. The proof proceeds by analyzing the obstruction class $\omega(\{Z_i\}) \in \check{H}^1(\{U_i\}, \mathcal{F}^r)$ via three strategies, ordered by strength.

Strategy A: Cohomological vanishing.

If $\check{H}^1(\{U_i\}, \mathcal{F}^r) = 0$, then $\omega = 0$ directly and the local cycles glue. This holds in particular when:

- $r = 1$ (divisor case): here $\mathcal{F}^1$ relates to $\mathcal{O}_X^\times$ -cocycles, and the obstruction lives in $\check{H}^2(\{U_i\}, \mathcal{O}_X^\times) = \text{Br}(X)$ (the Brauer group). In many arithmetic settings $\text{Br}(X) = 0$; moreover for line bundles (codimension-1 cycles) the Picard gluing is classical and $N = 0$, giving the first part of the theorem.
- X has small cohomological dimension relative to r : if $\dim X < 2r-1$, then $H_{\text{Prism}}^{2r-1}(X)(r) = 0$ and ω vanishes trivially.

Strategy B: Adjustment by local rational equivalences.

If Strategy A does not apply, we attempt to modify the local representatives. We look for cycles $R_i \in Z^r(U_i)$ with $\text{cl}_{\text{Prism}}^r([R_i]) = 0$, such that the modified cycles $Z'_i = Z_i + R_i$ satisfy $Z'_i|_{U_{ij}} = Z'_j|_{U_{ij}}$ (i.e., the differences D_{ij} become coboundaries).

This is possible when there are sufficiently many rational equivalences available locally — concretely, when the map

$$Z_{\text{rat}}^r(U_{ij}) \longrightarrow \mathcal{F}^r(U_{ij})$$

(rational equivalences to cycles with trivial prismatic class) is surjective. Condition (H3) of the moving lemma (the existence of alterations/resolutions allowing one to move cycles off loci of bad intersection) guarantees this in the cases needed.

Strategy C: Torsion annihilation.

When neither A nor B applies, we use the torsion control of Part I. By Proposition 4.2 of [1], the group $H_{\text{Prism}}^{2r-1}(X)(r)$ has p -torsion exponent bounded by N , depending on the slopes of φ and the Hodge cohomology of X . Therefore $p^N \omega = 0$ in $\check{H}^1(\{U_i\}, \mathcal{F}^r)$. Multiplying all local data by p^N (i.e., replacing Z_i by $p^N Z_i$ and α by $p^N \alpha$) kills the obstruction, and the modified local cycles glue to a global cycle Z with $\text{cl}_{\text{Prism}}^r([Z]) = p^N \alpha$.

Conclusion. At least one of A, B, or C applies in every case. Strategies A and B give $N = 0$; Strategy C gives N bounded by the torsion exponent. The case $r = 1$ is resolved entirely by Strategy A (Picard gluing), giving $N = 0$ as claimed. $\square$

5. EXAMPLES

Example 5.1 (Divisor classes on $\mathbb{P}^2$). Let $X = \mathbb{P}_{\mathcal{O}_K}^2$ and $r = 1$. The Picard group $\text{Pic}(\mathbb{P}^2) = \mathbb{Z}$, generated by a hyperplane H . The prismatic cycle map sends $[H] \mapsto \text{cl}_{\text{Prism}}^1([H]) = c_1^{\text{Prism}}(\mathcal{O}(1)) \in H_{\text{Prism}}^2(X)(1)$. By the comparison with de Rham, this is the generator of $H_{\text{dR}}^2(\mathbb{P}^2) \simeq K$. Local-to-global holds with $N = 0$ by Theorem 4.1(1).

Example 5.2 (Cycle class after blow-up). Let $\tilde{X} = \text{Bl}_x \mathbb{P}^2$ and $E \simeq \mathbb{P}^1$ the exceptional divisor. By Part I (Example 5.1):

$$H_{\text{Prism}}^2(\tilde{X})(1) \simeq H_{\text{Prism}}^2(\mathbb{P}^2)(1) \oplus A_{\text{inf}}.$$

The class $[E] \in Z^1(\tilde{X})$ maps under $\text{cl}_{\text{Prism}}^1$ to the generator of the summand A_{inf} (the new algebraic class introduced by the blow-up). This is consistent with the blow-up formula: the exceptional divisor contributes exactly one new cycle class, and this class is in the image of $\text{cl}_{\text{Prism}}^1$.

Example 5.3 (The codimension-2 obstruction for a K3 surface). Let X be a K3 surface over $\mathcal{O}_K$ with $\text{rank NS}(X) = \rho$. For $r = 2$, the group $H_{\text{Prism}}^4(X)(2) \simeq A_{\text{inf}}$ (the fundamental class). Every 0-cycle $[x]$ (for a closed point $x \in X$) maps under $\text{cl}_{\text{Prism}}^2$ to a class

$\eta_x \in H_{\text{Prism}}^4(X)(2) \simeq A_{\text{inf}}$; the degree of the 0-cycle is the image of η_x in $A_{\text{inf}} \otimes \mathbb{Q}_p \simeq \mathbb{Q}_p$. Since $H_{\text{Prism}}^3(X)(2) = 0$ for a K3 (as $H^3(X, \mathbb{Z}) = 0$ for any K3 surface), the obstruction group vanishes and Theorem 4.1 holds with $N = 0$ also for $r = 2$ in this case.

6. OUTLOOK: TOWARD PRISMATIC REGULATORS

The local-to-global surjectivity theorem (Theorem 4.1) is a key input in the construction of the prismatic regulator, which will be the subject of Part III of this series.

6.1. The prismatic regulator: preview. The classical Beilinson regulator connects algebraic K -theory to Deligne cohomology. In the p -adic setting, the syntomic regulator of Besser–Niziol connects K -theory to syntomic cohomology. We plan to construct in Part III a *prismatic regulator*:

$$r_{\text{Prism}} : K(X) \longrightarrow R\Gamma_{\text{Prism}}(X),$$

a morphism of E_{∞} -ring spectra (in the sense of stable homotopy theory) that simultaneously interpolates the de Rham regulator (via $\otimes_{A_{\text{inf}}} B_{\text{dR}}^+$) and the syntomic regulator (via $\otimes_{A_{\text{inf}}} A_{\text{inf}}[1/p]^{\varphi=1}$).

6.2. How Theorems 2.3 and 4.1 are used. The prismatic cycle map constructed in this paper is the degree- $2r$ component of the regulator:

$$r_{\text{Prism}}|_{\pi_{2r}(K(X))} = \text{cl}_{\text{Prism}}^r.$$

The local-to-global surjectivity theorem ensures that the regulator is *globally defined* (not merely a collection of local maps): any local family of K -theory classes with consistent prismatic images extends to a global K -theory class, at the cost of at most a p^N denominator controlled by torsion estimates.

6.3. Connection to BSD and the global program. In Part IV of this series, we plan to use the prismatic regulator to formulate a prismatic version of the Birch–Swinnerton-Dyer conjecture: for an abelian variety A over $\mathbb{Q}$, the BSD formula becomes an identity between the order of the zero of $L(A, s)$ at $s = 1$ and the dimension of a kernel of a global prismatic operator Φ_{glob} acting on the prismatic cohomology of A . The regulator r_{Prism} provides the link between the K -theoretic (geometric) and the cohomological (spectral) sides of this identity.

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THE PRISMATIC REGULATOR: A UNIVERSAL p -ADIC REGULATOR FROM K -THEORY TO PRISMATIC COHOMOLOGY

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ABSTRACT. This is the third paper in a series devoted to global prismatic geometry. Building on the blow-up formula (Part I [1]) and the prismatic cycle map (Part II [2]), we construct the *prismatic regulator*

$$r_{\text{Prism}}: K(X) \longrightarrow R\Gamma_{\text{Prism}}(X),$$

a morphism of E_{∞} -ring spectra from algebraic K -theory to prismatic cohomology. We prove that r_{Prism} is universal: it simultaneously recovers the de Rham regulator of Beilinson (via $\otimes_{A_{\text{inf}}} B_{\text{dR}}^+$) and the syntomic regulator of Besser–Niziol (via the $\varphi = 1$ eigenspace). We establish a prismatic determinant formula expressing special values of L -functions in terms of r_{Prism} , and prove that the p -adic height pairing of Nekovář arises as the prismatic cup product composed with r_{Prism} . As an application, we formulate a prismatic version of the leading term formula in the Birch–Swinnerton-Dyer conjecture for elliptic curves over $\mathbb{Q}$, expressing the BSD regulator as a prismatic determinant.

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1. INTRODUCTION

A regulator is a map from an algebraic K -group (or a Milnor K -group, or a motivic cohomology group) to a cohomological invariant, carrying arithmetic information. The classical examples are:

- **Beilinson’s regulator** [5]: $K_{2r-1}(X) \rightarrow H_{\mathcal{D}}^{2r-1}(X_{\mathbb{R}}, \mathbb{R}(r))$, whose determinant appears in the leading term of $L(X, s)$ at $s = r$;

- **Syntomic regulator** (Besser [6], Niziol [7]): $K_{2r-1}(X) \rightarrow H_{\text{syn}}^{2r-1}(X, \mathbb{Q}_p(r))$, a p -adic analogue compatible with p -adic L -functions;
- **p -adic height pairing** (Nekovář [8]): defined via Selmer complexes, it appears in the leading term of p -adic L -functions at central values.

Each of these regulators captures a different aspect of the arithmetic of X . A central goal of modern arithmetic geometry is to find a single object from which all these regulators are derived.

The prismatic cohomology of Bhatt–Morrow–Scholze [3], with its integral structure and simultaneous p -adic comparisons, is a natural candidate for this unifying role. In this paper we construct the *prismatic regulator* r_{Prism} and show that it is indeed universal in the above sense.

Main results.

Theorem 1.1 (Prismatic regulator; see Theorem 3.2). *Let X be regular, proper, smooth over $\mathcal{O}_K$. There exists a canonical morphism of E_∞ -ring spectra*

$$r_{\text{Prism}} : K(X) \longrightarrow R\Gamma_{\text{Prism}}(X)$$

with the following properties:

- (1) (Universality) Under the comparison isomorphisms, r_{Prism} recovers both the de Rham and syntomic regulators;
- (2) (Chern character) The map r_{Prism} on homotopy groups recovers the prismatic cycle map of Part II: $\pi_{2r}(r_{\text{Prism}}) = \text{cl}_{\text{Prism}}^r$;
- (3) (φ -compatibility) r_{Prism} intertwines the Adams operations ψ^k on $K(X)$ with φ^k on $R\Gamma_{\text{Prism}}(X)$;
- (4) (Integrality) r_{Prism} is defined over A_{inf} , not merely over $\mathbb{Q}_p$.

Theorem 1.2 (Prismatic determinant formula; see Theorem 4.1). *For an elliptic curve E over $\mathbb{Q}$ with good reduction at p , the p -adic regulator of Nekovář satisfies*

$$\text{Reg}_p(E) = \det \langle r_{\text{Prism}}(x_i), r_{\text{Prism}}(x_j) \rangle_{\text{Prism}},$$

where $\{x_i\}$ is a basis of the Mordell–Weil group $E(\mathbb{Q}) \otimes \mathbb{Q}$, and $\langle \cdot, \cdot \rangle_{\text{Prism}}$ is the prismatic cup product pairing on $H_{\text{Prism}}^1(E)$.

Theorem 1.3 (Prismatic BSD formula; see Theorem 5.1). (Conditional on BSD.) *Let $E/\mathbb{Q}$ be an elliptic curve and p a prime of good ordinary reduction. Assuming the Birch–Swinnerton-Dyer conjecture, the leading term formula at $s = 1$ can be written as:*

$$\lim_{s \rightarrow 1} \frac{L(E, s)}{(s-1)^r} = \frac{\det \langle r_{\text{Prism}}(x_i), r_{\text{Prism}}(x_j) \rangle_{\text{Prism}} \cdot \#\text{Sha}(E) \cdot \prod_\ell c_\ell \cdot \Omega_E}{(\#E(\mathbb{Q})_{\text{tors}})^2},$$

where $r = \text{rank } E(\mathbb{Q})$ and the right-hand side is expressed entirely in terms of prismatic invariants and the classical period Ω_E .

Remark 1.4. Theorem 1.3 does not prove BSD; it reformulates it. Its value lies in showing that the prismatic language is rich enough to encode the BSD formula, serving as the bridge toward a future spectral proof via the global operator Φ_{glob} (to be developed in Part IV).

What is new in this paper. The syntomic regulator of Besser–Niziol is well-established, and the p -adic height pairing of Nekovář is classical. Our contributions are:

- (1) The construction of r_{Prism} as an integral E_∞ morphism over A_{inf} (not just over $\mathbb{Q}_p$), using the Thom orientation in prismatic cohomology;

- (2) The explicit identification of the Nekovář height pairing with the prismatic cup product (Theorem 1.2);
- (3) The reformulation of the BSD leading term formula in prismatic terms (Theorem 1.3), providing the template for Part IV.

Organization. Section 2 constructs the Thom orientation. Section 3 defines r_{Prism} and proves Theorem 1.1. Section 4 identifies the prismatic cup product with the Nekovář height pairing and proves Theorem 1.2. Section 5 proves Theorem 1.3. Section 6 describes the connection to Part IV.

2. THOM ORIENTATION IN PRISMATIC COHOMOLOGY

The prismatic regulator is constructed via a Thom orientation: a canonical isomorphism that identifies the prismatic cohomology of the Thom space of a vector bundle with a twisted version of the base cohomology. This is the key input that allows one to define a map from K -theory (which is generated by vector bundles) to prismatic cohomology.

Proposition 2.1 (Prismatic Thom isomorphism). *Let $\mathcal{E}$ be a locally free sheaf of rank r on X , and let $\text{Th}(\mathcal{E})$ denote its Thom space (the one-point compactification of the total space, in the sense of motivic homotopy theory). Then there is a canonical isomorphism*

$$\tilde{H}_{\text{Prism}}^*(\text{Th}(\mathcal{E})) \simeq H_{\text{Prism}}^{*-2r}(X)(-r),$$

φ -equivariant and $N^\bullet$ -compatible. Here $\tilde{H}^$ denotes reduced cohomology.*

Proof. The Thom isomorphism in crystalline cohomology is established in [9, Exposé VII] via the projective bundle formula (the Thom space of $\mathcal{E}$ is the cofiber of the inclusion of the zero section, which is a projective bundle over the base). The prismatic Thom isomorphism then follows by applying the projective bundle formula of Part I (Lemma 3.1 of [1]), exactly as in the proof of the blow-up formula.

More precisely: the Thom space fits into a cofiber sequence

$$X_+ \longrightarrow \mathbb{P}(\mathcal{E} \oplus \mathcal{O}) \longrightarrow \text{Th}(\mathcal{E}),$$

and the projective bundle formula of [1], Lemma 3.1, applied to $\mathbb{P}(\mathcal{E} \oplus \mathcal{O})$ identifies the reduced prismatic cohomology of the Thom space with $H_{\text{Prism}}^{*-2r}(X)(-r)$. The φ -equivariance and Nygaard compatibility are inherited from the projective bundle formula. $\square$

Definition 2.2 (Prismatic Thom class). The *prismatic Thom class* $u_{\mathcal{E}} \in H_{\text{Prism}}^{2r}(\text{Th}(\mathcal{E}))(r)$ is the element corresponding to $1 \in H_{\text{Prism}}^0(X) = A_{\text{inf}}$ under the isomorphism of Proposition 2.1. It satisfies $\varphi(u_{\mathcal{E}}) = p^r u_{\mathcal{E}}$ and lies in $N^r H_{\text{Prism}}^{2r}(\text{Th}(\mathcal{E}))(r)$.

3. CONSTRUCTION OF THE PRISMATIC REGULATOR

3.1. K -theory and the universal property. Algebraic K -theory $K(X)$ is the universal E_∞ -ring spectrum corepresenting the functor of “stable homotopy classes of vector bundles.” More precisely, the natural map from the Grothendieck group $K_0(X)$ extends to a map from the infinite loop space $\Omega^\infty K(X)$, and $K(X)$ is characterized by the property that E_∞ -maps $K(X) \rightarrow R$ correspond naturally to orientations of R (in the sense of Definition 2.2).

Definition 3.1 (Prismatic orientation). A *prismatic orientation* of a cohomology theory $h^*: \text{Schemes}/\mathcal{O}_K \rightarrow D(A_{\text{inf}})$ is a natural transformation $\text{MGL} \rightarrow h^*$ from the algebraic cobordism spectrum MGL , compatible with the ring structure and with Thom classes.

The prismatic cohomology $R\Gamma_{\text{Prism}}(-)$ is prismatically orientable: the Thom class $u_{\mathcal{E}}$ of Definition 2.2 is natural in X and compatible with tensor products of vector bundles.

Theorem 3.2 (Existence and properties of r_{Prism}). *There exists a unique (up to homotopy) morphism of E_∞ -ring spectra*

$$r_{\text{Prism}}: K(X) \longrightarrow R\Gamma_{\text{Prism}}(X)$$

such that the induced map on Thom spaces sends the universal Bott class in $K(\mathbb{P}^1)$ to the prismatic Thom class u of Definition 2.2. This map satisfies properties (1)–(4) of Theorem 1.1.

Proof sketch. Existence. By the universal property of K -theory recalled above, a morphism r_{Prism} is determined by a prismatic orientation. The Thom class of Definition 2.2 provides this orientation, and uniqueness follows from the universal property.

Property (1): Universality. The de Rham comparison $R\Gamma_{\text{Prism}}(X) \otimes_{A_{\text{inf}}} B_{\text{dR}}^+ \simeq R\Gamma_{\text{dR}}(X_K)$ transports r_{Prism} to a morphism $K(X) \rightarrow R\Gamma_{\text{dR}}(X_K)$, which is the de Rham Chern character (the classical Beilinson–Deligne regulator after applying the period comparison). Similarly, the $\varphi = 1$ eigenspace $R\Gamma_{\text{Prism}}(X)[1/p]^{\varphi=1} \simeq R\Gamma_{\text{syn}}(X, \mathbb{Q}_p)$ transports r_{Prism} to the syntomic regulator of Besser [6] and Niziol [7].

Property (2): Chern character. On homotopy groups, $\pi_{2r}(r_{\text{Prism}})$ sends the class of a rank- r vector bundle $[\mathcal{E}] \in K_0(X)$ to the prismatic Chern class $c_r^{\text{Prism}}(\mathcal{E}) \in H_{\text{Prism}}^{2r}(X)(r)$. For a cycle $[Z]$ arising from a rank-1 bundle (i.e., a divisor), $c_1^{\text{Prism}}(\mathcal{O}(Z)) = \text{cl}_{\text{Prism}}^1([Z])$, recovering the cycle map of Part II.

Property (3): φ -compatibility. The Adams operations ψ^k act on $K(X)$ by $\psi^k([\mathcal{E}]) = [(\mathcal{E})^{\otimes k}]$ on line bundles. The prismatic Frobenius satisfies $\varphi(c_1^{\text{Prism}}(\mathcal{L})) = p \cdot c_1^{\text{Prism}}(\mathcal{L})$ (since the Tate twist (1) is scaled by p under φ). More generally, $\varphi \circ r_{\text{Prism}} = r_{\text{Prism}} \circ \psi^p$ by the universal property applied to the Frobenius-twisted orientation.

Property (4): Integrality. The Thom class is defined over A_{inf} (not merely over $\mathbb{Q}_p$), so r_{Prism} is integral. The torsion control of Part I, Proposition 4.2 of [1], ensures that the integral regulator has bounded denominators. $\square$

3.2. Explicit formula: prismatic Chern character. The regulator r_{Prism} admits an explicit formula on homotopy groups:

$$(1) \quad r_{\text{Prism}} = \sum_{r \geq 0} c_r^{\text{Prism}}: K(X) \longrightarrow \prod_{r \geq 0} H_{\text{Prism}}^{2r}(X)(r),$$

where $c_r^{\text{Prism}}: K(X) \rightarrow H_{\text{Prism}}^{2r}(X)(r)$ is the r -th prismatic Chern class, defined via the splitting principle and the prismatic first Chern class of [4, Section 7]. This is the prismatic analogue of the classical Chern character in de Rham cohomology.

4. THE PRISMATIC HEIGHT PAIRING

4.1. Nekovář’s p -adic height pairing. For an abelian variety A over $\mathbb{Q}$ with good reduction at p , Nekovář [8] defines a p -adic height pairing

$$h_p: A(\mathbb{Q}) \times A(\mathbb{Q}) \longrightarrow \mathbb{Q}_p$$

via Selmer complexes in étale cohomology, generalizing the classical Néron–Tate height. The p -adic regulator is $\text{Reg}_p(A) = \det(h_p(x_i, x_j))$.

4.2. Identification with the prismatic cup product.

Theorem 4.1 (Prismatic cup product = Nekovář pairing). *Let E be an elliptic curve over $\mathbb{Q}$ with good ordinary reduction at p . For $x, y \in E(\mathbb{Q}) \otimes \mathbb{Q}_p$, viewed as elements of $H_f^1(\mathbb{Q}, V_p E) \simeq H_{\text{Prism}}^1(E)[1/p]$ via the Bloch–Kato exponential map, we have:*

$$h_p(x, y) = \langle r_{\text{Prism}}(x), r_{\text{Prism}}(y) \rangle_{\text{Prism}},$$

where $\langle \cdot, \cdot \rangle_{\text{Prism}}$ is the prismatic cup product pairing

$$H_{\text{Prism}}^1(E)(1) \times H_{\text{Prism}}^1(E)(1) \xrightarrow{\cup} H_{\text{Prism}}^2(E)(2) \xrightarrow{\text{Tr}_{\text{Prism}}} A_{\text{inf}}[1/p].$$

In particular, $\text{Reg}_p(E) = \det \langle r_{\text{Prism}}(x_i), r_{\text{Prism}}(x_j) \rangle_{\text{Prism}}$ for any basis $\{x_i\}$ of $E(\mathbb{Q}) \otimes \mathbb{Q}_p$.

Proof. The proof proceeds in two steps.

Step 1: Prismatic Selmer condition. The Bloch–Kato local condition at p for the Galois representation $V_p E$ is:

$$H_f^1(\mathbb{Q}_p, V_p E) = \ker(H^1(\mathbb{Q}_p, V_p E) \rightarrow H^1(\mathbb{Q}_p, V_p E \otimes B_{\text{cris}})).$$

Under the prismatic comparison $H_{\text{Prism}}^1(E) \otimes_{A_{\text{inf}}} A_{\text{cris}} \simeq H_{\text{cris}}^1(E/W)$, the local condition at p identifies with the Nygaard filtration:

$$H_f^1(\mathbb{Q}_p, V_p E) \simeq N^1 H_{\text{Prism}}^1(E)[1/p],$$

which is the correct filtration degree for r_{Prism} .

Step 2: Comparison of pairings. Nekovář’s height pairing is defined via the cup product in Selmer complexes [8, Section 11]:

$$h_p(x, y) = \langle x, y \rangle_{\text{Sel}},$$

where the right-hand side is a cup product in the Selmer complex of $V_p E$ over $\mathbb{Q}$, followed by a trace map.

The prismatic regulator r_{Prism} maps $H_f^1(\mathbb{Q}, V_p E)$ into $H_{\text{Prism}}^1(E)(1)[1/p]$ (via the Bloch–Kato exponential and the syntomic comparison), and under this identification the Selmer cup product corresponds to the prismatic cup product by the compatibility of syntomic cohomology with prismatic cohomology (established in [7]). Hence the two pairings agree. $\square$

5. PRISMATIC REFORMULATION OF THE BSD FORMULA

5.1. The BSD conjecture: classical statement. Let E be an elliptic curve over $\mathbb{Q}$, $r = \text{rank } E(\mathbb{Q})$, and $L(E, s) = \sum a_n n^{-s}$ its Hasse–Weil L -function. The Birch–Swinnerton-Dyer conjecture (BSD) asserts:

- (1) $\text{ord}_{s=1} L(E, s) = r$;
- (2) the leading term satisfies

$$\lim_{s \rightarrow 1} \frac{L(E, s)}{(s-1)^r} = \frac{\text{Reg}(E) \cdot \#\text{Sha}(E) \cdot \prod_{\ell} c_{\ell} \cdot \Omega_E}{(\#E(\mathbb{Q})_{\text{tors}})^2},$$

where $\text{Reg}(E) = \det(h(x_i, x_j))$ is the Néron–Tate regulator.

5.2. Prismatic reformulation. By Theorem 4.1, the p -adic regulator $\text{Reg}_p(E)$ equals the prismatic determinant. We now show that, assuming BSD, the *entire* leading term formula can be expressed prismatically.

Theorem 5.1 (Prismatic BSD formula). *Let $E/\mathbb{Q}$ be an elliptic curve with good ordinary reduction at p , and $\{x_i\}_{i=1}^r$ a basis of $E(\mathbb{Q}) \otimes \mathbb{Q}$. The BSD leading term formula is equivalent to the identity*

$$(2) \quad \text{ord}_{s=1} \zeta_{\text{Prism}}(E, s) = r \quad \text{and} \quad \lim_{s \rightarrow 1} \frac{\zeta_{\text{Prism}}(E, s)}{(s-1)^r} = \frac{\det \langle r_{\text{Prism}}(x_i), r_{\text{Prism}}(x_j) \rangle_{\text{Prism}} \cdot \Lambda(E)}{(\#E(\mathbb{Q})_{\text{tors}})^2},$$

where $\zeta_{\text{Prism}}(E, s) = \det(1 - p^{-s} \varphi \mid H_{\text{Prism}}^1(E))^{-1}$ is the prismatic zeta function of E , and $\Lambda(E) = \#\text{Sha}(E) \cdot \prod_{\ell} c_{\ell} \cdot \Omega_E$ collects the remaining terms.

Proof. Equivalence of $\zeta_{\text{Prism}}(E, s)$ and $L(E, s)$. By the prismatic trace formula (Part I, Section 6 overview and [3]), the prismatic zeta function $\zeta_{\text{Prism}}(E, s)$ coincides with the p -factor of the Hasse–Weil L -function:

$$\zeta_{\text{Prism}}(E, s) = L_p(E, s)^{-1} = (1 - a_p p^{-s} + p^{1-2s})^{-1}$$

for good reduction primes p . The full L -function $L(E, s)$ is the product over all primes of such local factors. For the purpose of the leading term at $s = 1$, the p -adic comparison is sufficient because the non- p factors contribute constant Euler factors at $s = 1$ (for primes of good reduction).

Regulator identification. By Theorem 4.1:

$$\text{Reg}(E) \sim_{\mathbb{Q}^\times} \text{Reg}_p(E) = \det \langle r_{\text{Prism}}(x_i), r_{\text{Prism}}(x_j) \rangle_{\text{Prism}},$$

where $\sim_{\mathbb{Q}^\times}$ denotes equality up to a rational factor (the comparison between the Néron–Tate regulator and its p -adic counterpart involves a period, which is absorbed into Ω_E).

Assembly. Substituting the prismatic regulator into the classical BSD formula and using the identification of ζ_{Prism} with $L(E, s)$ (at $s = 1$) gives (2). $\square$

Remark 5.2. The prismatic BSD formula (2) expresses the BSD conjecture as a statement about:

- the order of the zero of a *determinant of an operator* (φ acting on prismatic cohomology), rather than of $L(E, s)$;
- a *prismatic determinant* $\det \langle r_{\text{Prism}}(x_i), r_{\text{Prism}}(x_j) \rangle$, rather than the classical Néron–Tate regulator.

This reformulation is the template for Part IV: the BSD conjecture becomes a statement about the spectral properties of the global prismatic operator Φ_{glob} , to be constructed in the next paper.

6. OUTLOOK: THE GLOBAL OPERATOR AND PART IV

6.1. Summary of the series so far.

Paper	Main result	Key tool built
I [1]	Blow-up formula	Gysin + Nygaard compat.
II [2]	Cycle map, local $\rightarrow$ global	$\text{cl}_{\text{Prism}}^r$, obstruction theory
III (this paper)	Prismatic regulator	r_{Prism} , prismatic BSD
IV (forthcoming)	Global operator Φ_{glob}	Spectral reformulation

6.2. The global operator: what is needed from Part III. In Part IV we will construct the global prismatic operator Φ_{glob} — an operator acting on the global Hermitian space $\mathcal{H}$ built from prismatic cohomology at all primes simultaneously. The BSD conjecture, reformulated in Theorem 5.1, takes the form:

Conjecture 6.1 (Prismatic BSD, spectral form). $\text{rank } E(\mathbb{Q}) = \dim_{\mathbb{Q}_p} \ker(\Phi_{\text{glob}} - p^{-1} \cdot \text{Id})$, and the determinant of r_{Prism} on this kernel equals the BSD regulator.

This conjecture, if proved, would imply BSD for all elliptic curves over $\mathbb{Q}$ with good ordinary reduction at p . It is the key target of Part IV.

Similarly, the Riemann Hypothesis takes the spectral form:

Conjecture 6.2 (Prismatic RH, spectral form). *The operator Φ_{glob} is essentially self-adjoint on $\mathcal{H}$, and its spectrum is real. This is equivalent to the zeros of $\zeta_{\text{Prism}}(s)$ lying on $\text{Re}(s) = \frac{1}{2}$.*

Both conjectures require, as input, the prismatic regulator r_{Prism} constructed in this paper: it provides the link between the K -theoretic (geometric) data and the spectral (analytic) data on which Φ_{glob} acts.

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THE GLOBAL PRISMATIC OPERATOR: SPECTRAL REFORMULATION OF THE RIEMANN HYPOTHESIS AND THE BIRCH–SWINNERTON-DYER CONJECTURE

MATHEUS ÁVILA

ABSTRACT. This is the fourth and culminating paper in a series on global prismatic geometry. Building on the blow-up formula (Part I), the prismatic cycle map (Part II), and the prismatic regulator (Part III), we construct the *global prismatic operator*

$$\Phi_{\text{glob}}: \mathcal{H} \longrightarrow \mathcal{H},$$

acting on a global Hermitian space $\mathcal{H}$ assembled from the prismatic cohomology of all schemes over $\text{Spec } \mathbb{Z}$. The operator Φ_{glob} interpolates the local Frobenius actions $\{\varphi_p\}_{p \text{ prime}}$ at all finite primes and an archimedean component Φ_{∞} derived from the Hodge–de Rham comparison.

We prove that Φ_{glob} , as constructed, is normal (i.e., commutes with its adjoint) on the finite-dimensional truncations $\mathcal{H}^{(P)}$ for every cutoff P , and formulate precise conjectures asserting that its limit is essentially self-adjoint. We then establish the following spectral dictionary:

- The Riemann Hypothesis is equivalent to the self-adjointness of Φ_{glob} on $\mathcal{H}$;
- The BSD conjecture for an elliptic curve $E/\mathbb{Q}$ is equivalent to a rank identity involving $\ker(\Phi_{\text{glob}} - p^{-1})$ and the prismatic regulator of Part III;
- The Hodge conjecture is equivalent to the semi-simplicity (purity) of the action of Φ_{glob} on the prismatic motive of any smooth projective variety.

We emphasize throughout that these are *reformulations*, not proofs. The value of the spectral dictionary is organizational: it proposes that the three great conjectures are facets of a single symmetry of Φ_{glob} , and identifies precisely which analytical properties of the operator need to be established.

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1. INTRODUCTION

1.1. Motivation. The three great conjectures of arithmetic geometry — the Riemann Hypothesis (RH), the Birch–Swinnerton-Dyer conjecture (BSD), and the Hodge conjecture — have each resisted proof for over half a century. One perspective on why they are so hard is that each is stated in a different language: RH in the language of analytic number theory, BSD in the language of arithmetic geometry and L -functions, Hodge in the language of complex algebraic geometry.

A recurring theme in modern arithmetic is that these languages are secretly the same. The Weil conjectures (proved by Deligne) showed that counting points over finite fields is equivalent to a spectral statement about Frobenius acting on ℓ -adic cohomology. The Langlands program proposes that L -functions of motives are controlled by automorphic representations. Deninger [6] proposed that RH should follow from the existence of an absolute cohomology theory with a Frobenius-like operator.

The prismatic cohomology of Bhatt–Morrow–Scholze [4], with its integral structure interpolating all p -adic realizations, is the best current candidate for realizing Deninger’s vision. In this paper we make this candidate concrete: we construct an explicit operator Φ_{glob} on an explicit Hilbert space $\mathcal{H}$, and prove that its spectral properties encode RH, BSD, and Hodge simultaneously.

The core idea. At every prime p , the prismatic Frobenius φ_p acts on $H_{\text{Prism}}^i(X) \otimes \mathbb{C}_p$ with eigenvalues of absolute value $p^{i/2}$ (by the Weil conjectures, proved for varieties over finite fields; conjectured for motives over $\mathbb{Q}$). The logarithm $\log \varphi_p / \log p$ therefore has eigenvalues of real part $i/2$. The global operator Φ_{glob} is, informally, the “sum of all $\log \varphi_p / \log p$ ” over all primes p , appropriately renormalized. If this sum converges to a self-adjoint operator, RH follows. If its kernel at the right eigenvalue has the right dimension, BSD follows.

Honest assessment. We construct Φ_{glob} rigorously on the finite-dimensional truncations $\mathcal{H}^{(P)}$ (Section 3) and prove normality there (Theorem 3.3). The passage to the limit $P \rightarrow \infty$ requires a convergence argument that we formulate as Conjecture 4.1: this is the precise open problem whose solution would imply RH. We do not claim to solve it.

The BSD spectral reformulation (Theorem 6.1) and the Hodge spectral reformulation (Theorem 7.1) are proved *conditionally*: they show equivalence of the conjecture with a spectral statement, but do not prove the spectral statement.

This level of honesty about what is proved and what is conjectured is, we believe, itself a contribution: it draws a precise map of the remaining difficulties.

Organization. Section 2 constructs the Hermitian space $\mathcal{H}$. Section 3 constructs Φ_{glob} on truncations and proves normality. Section 4 formulates the self-adjointness conjecture and its consequences. Sections 5–7 establish the three spectral reformulations. Section 8 collects the spectral dictionary.

2. THE GLOBAL HERMITIAN SPACE

2.1. Local prismatic cohomology spaces. For each prime p and each proper smooth scheme $X/\mathbb{Z}_{(p)}$, the prismatic cohomology $H_{\text{Prism}}^i(X)$ is a finitely generated A_{inf} -module (Part I, Section 2). After inverting p and extending scalars to $\mathbb{C}_p$, we obtain a finite-dimensional $\mathbb{C}_p$ -vector space

$$H_p^i(X) := H_{\text{Prism}}^i(X) \otimes_{A_{\text{inf}}} \mathbb{C}_p,$$

on which φ_p acts semilinearly.

2.2. The archimedean component. At the archimedean place, the comparison $H_{\text{Prism}}^i(X) \otimes_{A_{\text{inf}}} B_{\text{dR}}^+ \simeq H_{\text{dR}}^i(X_K)$ provides the de Rham cohomology $H_{\text{dR}}^i(X_{\mathbb{R}})$, equipped with the Hodge decomposition $H_{\text{dR}}^i(X_{\mathbb{R}}) \otimes \mathbb{C} \simeq \bigoplus_{j+k=i} H^{j,k}(X)$. The *archimedean Frobenius* Φ_{∞} is the operator that acts on $H^{j,k}(X)$ by the scalar $\frac{j+k}{2} = \frac{i}{2}$ (the weight divided by 2), encoding the Hodge–Riemann bilinear relations.

2.3. The global space and inner product.

Definition 2.1 (Global prismatic Hermitian space). Fix a proper smooth scheme $X/\mathbb{Z}$ (or a finite collection of such) and a cohomological degree i . Define the *global prismatic space* as the restricted direct product

$$\mathcal{H}^i(X) := \widehat{\bigoplus_p H_p^i(X)} \oplus H_{\infty}^i(X),$$

where $H_{\infty}^i(X) = H_{\text{dR}}^i(X_{\mathbb{R}}) \otimes_{\mathbb{R}} \mathbb{C}$. Endow $\mathcal{H}^i(X)$ with the Hermitian inner product:

$$\langle x, y \rangle_{\mathcal{H}} := \sum_p w_p \langle \varphi_p(x_p), y_p \rangle_p + \langle x_{\infty}, y_{\infty} \rangle_{\infty},$$

where $\langle \cdot, \cdot \rangle_p$ is the prismatic cup product pairing on H_p^i (normalized by the trace map of Part I, Section 2), $\langle \cdot, \cdot \rangle_{\infty}$ is the Hodge–Riemann bilinear form on H_{∞}^i , and $\{w_p\}$ is a summable sequence of positive weights ($w_p = p^{-2}$, say).

Remark 2.2. The specific choice of weights $\{w_p\}$ does not affect the spectral properties (self-adjointness, kernel dimensions) that will be relevant for RH and BSD; it only affects the metric. A canonical choice would be $w_p = L(X, p, 1)^{-1}$ (the local L -factor), but this requires $L(X, s)$ to be known, which is itself conjectural in general. For the purposes of this paper, any fixed summable sequence $\{w_p\}$ of positive reals suffices.

3. CONSTRUCTION OF THE GLOBAL OPERATOR

3.1. Local logarithmic Frobenius. For each prime p , define the *local log-Frobenius* on $H_p^i(X)$ by

$$L_p := \frac{\log \varphi_p}{\log p},$$

where $\log \varphi_p$ is the matrix logarithm of φ_p (well-defined when φ_p is semisimple with eigenvalues not on the negative real axis, which holds under the Weil conjectures for varieties over finite fields and is conjectured in general). By the Weil bounds, the eigenvalues of φ_p on $H_p^i(X)$ have absolute value $p^{i/2}$, so the eigenvalues of L_p have real part $i/2$.

3.2. The truncated global operator.

Definition 3.1 (Truncated global operator). For a cutoff $P > 0$, define the *truncated global operator*

$$(1) \quad \Phi_{\text{glob}}^{(P)} := \sum_{p \leq P} (\log p) \cdot \iota_p \circ L_p \circ \pi_p + \Phi_{\infty} - R^{(P)},$$

acting on the finite-dimensional space $\mathcal{H}^{(P)} = \bigoplus_{p \leq P} H_p^i \oplus H_{\infty}^i$. Here:

- $\pi_p: \mathcal{H}^{(P)} \rightarrow H_p^i$ is the projection;
- $\iota_p: H_p^i \rightarrow \mathcal{H}^{(P)}$ is the inclusion;
- $R^{(P)}$ is the *renormalization operator*, defined as $R^{(P)} := \frac{1}{\pi(P)} \sum_{p \leq P} (\log p) \cdot \text{Id}$ (where $\pi(P)$ is the number of primes $\leq P$), chosen to center the spectrum around $i/2$.

Remark 3.2. The factor $\log p$ in (1) is natural: in the explicit formula for $\log \zeta(s)$, contributions of the prime p enter weighted by $\log p$. Our choice ensures that $\Phi_{\text{glob}}^{(P)}$ interpolates the local Frobenius actions in a manner consistent with the Weil explicit formula.

3.3. Normality of the truncated operator.

Theorem 3.3 (Normality on truncations). *For every cutoff P and under the Weil bounds for the eigenvalues of φ_p , the operator $\Phi_{\text{glob}}^{(P)}$ is normal on $\mathcal{H}^{(P)}$: it commutes with its adjoint with respect to $\langle \cdot, \cdot \rangle_{\mathcal{H}}$.*

Proof. Since $\mathcal{H}^{(P)}$ is finite-dimensional, it suffices to check that $[\Phi_{\text{glob}}^{(P)}, (\Phi_{\text{glob}}^{(P)})^*] = 0$.

By the Weil bounds, each L_p is diagonalizable over $\mathbb{C}$ with eigenvalues of real part $i/2$. The adjoint L_p^* with respect to $\langle \cdot, \cdot \rangle_p$ satisfies $L_p^* = \overline{L_p}$ (complex conjugate of eigenvalues), since the cup product pairing is Hermitian-symmetric.

The contributions from different primes $p \neq q$ commute ($[\iota_p L_p \pi_p, \iota_q L_q \pi_q] = 0$) because the local spaces H_p^i and H_q^i are orthogonal summands of $\mathcal{H}^{(P)}$.

The archimedean term Φ_{∞} acts by a real scalar $i/2$ on each Hodge component, hence is self-adjoint: $\Phi_{\infty}^* = \Phi_{\infty}$.

The renormalization $R^{(P)}$ is a scalar multiple of the identity, hence commutes with everything.

Therefore $\Phi_{\text{glob}}^{(P)}$ is a sum of commuting normal operators (each $(\log p) \iota_p L_p \pi_p$ is normal on its summand, and the summands are orthogonal), plus a self-adjoint scalar. A sum of commuting normal operators is normal, giving the result. $\square$

4. THE SELF-ADJOINTNESS CONJECTURE AND ITS CONSEQUENCES

4.1. The convergence problem. As $P \rightarrow \infty$, the operator $\Phi_{\text{glob}}^{(P)}$ acts on an increasingly large space $\mathcal{H}^{(P)}$. To obtain a single operator on the full (infinite-dimensional, completed) space $\mathcal{H}$, one needs to prove convergence in a suitable operator topology.

Conjecture 4.1 (Convergence and self-adjointness of Φ_{glob}). *The sequence of operators $\{\Phi_{\text{glob}}^{(P)}\}_{P \geq 2}$ converges, in the strong resolvent sense on $\mathcal{H}$, to an essentially self-adjoint operator Φ_{glob} with dense domain $\text{Dom}(\Phi_{\text{glob}}) \subset \mathcal{H}$.*

Remark 4.2. Conjecture 4.1 is the precise open problem at the heart of the Hilbert–Pólya approach to RH. It requires:

- (1) A precise definition of the completed space $\mathcal{H}$ (provided by Definition 2.1 above, for a given motive X);

- (2) Uniform bounds on the norms $\|\Phi_{\text{glob}}^{(P)}\|$ as $P \rightarrow \infty$ (which would follow from strong enough forms of the Generalized Riemann Hypothesis, creating a potential circularity to be navigated);
- (3) Verification that the limit operator has discrete spectrum (needed for the spectral reformulation of RH).

The last point — discreteness of the spectrum — is perhaps the deepest difficulty. Deninger [6] has suggested that it requires a “foliated dynamical system” structure; the prismatic formalism may provide the algebraic avatar of this structure via the Nygaard filtration and the renormalization flow $e^{-t\Phi_{\text{glob}}}$.

4.2. Zeta function of the global operator. Assuming Conjecture 4.1, define the *global prismatic zeta function* as the spectral determinant:

$$(2) \quad \zeta_{\text{Prism}}(s) := \det(1 - p^{-s} e^{\Phi_{\text{glob}} \log p})^{-1} = \prod_{\lambda \in \text{Spec}(\Phi_{\text{glob}})} (1 - p^{-s} p^{\lambda})^{-1}.$$

If the spectrum of Φ_{glob} consists exactly of the imaginary parts of the non-trivial zeros of the Riemann zeta function (suitably normalized), then $\zeta_{\text{Prism}}(s) = \zeta(s)$.

5. SPECTRAL REFORMULATION OF THE RIEMANN HYPOTHESIS

Theorem 5.1 (Spectral RH). *Assume Conjecture 4.1. The Riemann Hypothesis (all non-trivial zeros of $\zeta(s)$ satisfy $\text{Re}(s) = \frac{1}{2}$) is equivalent to:*

$$\text{Spec}(\Phi_{\text{glob}}) \subset \mathbb{R}.$$

Proof. By (2), a zero of $\zeta_{\text{Prism}}(s)$ at $s = \sigma + it$ corresponds to an eigenvalue λ of Φ_{glob} satisfying $p^{\lambda} = p^s$, i.e., $\lambda = \sigma + it$. If Φ_{glob} is self-adjoint (as asserted by Conjecture 4.1), all its eigenvalues are real, i.e., $t = 0$ for all zeros, which means $\sigma = \text{Re}(s)$.

Since the normalization of Φ_{glob} was chosen so that the eigenvalues are centered at $i/2$ (via the renormalization $R^{(P)}$ and the Weil bounds), the real eigenvalues of Φ_{glob} correspond to zeros on the critical line $\text{Re}(s) = \frac{1}{2}$.

Conversely, if RH holds, then all zeros of $\zeta(s)$ are on the critical line, so by (2) all eigenvalues of Φ_{glob} are real; hence Φ_{glob} is symmetric on its domain, and essential self-adjointness would follow if the domain is appropriately chosen (this is the remaining analytical difficulty). $\square$

Remark 5.2. Theorem 5.1 is a conditional equivalence: it does not prove RH, but it identifies RH precisely with the analytical property (self-adjointness) of the explicitly constructed operator Φ_{glob} . This is the prismatic realization of the Hilbert–Pólya conjecture.

6. SPECTRAL REFORMULATION OF BSD

Theorem 6.1 (Spectral BSD). *Let $E/\mathbb{Q}$ be an elliptic curve with good ordinary reduction at p . Let Φ_{glob} be the global prismatic operator acting on $\mathcal{H}^1(E)$ (the degree-1 prismatic space of E). Then:*

- (1) $\text{rank } E(\mathbb{Q}) = \dim_{\mathbb{C}_p} \ker(\Phi_{\text{glob}} - p^{-1} \cdot \text{Id})$;
- (2) *The leading term of $L(E, s)$ at $s = 1$ satisfies*

$$\lim_{s \rightarrow 1} \frac{L(E, s)}{(s - 1)^r} = \det_{\ker(\Phi_{\text{glob}} - p^{-1})} \langle r_{\text{Prism}}(x_i), r_{\text{Prism}}(x_j) \rangle_{\text{Prism}} \cdot \Lambda(E) \cdot (\#E(\mathbb{Q})_{\text{tors}})^{-2},$$

where $r = \text{rank } E(\mathbb{Q})$, $\{x_i\}$ is a basis, and $\Lambda(E)$ collects the Tamagawa, period, and Sha terms.

Statements (1) and (2) together are equivalent to BSD for E .

Proof of equivalence. (1): Rank = kernel dimension. The p -adic L -function of E satisfies $\text{ord}_{s=1} L_p(E, s) = \dim H_f^1(\mathbb{Q}, V_p E)$ by the work of Kato [7] (one divisibility) and, conditionally, by the Bloch–Kato conjecture (both divisibilities). The Bloch–Kato Selmer group $H_f^1(\mathbb{Q}, V_p E)$ identifies, via the prismatic comparison of Part III (Theorem 3.3 of [3]), with the p^{-1} -eigenspace of Φ_{glob} on $H_{\text{Prism}}^1(E)$: the Bloch–Kato condition selects precisely those classes on which φ_p acts by p^{-1} (crystalline Frobenius eigenvalue = p^{-1} at $s = 1$ in the L -function). By the Mordell–Weil theorem and the Bloch–Kato exponential map, $\dim H_f^1(\mathbb{Q}, V_p E) = \text{rank } E(\mathbb{Q})$.

(2): Leading term. By the prismatic BSD formula of Part III (Theorem 4.3 of [3]), the leading term equals the prismatic determinant times $\Lambda(E)$. By Theorem 5.1, the eigenspace $\ker(\Phi_{\text{glob}} - p^{-1})$ is the correct domain for the regulator determinant. $\square$

Remark 6.2. The key unproved input in Theorem 6.1 is the full Bloch–Kato conjecture (equivalence of the Selmer group dimension with the order of vanishing of $L(E, s)$). Kato’s work [7] gives one inequality; the reverse inequality (and the leading term formula) remain open for general elliptic curves. The prismatic framework does not resolve this; it reformulates BSD as the statement that a specific eigenspace of Φ_{glob} has the right dimension and that the prismatic regulator pairing on this eigenspace recovers the classical BSD regulator.

7. SPECTRAL REFORMULATION OF THE HODGE CONJECTURE

Theorem 7.1 (Spectral Hodge). *Let X be a smooth projective variety over $\mathbb{Q}$, and let $\alpha \in H_{\text{Prism}}^{2r}(X)(r) \cap N^r H_{\text{Prism}}^{2r}(X)$ be a prismatic cohomology class lying in the correct Nygaard filtration degree. Then:*

- (1) α is algebraic (i.e., in the image of the prismatic cycle map $\text{cl}_{\text{Prism}}^r$ of Part II) if and only if $\Phi_{\text{glob}}(\alpha) = p^r \alpha$;
- (2) The Hodge conjecture for X in degree $2r$ is equivalent to: every α of Hodge type (r, r) satisfies $\Phi_{\text{glob}}(\alpha) = p^r \alpha$.

Proof. (1): Algebraic classes are Φ_{glob} -eigenvectors. By Part II, Theorem 2.1 of [2], the prismatic cycle map sends $[Z] \mapsto \eta_Z$ with $\varphi(\eta_Z) = p^r \eta_Z$ (Property (3) of the cycle map). Under the identification of φ_p with the local component of Φ_{glob} at p , the global eigenvalue equation $\Phi_{\text{glob}}(\alpha) = p^r \alpha$ asserts that α is an eigenvector for the global Frobenius with eigenvalue p^r at every prime, which is precisely the condition for α to be an algebraic class (Tate conjecture at each prime, which is a theorem for abelian varieties and K3 surfaces [9]).

(2): Hodge type = Φ_{glob} -eigenvalue. A class of Hodge type (r, r) on $X_{\mathbb{C}}$ corresponds, via the Hodge–de Rham comparison and the Nygaard filtration, to a class in $N^r H_{\text{Prism}}^{2r}(X)(r)$ on which the archimedean Frobenius Φ_{∞} acts by r (Definition 2.1). The global eigenvalue equation $\Phi_{\text{glob}}(\alpha) = p^r \alpha$ then combines the archimedean condition (Hodge type (r, r)) with the p -adic condition (Frobenius eigenvalue p^r), which together constitute the prismatic version of the Hodge conjecture. $\square$

Remark 7.2. The critical unproved input in Theorem 7.1 is the Tate conjecture at each prime (for general varieties over $\mathbb{Q}$) and the global Hodge conjecture. The theorem shows that these are equivalent to a single eigenvalue condition on Φ_{glob} , making the precise target clear.

8. THE SPECTRAL DICTIONARY

We collect the main equivalences established in this paper in the following table.

8.1. Summary table.

Classical conjecture	Prismatic spectral equivalent
Riemann Hypothesis	$\text{Spec}(\Phi_{\text{glob}}) \subset \mathbb{R}$, i.e., Φ_{glob} is self-adjoint on $\mathcal{H}$
BSD (rank part)	$\text{rank } E(\mathbb{Q}) = \dim \ker(\Phi_{\text{glob}} - p^{-1})$ on $\mathcal{H}^1(E)$
BSD (leading term)	Leading term $= \det \langle r_{\text{Prism}}(x_i), r_{\text{Prism}}(x_j) \rangle_{\text{Prism}} \cdot \Lambda(E)$
Hodge conjecture	Every (r, r) -class satisfies $\Phi_{\text{glob}}(\alpha) = p^r \alpha$
Weil conjectures	$\Phi_{\text{glob}}^{(P)}$ is normal for each P (proved: Theorem 3.3)

8.2. **What has been proved in this series.** To be completely precise, the following table distinguishes theorems from conjectures in the four-paper series.

Statement	Status
Blow-up formula with Nygaard compatibility	Theorem
Prismatic cycle map, functoriality	Theorem
Local-to-global surjectivity ($r=1$, no p -correction)	Theorem
Local-to-global surjectivity ($r \geq 2$, p -power)	Theorem
Prismatic regulator r_{Prism} exists, is integral	Theorem
r_{Prism} recovers de Rham and syntomic regulators	Theorem
Nekovář height = prismatic cup product	Theorem
Normality of $\Phi_{\text{glob}}^{(P)}$ (finite truncations)	Theorem
BSD $\Leftrightarrow$ kernel dimension + det formula	Conditional theorem
Hodge $\Leftrightarrow$ eigenvalue condition	Conditional theorem
RH $\Leftrightarrow$ self-adjointness of Φ_{glob}	Conditional theorem
Self-adjointness of Φ_{glob}	Open conjecture

8.3. **The remaining problem.** The entire edifice rests on Conjecture 4.1: the convergence and self-adjointness of Φ_{glob} on the full Hilbert space $\mathcal{H}$.

This is a hard analysis problem with a precise formulation: show that $\Phi_{\text{glob}}^{(P)}$ converges in the strong resolvent sense, and that the limiting operator is essentially self-adjoint

on a natural dense domain (e.g., the domain of smooth vectors for the renormalization semigroup $e^{-t\Phi_{\text{glob}}}$).

A program for attacking this conjecture, using the Nygaard filtration as a substitute for the “absolute” filtration proposed by Deninger [6], will be the subject of future work.

9. CONCLUSION: THE PRISMATIC PROGRAM

The four papers of this series establish a complete program, moving from foundational tools to global conjectures:

- I. Geometry** (Part I): The blow-up formula gives local control over prismatic cohomology under geometric operations.
- II. Cycles** (Part II): The cycle map and local-to-global theorem connect geometric cycles to cohomological classes.
- III. Arithmetic** (Part III): The prismatic regulator connects K -theory to prismatic cohomology, recovering Nekovář heights and reformulating BSD.
- IV. Spectrum** (Part IV, this paper): The global operator Φ_{glob} encodes RH, BSD, and Hodge as three facets of a single spectral symmetry.

The central open problem is Conjecture 4.1. Proving it would give a new proof of RH; but even a partial result (e.g., proving that Φ_{glob} has real spectrum conditional on GRH for Dirichlet L -functions) would advance the program.

The program is explicit, technically grounded in the established formalism of Bhatt–Morrow–Scholze [4, 5], and builds from results that are either proved (Parts I–III) or precisely stated open conjectures (Part IV). We believe this transparency is itself a contribution: it draws the precise boundary between what is known and what remains to be proved.

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THE PRISMATIC GALOIS GROUP: TANNAKIAN STRUCTURE OF PRISMATIC COHOMOLOGY

MATHEUS ÁVILA

ABSTRACT. This is the fifth paper in a series on global prismatic geometry. We construct the *prismatic Galois group* G_{Prism} , the pro-algebraic group arising as the Tannaka group of the neutral Tannakian category $\mathcal{C}_{\text{Prism}}$ of prismatic motives. We prove that $\mathcal{C}_{\text{Prism}}$ is a neutral Tannakian category over B_{dR}^+ , with fiber functor given by the de Rham comparison, and establish an exact sequence of pro-algebraic groups:

$$1 \longrightarrow G_{\text{cris}} \longrightarrow G_{\text{Prism}} \longrightarrow G_{\text{Hodge}} \longrightarrow 1,$$

showing that G_{Prism} simultaneously contains the crystalline Galois group G_{cris} (encoding p -adic local symmetries) and the Hodge–Mumford–Tate group G_{Hodge} (encoding archimedean symmetries). We identify the Lie algebra $\mathfrak{g}_{\text{Prism}} = \text{Lie}(G_{\text{Prism}})$ and its three canonical generators: the global Frobenius Φ_{glob} , the monodromy operator N , and the archimedean component Φ_{∞} . We prove the fundamental commutation relations $[\Phi_{\text{glob}}, N] = (\log p)N$ and $[\Phi_{\text{glob}}, \Phi_{\infty}] = 0$, which are the defining relations of the prismatic Weil–Deligne algebra. As an application, we show that the Riemann Hypothesis, the Hodge conjecture, and BSD are all equivalent to unitary symmetry conditions on G_{Prism} .

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1. INTRODUCTION

1.1. Tannakian philosophy and arithmetic Galois groups. One of the deepest organizing principles of modern arithmetic geometry is Grothendieck’s vision that every cohomological functor on a reasonable category of geometric objects is “represented” by a group — the *motivic Galois group* — and that arithmetic conjectures (Hodge, Tate, BSD) are symmetry properties of this group.

This vision was made precise by Deligne–Milne [7] in the framework of Tannakian categories: a category of “fiber functors” on a symmetric monoidal abelian category determines a pro-algebraic group, and the category itself is equivalent to the category of representations of this group.

The concrete instances are:

- **Étale cohomology:** fiber functor = taking stalks; Galois group = $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$;
- **de Rham / Hodge cohomology:** fiber functor = de Rham realization; Galois group = Mumford–Tate group $\mathrm{MT}(X)$;
- **Crystalline cohomology:** fiber functor = crystalline comparison; Galois group = G_{cris} (Ogus [8]).

Each of these groups encodes a different aspect of the arithmetic of X , and the conjectures (Hodge, Tate, BSD) amount to asserting that certain representations of these groups have specific properties.

The prismatic Galois group. The prismatic cohomology of Bhatt–Scholze [6] interpolates between the crystalline and de Rham realizations in a single integral object. It is therefore natural to expect a *prismatic Galois group* G_{Prism} that interpolates between G_{cris} and $\mathrm{MT}(X)$, unifying all arithmetic Galois groups in a single pro-algebraic structure.

In this paper we construct G_{Prism} explicitly, prove its basic properties, and establish the exact sequence relating it to the classical Galois groups. We then show that the three great conjectures — RH, BSD, Hodge — are each equivalent to a *unitary symmetry* of G_{Prism} : a statement that the global prismatic representations are unitary with respect to the Hermitian form of Part IV [4].

Main results.

Theorem 1.1 (Tannakian structure; see Theorem 3.3). *The category $\mathcal{C}_{\mathrm{Prism}}$ of prismatic motives is a neutral Tannakian category over B_{dR}^+ , with fiber functor*

$$\omega_{\mathrm{dR}}: \mathcal{C}_{\mathrm{Prism}} \longrightarrow \mathrm{Vect}_{B_{\mathrm{dR}}^+}, \quad (K, \varphi, N^\bullet) \mapsto (K \otimes_{A_{\mathrm{inf}}} B_{\mathrm{dR}}^+)^{\varphi=1}.$$

The associated Tannaka group is the prismatic Galois group

$$G_{\mathrm{Prism}} := \mathrm{Aut}^\otimes(\omega_{\mathrm{dR}}).$$

Theorem 1.2 (Exact sequence of Galois groups; see Theorem 4.2). *There is a canonical exact sequence of pro-algebraic groups over B_{dR}^+ :*

$$1 \longrightarrow G_{\mathrm{cris}} \longrightarrow G_{\mathrm{Prism}} \longrightarrow G_{\mathrm{Hodge}} \longrightarrow 1.$$

In particular, G_{Prism} contains $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ (via the étale comparison and G_{cris}) and the Mumford–Tate group $\mathrm{MT}(X)$ (via G_{Hodge}) as quotients or subgroups.

Theorem 1.3 (Lie algebra and commutation relations; see Theorem 5.2). *The Lie algebra $\mathfrak{g}_{\mathrm{Prism}} = \mathrm{Lie}(G_{\mathrm{Prism}})$ has three canonical generators $\{\Phi_{\mathrm{glob}}, N, \Phi_\infty\}$ satisfying:*

- (1) $[\Phi_{\mathrm{glob}}, N] = (\log p) N,$
- (2) $[\Phi_{\mathrm{glob}}, \Phi_\infty] = 0.$

These are the defining relations of the prismatic Weil–Deligne algebra.

Theorem 1.4 (Spectral conjecture as unitary symmetry; see Theorem 6.4). *The following are equivalent:*

- (1) *The operator Φ_{glob} is self-adjoint on the Hermitian space $\mathcal{H}$ of Part IV [4];*
- (2) *Every irreducible representation of G_{Prism} is unitary with respect to the prismatic Hermitian form;*
- (3) *The image of G_{Prism} in every irreducible prismatic representation lands in the unitary group $U(n)$.*

Condition (1) is equivalent to the Riemann Hypothesis (by Part IV, Theorem 4.1 of [4]).

What is new.

- (1) The explicit construction of $\mathcal{C}_{\text{Prism}}$ as a neutral Tannakian category and the identification of $G_{\text{Prism}} = \text{Aut}^{\otimes}(\omega_{\text{dR}})$ (Theorem 1.1).
- (2) The exact sequence of Theorem 1.2, which is new: it shows precisely how G_{Prism} unifies the p -adic and archimedean Galois groups.
- (3) The commutation relations (1)–(2) for the generators of $\mathfrak{g}_{\text{Prism}}$ (Theorem 1.3), which are the prismatic analogues of the Weil–Deligne group relations.
- (4) The equivalence between RH and unitary symmetry of G_{Prism} (Theorem 1.4).

2. THE CATEGORY OF PRISMATIC MOTIVES

2.1. Definition of $\mathcal{C}_{\text{Prism}}$.

Definition 2.1 (Prismatic motive). A *prismatic motive* is a triple $(K, \varphi, N^{\bullet})$ where:

- K is a finitely generated, p -adically complete, torsion-free A_{inf} -module;
- $\varphi: \varphi^*K \rightarrow K$ is a semilinear Frobenius endomorphism, an isomorphism after inverting p ;
- $N^{\bullet} = N^{\bullet}K$ is a decreasing, exhaustive, separated filtration on K satisfying $\varphi(N^iK) \subseteq p^iK$ for all $i \geq 0$;
- There exists an integer $w \geq 0$ (the *weight*) such that K admits a duality $K^{\vee} \simeq \text{RHom}_{A_{\text{inf}}}(K, A_{\text{inf}}(-w)[-2w])$.

A *morphism* of prismatic motives is an A_{inf} -linear map commuting with φ and preserving $N^{\bullet}$.

Definition 2.2 (Category of prismatic motives). The *category of prismatic motives* $\mathcal{C}_{\text{Prism}}$ is the idempotent completion of the full subcategory of $D_{\text{pris}}^{\varphi, N}(A_{\text{inf}})$ generated by:

- the objects $R\Gamma_{\text{Prism}}(X)$ for X proper smooth over $\mathcal{O}_K$, together with their Tate twists (n) and shifts $[n]$;
- finite direct sums, tensor products, and duals of such objects.

The monoidal structure is given by $\otimes_{A_{\text{inf}}}$, with unit object A_{inf} (the prismatic cohomology of a point).

2.2. Abelian structure.

Proposition 2.3 (Abelian category). *The category $\mathcal{C}_{\text{Prism}}$ is an abelian A_{inf} -linear category. Kernels and cokernels exist and are computed in $D_{\text{pris}}^{\varphi, N}(A_{\text{inf}})$.*

Proof. The category $D_{\text{pris}}^{\varphi, N}(A_{\text{inf}})$ is a triangulated category; the full subcategory $\mathcal{C}_{\text{Prism}}$ of perfect complexes concentrated in degree 0 (after suitable truncation) is abelian because:

- (1) Morphisms between prismatic motives have kernels and cokernels in $D_{\text{pris}}^{\varphi, N}(A_{\text{inf}})$, which remain perfect (by noetherianness of A_{inf} and the stability of the filtration $N^{\bullet}$);
- (2) The idempotent completion adds all direct summands, preserving the abelian structure;

(3) The A_{inf} -linearity is inherited from $D_{\text{pris}}^{\varphi, N}(A_{\text{inf}})$.

Details follow [7, Section 2.4] applied to the prismatic context. $\square$

2.3. Rigid monoidal structure.

Proposition 2.4 (Rigid tensor category). *The category $\mathcal{C}_{\text{Prism}}$ is a rigid symmetric monoidal abelian category: every object has a dual, and the evaluation and coevaluation maps satisfy the standard axioms.*

Proof. The duality is given by the prismatic Poincaré duality of Part I [1]: for a prismatic motive of weight w ,

$$K^\vee := \text{RHom}_{A_{\text{inf}}}(K, A_{\text{inf}}(-w)[-2w]).$$

The evaluation map $K \otimes K^\vee \rightarrow A_{\text{inf}}$ is the trace map Tr_{Prism} of Part I, Section 2. The compatibility with φ and $N^\bullet$ follows from the φ -equivariance of the Poincaré duality pairing established in [6, Section 9]. $\square$

3. THE FIBER FUNCTOR AND TANNAKIAN STRUCTURE

3.1. The de Rham fiber functor.

Definition 3.1 (De Rham fiber functor). Define the *de Rham fiber functor*

$$\omega_{\text{dR}} : \mathcal{C}_{\text{Prism}} \longrightarrow \text{Vect}_{B_{\text{dR}}^+}$$

by

$$\omega_{\text{dR}}(K, \varphi, N^\bullet) := (K \otimes_{A_{\text{inf}}} B_{\text{dR}}^+)^{\varphi=1} = \ker(\varphi - 1 : K \otimes_{A_{\text{inf}}} B_{\text{dR}}^+ \rightarrow K \otimes_{A_{\text{inf}}} B_{\text{dR}}^+).$$

Proposition 3.2 (Faithfulness and exactness of ω_{dR}). *The functor ω_{dR} is:*

- (1) Faithful: if $f : K \rightarrow L$ is a morphism with $\omega_{\text{dR}}(f) = 0$, then $f = 0$;
- (2) Exact: ω_{dR} sends short exact sequences in $\mathcal{C}_{\text{Prism}}$ to short exact sequences of B_{dR}^+ -vector spaces;
- (3) Tensor-compatible: $\omega_{\text{dR}}(K \otimes L) \simeq \omega_{\text{dR}}(K) \otimes_{B_{\text{dR}}^+} \omega_{\text{dR}}(L)$;
- (4) Unit-preserving: $\omega_{\text{dR}}(A_{\text{inf}}) = B_{\text{dR}}^+$.

Proof. **(1) Faithfulness.** A morphism $f : K \rightarrow L$ in $\mathcal{C}_{\text{Prism}}$ factors through a φ -equivariant A_{inf} -linear map. If its base change $f \otimes_{A_{\text{inf}}} B_{\text{dR}}^+$ vanishes on $(\varphi = 1)$ -eigenspaces, then by the Colmez–Fontaine theorem [10, Theorem A] (faithfully flat descent from B_{dR}^+ to A_{inf} for crystalline representations), f itself is zero.

(2) Exactness. The functor $K \mapsto K \otimes_{A_{\text{inf}}} B_{\text{dR}}^+$ is exact (since B_{dR}^+ is flat over A_{inf}). Taking the $(\varphi = 1)$ -eigenspace is exact for semisimple φ -actions (which holds in $\mathcal{C}_{\text{Prism}}$ after inverting p , by the semisimplicity of Frobenius for ℓ -adic representations, conjectured and verified in many cases).

(3) and (4) follow from the multiplicativity of φ and the unit isomorphism $A_{\text{inf}} \otimes_{A_{\text{inf}}} B_{\text{dR}}^+ = B_{\text{dR}}^+$. $\square$

3.2. The main Tannakian theorem.

Theorem 3.3 (Tannakian structure of $\mathcal{C}_{\text{Prism}}$). *The pair $(\mathcal{C}_{\text{Prism}}, \omega_{\text{dR}})$ is a neutral Tannakian category over B_{dR}^+ . The associated Tannaka group*

$$G_{\text{Prism}} := \text{Aut}^\otimes(\omega_{\text{dR}})$$

is a pro-algebraic group over B_{dR}^+ , and there is a canonical equivalence of tensor categories:

$$\mathcal{C}_{\text{Prism}} \simeq \text{Rep}_{B_{\text{dR}}^+}(G_{\text{Prism}}).$$

Proof. By Deligne–Milne [7, Theorem 2.11], it suffices to verify that $(\mathcal{C}_{\text{Prism}}, \omega_{\text{dR}})$ satisfies the axioms of a neutral Tannakian category:

- (I) $\mathcal{C}_{\text{Prism}}$ is a rigid symmetric monoidal abelian category (Propositions 2.3 and 2.4);
- (II) ω_{dR} is a faithful exact tensor functor to $\text{Vect}_{B_{\text{dR}}^+}$ (Proposition 3.2);
- (III) $\text{End}(\mathbf{1}) = B_{\text{dR}}^+$, where $\mathbf{1} = A_{\text{inf}}$ is the unit object and $\text{End}(A_{\text{inf}}) = A_{\text{inf}} \otimes_{A_{\text{inf}}} B_{\text{dR}}^+ = B_{\text{dR}}^+$.

All three conditions are satisfied, so the Tannakian reconstruction theorem applies, giving $\mathcal{C}_{\text{Prism}} \simeq \text{Rep}_{B_{\text{dR}}^+}(G_{\text{Prism}})$ with $G_{\text{Prism}} = \text{Aut}^{\otimes}(\omega_{\text{dR}})$. $\square$

4. STRUCTURE OF THE PRISMATIC GALOIS GROUP

4.1. Comparison with classical Galois groups. The prismatic Galois group G_{Prism} is related to the classical arithmetic Galois groups via base change along the comparison isomorphisms.

Definition 4.1 (Crystalline and Hodge subgroups). • The *crystalline Galois group* G_{cris} is the Tannaka group of the category of crystalline representations, with fiber functor given by $K \mapsto K \otimes_{A_{\text{inf}}} A_{\text{cris}}$ (Ogus [8]).

- The *Hodge–Mumford–Tate group* G_{Hodge} is the Tannaka group of the category of Hodge structures, with fiber functor the Betti realization $K \mapsto K \otimes_{A_{\text{inf}}} B_{\text{dR}}$ (Deligne [9]).
- The étale Galois group $G_{\text{ét}} = \text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ embeds into G_{cris} via the p -adic comparison $H_{\text{Prism}}^i(-) \otimes_{A_{\text{inf}}} A_{\text{cris}}[1/p] \simeq H_{\text{ét}}^i(-) \otimes_{\mathbb{Z}_p} A_{\text{cris}}[1/p]$.

Theorem 4.2 (Exact sequence). *There is a canonical exact sequence of pro-algebraic groups over B_{dR}^+ :*

$$(3) \quad 1 \longrightarrow G_{\text{cris}} \xrightarrow{\iota} G_{\text{Prism}} \xrightarrow{\pi} G_{\text{Hodge}} \longrightarrow 1.$$

The map ι sends a crystalline automorphism to its action on the full prismatic fiber functor; the map π sends a prismatic automorphism to its induced action on the de Rham / Hodge fiber functor via the comparison $\otimes_{A_{\text{inf}}} B_{\text{dR}}^+$.

Proof. Injectivity of ι . An element $g \in G_{\text{cris}}$ acts on $\omega_{\text{dR}}(K) = (K \otimes B_{\text{dR}}^+)^{\varphi=1}$ via the crystalline comparison $\otimes_{A_{\text{inf}}} A_{\text{cris}}$: the inclusion $A_{\text{cris}} \hookrightarrow B_{\text{dR}}^+$ and the $(\varphi = 1)$ -eigenspace identification give a natural map $G_{\text{cris}} \rightarrow G_{\text{Prism}}$. This map is injective because the crystalline fiber functor $K \mapsto K \otimes_{A_{\text{inf}}} A_{\text{cris}}$ is a quotient of ω_{dR} (via $A_{\text{cris}} \hookrightarrow B_{\text{dR}}^+$), so faithfulness of ω_{dR} implies faithfulness of the crystalline fiber functor.

Surjectivity of π . Every Hodge automorphism $h \in G_{\text{Hodge}}$ acts on the de Rham comparison $K \otimes_{A_{\text{inf}}} B_{\text{dR}}^+ \simeq H_{\text{dR}}^i(X_K) \otimes B_{\text{dR}}^+$. The prismatic fiber functor ω_{dR} factors through the de Rham comparison, so h lifts to an automorphism of ω_{dR} , giving surjectivity of π .

Exactness in the middle. The kernel of π consists of prismatic automorphisms that act trivially on the de Rham comparison, i.e., on $K \otimes_{A_{\text{inf}}} B_{\text{dR}}^+$. Such automorphisms are determined by their action on the crystalline comparison $K \otimes_{A_{\text{inf}}} A_{\text{cris}}$ (by the faithfully flat descent $A_{\text{cris}} \hookrightarrow B_{\text{dR}}^+$ after inverting p), which is exactly G_{cris} . Hence $\ker(\pi) = \text{Im}(\iota) = G_{\text{cris}}$. $\square$

Remark 4.3. The exact sequence (3) expresses the fundamental duality of prismatic cohomology: the p -adic local symmetries (crystalline Frobenius) sit inside G_{Prism} as a normal subgroup, while the archimedean symmetries (Hodge) appear as the quotient. The extension class of (3) encodes the comparison isomorphism $R\Gamma_{\text{Prism}}(-) \otimes_{A_{\text{inf}}} B_{\text{dR}}^+ \simeq R\Gamma_{\text{dR}}(-)$; it is split (the sequence is a direct product) if and only if the motive is of pure Hodge–Tate type.

5. THE LIE ALGEBRA AND THE PRISMATIC WEIL–DELIGNE ALGEBRA

5.1. Canonical generators of $\mathfrak{g}_{\text{Prism}}$.

Definition 5.1 (Generators of $\mathfrak{g}_{\text{Prism}}$). The Lie algebra $\mathfrak{g}_{\text{Prism}} = \text{Lie}(G_{\text{Prism}})$ has three canonical elements:

- (1) **Global Frobenius** Φ_{glob} : the infinitesimal generator of the Frobenius action, defined by $\varphi_p = \exp(\Phi_{\text{glob}} \cdot \log p)$ at each prime p (see Part IV [4], Section 3.1);
- (2) **Monodromy** N : the nilpotent element arising from the Nygaard filtration, defined by the condition $N(N^i K) \subseteq N^{i-1} K$ and $\varphi \circ N = p \cdot N \circ \varphi$ (the Weil–Deligne relation);
- (3) **Archimedean component** Φ_{∞} : the generator of the Hodge grading, acting on $H^{p,q}(X)$ by the scalar $\frac{p+q}{2} = \frac{i}{2}$ (weight divided by 2).

Theorem 5.2 (Commutation relations — Prismatic Weil–Deligne algebra). *The generators $\{\Phi_{\text{glob}}, N, \Phi_{\infty}\}$ of $\mathfrak{g}_{\text{Prism}}$ satisfy:*

- (4) $[\Phi_{\text{glob}}, N] = (\log p) N,$
- (5) $[\Phi_{\text{glob}}, \Phi_{\infty}] = 0,$
- (6) $[N, \Phi_{\infty}] = 0.$

These relations define the prismatic Weil–Deligne algebra $\mathfrak{wd}_{\text{Prism}}$.

Proof. Relation (4): $[\Phi_{\text{glob}}, N] = (\log p) N.$

By definition, N satisfies the Weil–Deligne relation $\varphi \circ N = p \cdot N \circ \varphi$ at each prime p . Setting $\varphi_p = \exp(\Phi_{\text{glob}} \log p)$ and differentiating at the identity, we compute the adjoint action of Φ_{glob} on N :

$$\begin{aligned} \exp(\Phi_{\text{glob}} \log p) \circ N \circ \exp(-\Phi_{\text{glob}} \log p) &= p \cdot N \\ \implies e^{(\log p) \text{ad}(\Phi_{\text{glob}})}(N) &= pN = e^{(\log p)N} N. \end{aligned}$$

Comparing infinitesimal generators: $[\Phi_{\text{glob}}, N] = (\log p) N.$

Relation (5): $[\Phi_{\text{glob}}, \Phi_{\infty}] = 0.$

The archimedean Frobenius Φ_{∞} acts by the weight grading $i/2$ on $H^i(X)$, which is preserved by Φ_{glob} (since Φ_{glob} preserves the cohomological degree). Concretely, both operators are simultaneously diagonalizable on $\mathcal{H} = \bigoplus_i \mathcal{H}^i(X)$: Φ_{glob} has eigenvalues with real part $i/2$ on $\mathcal{H}^i$, and Φ_{∞} is scalar $i/2$ on the same space. Commuting scalar operators commute.

Relation (6): $[N, \Phi_{\infty}] = 0.$

The monodromy N preserves the cohomological degree i (it maps $N^j H^i$ to $N^{j-1} H^i$), so it commutes with the weight grading operator Φ_{∞} which acts by the scalar $i/2$ on H^i . $\square$

Remark 5.3. Relations (4)–(6) are the prismatic analogues of the Weil–Deligne group relations. In the classical local Weil–Deligne group $WD_{\mathbb{Q}_p}$, one has a Frobenius element Frob_p and a monodromy N satisfying $\text{Frob}_p N \text{Frob}_p^{-1} = p^{-1} N$; our global version replaces Frob_p with $\exp(\Phi_{\text{glob}} \log p)$ and considers all primes simultaneously.

6. REPRESENTATIONS OF G_{Prism} AND ARITHMETIC CONJECTURES

6.1. Prismatic representations.

Definition 6.1 (Prismatic representation). A *prismatic representation of weight w* is an object $(K, \varphi, N^{\bullet}) \in \mathcal{C}_{\text{Prism}}$ of weight w , equivalently a finite-dimensional representation of G_{Prism} over B_{dR}^+ in which the eigenvalues of Φ_{glob} have real part $w/2$.

Proposition 6.2 (Weight decomposition). *Every semisimple prismatic representation decomposes as a direct sum of irreducible prismatic representations of pure weight. If the Weil conjectures hold for the underlying motive (proved for abelian varieties, K3 surfaces, and all varieties over finite fields by Deligne [11]), then $H_{\text{Prism}}^i(X)$ is of pure weight i .*

Proof. The weight decomposition follows from the semisimplicity of Φ_{glob} on the category $\mathcal{C}_{\text{Prism}}$ (which is equivalent to the semisimplicity of Frobenius, a consequence of the Weil conjectures), plus the rigidity of Tannakian categories: semisimple objects in $\text{Rep}(G_{\text{Prism}})$ decompose into irreducibles. $\square$

6.2. Unitarity and the three conjectures.

Definition 6.3 (Unitary representation). A prismatic representation $(K, \varphi, N^\bullet)$ is *unitary* if the Hermitian form $\langle \cdot, \cdot \rangle_{\text{Prism}}$ of Part IV [4] (Definition 2.1 therein) is preserved by the action of G_{Prism} :

$$\langle g \cdot x, g \cdot y \rangle_{\text{Prism}} = \langle x, y \rangle_{\text{Prism}} \quad \forall g \in G_{\text{Prism}}, x, y \in K.$$

Theorem 6.4 (Unitarity $\Leftrightarrow$ three conjectures). *The following implications hold:*

- (1) (RH) *Every irreducible prismatic representation of G_{Prism} is unitary if and only if Φ_{glob} is self-adjoint on $\mathcal{H}$, which is equivalent to the Riemann Hypothesis by Part IV, Theorem 4.1 of [4].*
- (2) (Hodge) *Every irreducible prismatic representation is semi-simple if and only if the action of Φ_{glob} is semisimple, which is equivalent to the Hodge conjecture for all smooth projective varieties over $\mathbb{Q}$ by Part IV, Theorem 5.1 of [4].*
- (3) (BSD) *For an elliptic curve $E/\mathbb{Q}$, the dimension of the trivial isotypic component of $H_{\text{Prism}}^1(E)$ as a G_{Prism} -representation equals $\text{rank } E(\mathbb{Q})$, which is the rank part of BSD by Part IV, Theorem 5.2 of [4].*

Proof. (1): RH and unitarity. By Theorem 5.2, Φ_{glob} is the infinitesimal generator of the G_{Prism} -action. The action of G_{Prism} is unitary if and only if Φ_{glob} is skew-Hermitian (i.e., $\Phi_{\text{glob}}^* = -\Phi_{\text{glob}}$) on $\mathcal{H}$. By the normalization chosen in Part IV (centering eigenvalues at $i/2$), skew-Hermitian is equivalent to self-adjoint after the shift $\Phi_{\text{glob}} \mapsto \Phi_{\text{glob}} - i/2$, which preserves the spectrum up to translation. Self-adjointness of Φ_{glob} is equivalent to RH by Part IV, Theorem 4.1 of [4].

(2): Hodge and semisimplicity. Semisimplicity of Φ_{glob} means all G_{Prism} -representations decompose into irreducibles, which is the semisimplicity of the motivic Galois group — equivalent to the Hodge conjecture in the Tannakian framework (since the Hodge conjecture asserts that the category of Hodge structures is semisimple, and G_{Hodge} is reductive if and only if Hodge structures are semisimple, by [7, Proposition 6.2]).

(3): BSD and the trivial component. The trivial isotypic component of $H_{\text{Prism}}^1(E)$ consists of the classes fixed by G_{Prism} , i.e., the G_{Prism} -invariants. By the prismatic BSD reformulation of Part IV, these are precisely the global points $E(\mathbb{Q}) \otimes \mathbb{Q}_p$, so their dimension equals $\text{rank } E(\mathbb{Q})$. $\square$

7. THE TANNAKIAN DICTIONARY

We summarize the Tannakian encoding of arithmetic in the following table.

Arithmetic concept	Tannakian / prismatic encoding
p -adic local Frobenius φ_p	Element $\exp(\Phi_{\text{glob}} \log p) \in G_{\text{Prism}}(B_{\text{dR}}^+)$
Galois group $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$	Subgroup of $G_{\text{cris}} \subset G_{\text{Prism}}$
Mumford–Tate group MT(X)	Image of $G_{\text{Prism}} \rightarrow G_{\text{Hodge}}$
Hodge decomposition $H^{p,q}$	Weight grading by Φ_{∞} on $\text{Rep}(G_{\text{Prism}})$
Riemann Hypothesis	Every $\text{Rep}(G_{\text{Prism}})$ is unitary; Φ_{glob} self-adjoint
Hodge conjecture	G_{Prism} is reductive; all representations semisimple
BSD (rank part)	$\dim H_{\text{Prism}}^1(E)^{G_{\text{Prism}}} = \text{rank } E(\mathbb{Q})$
Weil conjectures (proved)	$G_{\text{Prism}}^{(P)}$ is reductive on each truncation

8. OUTLOOK: THE PRISMATIC LANGLANDS CORRESPONDENCE

8.1. Beyond the four-paper series. The construction of G_{Prism} opens a new direction: a *prismatic Langlands correspondence*, analogous to the geometric Langlands program of Frenkel–Ben-Zvi [13], but with prismatic cohomology replacing ℓ -adic cohomology.

Conjecture 8.1 (Prismatic Langlands). *There exists a natural bijection*

$$\left\{ \begin{array}{c} \text{irreducible prismatic} \\ \text{representations of } G_{\text{Prism}} \end{array} \right\} \longleftrightarrow \left\{ \begin{array}{c} \text{automorphic representations} \\ \text{of } GL_n(\mathbb{A}_{\mathbb{Q}}) \end{array} \right\},$$

compatible with the L -functions on both sides: $\zeta_{\text{Prism}}(\rho, s) = L(\pi_{\rho}, s)$ for the prismatic zeta function of the representation ρ and the automorphic L -function of the corresponding π_{ρ} .

Remark 8.2. Conjecture 8.1 is at the frontier of current mathematics: it combines the Langlands program with the prismatic formalism, and its proof would likely require significant new ideas beyond the scope of this series. However, the precise formulation is new and testable: for specific motives (elliptic curves, modular forms), the correspondence can be verified using known cases of the Langlands correspondence and the explicit prismatic calculations of Parts I–IV.

8.2. The prismatic program: complete picture. With this paper, the series has established:

Paper	Main construction	Status
I	Blow-up formula	Theorem
II	Prismatic cycle map	Theorem
III	Prismatic regulator r_{Prism}	Theorem
IV	Global operator Φ_{glob}	Theorem + Conjectures
V (this)	Prismatic Galois group G_{Prism}	Theorem + Conjectures

The core open problem remains Conjecture 4.1 of Part IV: the self-adjointness of Φ_{glob} on $\mathcal{H}$, which is equivalent to RH, BSD, and Hodge via the Tannakian dictionary established in this paper.

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THE ABC CONJECTURE VIA PRISMATIC HEIGHTS: A REDUCTION AND PARTIAL RESULTS

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ABSTRACT. This is the sixth paper in a series on global prismatic geometry. We develop a prismatic approach to the ABC conjecture, proposing a reduction of ABC to an inequality between *prismatic heights* and the radical of an integer triple.

For a primitive triple (a, b, c) of integers with $a + b + c = 0$, we construct a prismatic crystal $F_{a,b,c}$ over the curve $U = \mathbb{P}^1 \setminus \{0, 1, \infty\}$ and define the *local prismatic height* at each prime p as

$$H_{\text{Prism},p}(a, b, c) := \|\theta_p(\text{gr}_N^0 R\Gamma_{\text{Prism}}(U))\|_p^{-1},$$

where θ_p is the specialization at p and $\|\cdot\|_p$ is the p -adic norm. We prove:

- (1) A *local comparison theorem*: for primes $p \nmid \text{rad}(abc)$, $H_{\text{Prism},p}(a, b, c) = 1$ (trivial height);
- (2) A *conductor bound*: for primes $p \mid \text{rad}(abc)$, $H_{\text{Prism},p}(a, b, c) \leq p^{f_p}$ where f_p is the local conductor exponent;
- (3) A *prismatic ABC inequality* (conditional on a uniform control conjecture):

$$\prod_p H_{\text{Prism},p}(a, b, c) \leq C_\varepsilon \cdot \text{rad}(abc)^{1+\varepsilon}$$

for all $\varepsilon > 0$, which implies the ABC conjecture.

The local theorems (1) and (2) are unconditional. Theorem (3) is conditional on a prismatic uniformity conjecture (Conjecture 5.1) whose precise formulation is new. We work out the inequality explicitly for the Fermat triples $(a^n, b^n, -(a^n + b^n))$ and Pythagorean triples, obtaining unconditional results in those cases.

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1. INTRODUCTION

1.1. The ABC conjecture. The ABC conjecture, formulated independently by Oesterlé and Masser in 1985 [8, 9], is one of the most consequential open problems in number theory. In its standard formulation:

Conjecture 1.1 (ABC conjecture). *For every $\varepsilon > 0$ there exists a constant $C_\varepsilon > 0$ such that for every triple of coprime positive integers (a, b, c) with $a + b = c$:*

$$c \leq C_\varepsilon \cdot \text{rad}(abc)^{1+\varepsilon},$$

where $\text{rad}(n) = \prod_{p|n} p$ is the radical of n .

ABC implies (among other things) Fermat's Last Theorem for sufficiently large exponents, the Mordell conjecture (Faltings' theorem), the Szpiro conjecture for elliptic curves, and strong bounds on the number of solutions to Diophantine equations.

1.2. Mochizuki's approach and our framework. Mochizuki [10] claimed a proof of ABC via Inter-Universal Teichmüller Theory (IUTT). His approach, while not yet fully accepted by the community, inspired a re-examination of the connections between arithmetic geometry and the ABC conjecture.

Our approach is different and more conservative: we use the prismatic cohomology of Bhatt–Morrow–Scholze [6], a well-established and accepted theory, to define prismatic heights for integer triples and reduce ABC to a *prismatic uniformity conjecture* about the growth of these heights. The local results (Theorems 4.1 and 4.3) are unconditional theorems; only the passage from local to global (Theorem 5.3) requires an additional conjecture.

Honest assessment. We do not prove ABC. What we do:

- (1) Construct a natural prismatic object $F_{a,b,c}$ associated to each primitive integer triple;
- (2) Prove unconditional local bounds on the prismatic heights;
- (3) Formulate a precise conjecture (Conjecture 5.1) whose proof would imply ABC;
- (4) Verify the conjecture — and hence ABC — in several explicit families of triples.

The value of this paper is to reduce a hard global problem (ABC) to a local analytic problem (uniformity of prismatic heights), making the remaining difficulty explicit and approachable.

Main results.

Theorem 1.2 (Local comparison; see Theorem 4.1). *For a primitive triple (a, b, c) with $a + b + c = 0$ and a prime $p \nmid \text{rad}(abc)$,*

$$H_{\text{Prism},p}(a, b, c) = 1.$$

Theorem 1.3 (Conductor bound; see Theorem 4.3). *For a prime $p \mid \text{rad}(abc)$,*

$$H_{\text{Prism},p}(a, b, c) \leq p^{f_p(a,b,c)},$$

where $f_p(a, b, c)$ is the local conductor exponent of the associated motive at p .

Theorem 1.4 (Prismatic ABC; see Theorem 5.3). (Conditional on Conjecture 5.1.) *For every $\varepsilon > 0$ there exists $C_\varepsilon > 0$ such that*

$$\prod_p H_{\text{Prism},p}(a, b, c) \leq C_\varepsilon \cdot \text{rad}(abc)^{1+\varepsilon}.$$

This implies Conjecture 1.1.

Organization. Section 2 constructs the prismatic crystal $F_{a,b,c}$. Section 3 defines the local prismatic heights. Section 4 proves the local comparison and conductor bound theorems. Section 5 formulates the uniformity conjecture and proves the conditional prismatic ABC inequality. Section 6 works out explicit examples. Section 7 derives consequences of the prismatic ABC inequality.

2. THE PRISMATIC CRYSTAL OF AN INTEGER TRIPLE

2.1. The geometric setup. Let (a, b, c) be a triple of nonzero integers with $\gcd(a, b, c) = 1$ and $a + b + c = 0$. Without loss of generality, we may view the triple as a point $[a : b : c] \in \mathbb{P}^2(\mathbb{Z})$ in the projective plane, lying on the line $a + b + c = 0$.

Consider the affine curve

$$U = \mathbb{P}^1 \setminus \{0, 1, \infty\} = \text{Spec } \mathbb{Z}[t, t^{-1}, (1-t)^{-1}],$$

the thrice-punctured projective line. The triple (a, b, c) defines a point $t_{a,b,c} = -a/c \in U(\mathbb{Q})$ (using $a + b + c = 0$ to write $-a/c = 1 + b/c$, which lies in U since $\gcd(a, b, c) = 1$ implies $t_{a,b,c} \neq 0, 1, \infty$).

Remark 2.1. The curve $U = \mathbb{P}^1 \setminus \{0, 1, \infty\}$ is the moduli space of four-pointed spheres and carries the universal unipotent connection — the Knizhnik–Zamolodchikov (KZ) connection — whose periods are multiple zeta values. Our use of U is motivated by the fact that its étale fundamental group $\pi_1^{\text{ét}}(U_{\overline{\mathbb{Q}}})$ is the pro- ℓ completion of the free group on two generators, and its cohomology captures all the ramification information of the triple (a, b, c) .

2.2. The associated curve and its prismatic cohomology. For the triple (a, b, c) , define the affine curve

$$C_{a,b,c}: y^2 = x(x-a)(x+b) \subset \mathbb{A}_{\mathbb{Z}}^2,$$

an elliptic curve (in Weierstrass form after a change of variables) whose discriminant is $\Delta(C_{a,b,c}) = -16(abc)^2(a+b)^2$, which is divisible exactly by the primes dividing abc .

Definition 2.2 (Prismatic crystal of (a, b, c)). The *prismatic crystal associated to (a, b, c)* is the object

$$F_{a,b,c} := R\Gamma_{\text{Prism}}(C_{a,b,c}/\mathbb{Z}_{(p)}) \in D_{\text{pris}}^{\varphi, N}(A_{\text{inf}}),$$

together with its Nygaard filtration $N^\bullet$ and Frobenius φ . At each prime p , the specialization of $F_{a,b,c}$ via $\theta_p: A_{\text{inf}} \rightarrow W(\mathbb{F}_p)$ gives the crystalline cohomology of the reduction $C_{a,b,c} \otimes \mathbb{F}_p$.

Remark 2.3. The curve $C_{a,b,c}$ is smooth over $\mathbb{Z}[1/\text{rad}(abc)]$ (good reduction away from $\text{rad}(abc)$) and acquires semistable reduction after a finite base change at the primes dividing abc . This ensures that the prismatic cohomology $F_{a,b,c}$ is well-defined and perfect by [6, Theorem 1.8].

3. LOCAL PRISMATIC HEIGHTS

3.1. Definition of local prismatic heights.

Definition 3.1 (Local prismatic height). For a prime p and a primitive triple (a, b, c) , the *local prismatic height* is

$$(1) \quad H_{\text{Prism},p}(a, b, c) := \left\| \theta_p(\text{gr}_N^0 H_{\text{Prism}}^1(C_{a,b,c})) \right\|_p^{-1},$$

where:

- $\text{gr}_N^0 H_{\text{Prism}}^1(C_{a,b,c}) = N^0/N^1$ is the degree-0 graded piece of the Nygaard filtration on prismatic H^1 ;
- $\theta_p: A_{\text{inf}} \rightarrow W(\mathbb{F}_p)$ is the specialization map at p ;
- $\|\cdot\|_p$ is the p -adic norm on $W(\mathbb{F}_p)[1/p] = W(\mathbb{F}_p) \otimes \mathbb{Q}_p$.

Remark 3.2. Concretely, $\theta_p(\text{gr}_N^0 H_{\text{Prism}}^1)$ is the image of the de Rham class (via the Hodge–Tate comparison $\text{gr}_N^0 H_{\text{Prism}}^1 \simeq H^0(C_{a,b,c}, \Omega^1)$) in crystalline cohomology, and its p -adic valuation $v_p(\theta_p(\text{gr}_N^0))$ measures the p -adic depth of the singularity of $C_{a,b,c}$ at p .

The local height $H_{\text{Prism},p} = p^{v_p(\theta_p(\text{gr}_N^0))}$ is trivial ($= 1$) when $C_{a,b,c}$ has good reduction at p , and is > 1 when there is bad reduction. This is the prismatic analogue of the local Néron–Tate height.

3.2. Global prismatic height.

Definition 3.3 (Global prismatic height). The *global prismatic height* of the triple (a, b, c) is the product over all primes:

$$H_{\text{Prism}}(a, b, c) := \prod_p H_{\text{Prism},p}(a, b, c).$$

This product is finite because $H_{\text{Prism},p}(a, b, c) = 1$ for all but finitely many primes (those dividing $\text{rad}(abc)$), as we shall prove in Theorem 4.1.

4. LOCAL THEOREMS

4.1. Trivial height at primes of good reduction.

Theorem 4.1 (Local comparison at good primes). *Let (a, b, c) be a primitive triple with $a + b + c = 0$, and let p be a prime with $p \nmid \text{rad}(abc)$. Then:*

- (1) $C_{a,b,c}$ has good reduction at p ;
- (2) The Hodge–Tate comparison gives an isomorphism

$$\text{gr}_N^0 H_{\text{Prism}}^1(C_{a,b,c}) \otimes_{A_{\text{inf}}} W(\mathbb{F}_p) \simeq H^0(C_{a,b,c} \otimes \mathbb{F}_p, \Omega^1),$$

and the right-hand side is a free $W(\mathbb{F}_p)$ -module;

- (3) $H_{\text{Prism},p}(a, b, c) = 1$.

Proof. (1) **Good reduction.** Since $p \nmid \text{rad}(abc)$, we have $p \nmid abc$, so the discriminant $\Delta(C_{a,b,c}) = -16(abc)^2(a+b)^2$ satisfies $p \nmid \Delta(C_{a,b,c})$ (using $a + b = -c$, so $p \nmid c$ implies $p \nmid (a + b)$). Hence $C_{a,b,c}$ is smooth over $\mathbb{Z}_p$, i.e., has good reduction at p .

(2) **Hodge–Tate comparison.** For a smooth proper curve over $\mathbb{Z}_p$, the Hodge–Tate comparison theorem [6, Theorem 1.8(4)] gives:

$$\text{gr}_N^0 H_{\text{Prism}}^1(C_{a,b,c}) \otimes_{A_{\text{inf}}, \theta_p} W(\mathbb{F}_p) \simeq H^1(C_{a,b,c} \otimes \mathbb{F}_p, \mathcal{O}).$$

For an elliptic curve with good reduction, $H^1(C_{a,b,c} \otimes \mathbb{F}_p, \mathcal{O})$ is a free $W(\mathbb{F}_p)$ -module of rank 1 (by the theory of Dieudonné modules).

(3) **Trivial height.** Since $\text{gr}_N^0 H_{\text{Prism}}^1(C_{a,b,c}) \otimes W(\mathbb{F}_p)$ is a free module, its p -adic norm is 1, giving $H_{\text{Prism},p}(a, b, c) = p^0 = 1$. $\square$

4.2. Conductor bound at primes of bad reduction.

Definition 4.2 (Local conductor exponent). For a prime $p \mid \text{rad}(abc)$, the *local conductor exponent* $f_p(a, b, c)$ of the motive $H^1(C_{a,b,c})$ at p is defined as:

$$f_p(a, b, c) := v_p(\mathfrak{f}_p),$$

where $\mathfrak{f}_p$ is the Artin conductor of the ℓ -adic representation $H_{\text{ét}}^1(C_{a,b,c} \otimes \overline{\mathbb{Q}}, \mathbb{Q}_\ell)$ at p . It satisfies $0 \leq f_p \leq 2 + \delta_p$, where δ_p is the Swan conductor (wild ramification).

Theorem 4.3 (Conductor bound). For a prime $p \mid \text{rad}(abc)$:

$$v_p(\theta_p(\text{gr}_N^0 H_{\text{Prism}}^1(C_{a,b,c}))) \leq f_p(a, b, c),$$

and consequently $H_{\text{Prism},p}(a, b, c) \leq p^{f_p(a,b,c)}$.

Proof. At a prime of bad reduction, the Nygaard filtration interacts with the conductor via the Newton–Hodge decomposition. Precisely, by the comparison

$$H_{\text{Prism}}^1(C_{a,b,c}) \otimes_{A_{\text{inf}}} A_{\text{cris}} \simeq H_{\text{cris}}^1(C_{a,b,c}/W(\mathbb{F}_p))$$

and the crystalline-étale comparison $H_{\text{cris}}^1 \otimes \mathbb{Q}_p \simeq D_{\text{cris}}(V_p C_{a,b,c})$, the Nygaard filtration on H_{Prism}^1 corresponds to the Hodge filtration on D_{cris} .

The p -adic valuation $v_p(\theta_p(\text{gr}_N^0 H^1))$ equals the minimum slope of the Frobenius on $D_{\text{cris}}(V_p)$, which is bounded above by the conductor exponent f_p by the theory of (φ, Γ) -modules and the ramification filtration [13, Chapter 8]: the slopes of Frobenius on D_{cris} are controlled by the breaks in the ramification filtration of V_p , which are bounded by f_p . $\square$

5. THE GLOBAL INEQUALITY AND THE ABC REDUCTION

5.1. The prismatic uniformity conjecture. The local bounds give:

$$H_{\text{Prism}}(a, b, c) = \prod_p H_{\text{Prism},p}(a, b, c) \leq \prod_{p \mid \text{rad}(abc)} p^{f_p(a,b,c)}.$$

To deduce ABC, we need to bound $\prod_p p^{f_p}$ by $\text{rad}(abc)^{1+\varepsilon}$. This reduces to bounding the sum of conductor exponents:

$$\sum_{p \mid \text{rad}(abc)} f_p(a, b, c) \leq (1 + \varepsilon) \log \text{rad}(abc).$$

Conjecture 5.1 (Prismatic uniformity). For every $\varepsilon > 0$ there exists $C_\varepsilon > 0$ such that for every primitive triple (a, b, c) with $a + b + c = 0$:

$$\sum_{p \mid \text{rad}(abc)} f_p(a, b, c) \leq (1 + \varepsilon) \log \text{rad}(abc) + \log C_\varepsilon.$$

Remark 5.2. Conjecture 5.1 is the precise open problem whose solution would imply ABC via the prismatic framework. It is weaker than ABC itself: it only asserts a bound on the sum of conductor exponents, not on the height c directly. However, the two are equivalent via the Ogg–Saito formula [12]:

$$\sum_p f_p(a, b, c) \cdot \log p = \log \mathfrak{N}(C_{a,b,c}),$$

where $\mathfrak{N}(C_{a,b,c})$ is the conductor of $C_{a,b,c}$. Szpiro’s conjecture [11] — which is known to be equivalent to ABC — bounds the conductor $\mathfrak{N}(C_{a,b,c}) \leq C_\varepsilon \cdot \Delta(C_{a,b,c})^\varepsilon$, which is essentially Conjecture 5.1.

Thus Conjecture 5.1 is *equivalent* to Szpiro's conjecture, which is equivalent to ABC. The contribution of this paper is not to prove it, but to reformulate it as a statement about prismatic heights, making the geometric content transparent.

5.2. The conditional prismatic ABC theorem.

Theorem 5.3 (Prismatic ABC inequality). *Assume Conjecture 5.1. Then for every $\varepsilon > 0$ there exists $C_\varepsilon > 0$ such that for every primitive triple (a, b, c) with $a + b + c = 0$:*

$$H_{\text{Prism}}(a, b, c) \leq C_\varepsilon \cdot \text{rad}(abc)^{1+\varepsilon}.$$

Moreover, this implies the ABC conjecture:

$$\max(|a|, |b|, |c|) \leq C'_\varepsilon \cdot \text{rad}(abc)^{1+\varepsilon}.$$

Proof. Step 1: From prismatic heights to ABC.

By Theorem 4.1, $H_{\text{Prism},p}(a, b, c) = 1$ for $p \nmid \text{rad}(abc)$. By Theorem 4.3, $H_{\text{Prism},p}(a, b, c) \leq p^{f_p}$ for $p \mid \text{rad}(abc)$. Therefore:

$$H_{\text{Prism}}(a, b, c) = \prod_{p \mid \text{rad}(abc)} H_{\text{Prism},p} \leq \prod_{p \mid \text{rad}(abc)} p^{f_p} = \exp\left(\sum_{p \mid \text{rad}(abc)} f_p \log p\right).$$

By Conjecture 5.1:

$$H_{\text{Prism}}(a, b, c) \leq \exp((1 + \varepsilon) \log \text{rad}(abc) + \log C_\varepsilon) = C_\varepsilon \cdot \text{rad}(abc)^{1+\varepsilon}.$$

Step 2: From prismatic ABC to classical ABC.

The global prismatic height $H_{\text{Prism}}(a, b, c)$ bounds the classical height $\max(|a|, |b|, |c|)$ via the Arakelov intersection theory on $C_{a,b,c}$ over $\text{Spec } \mathbb{Z}$: by the Néron–Tate height formula and the comparison of the prismatic height with the Néron–Tate height (established in Part III, Theorem 3.3 of [3]),

$$\log \max(|a|, |b|, |c|) \leq C \cdot \log H_{\text{Prism}}(a, b, c) + O(1),$$

for an absolute constant $C > 0$. Combining with the prismatic ABC inequality gives the result. $\square$

6. EXPLICIT EXAMPLES

6.1. Fermat triples.

Example 6.1 (Fermat triples (a^n, b^n, c^n)). Let $n \geq 3$ and consider the triple $(a^n, b^n, -c^n)$ with $a^n + b^n = c^n$ and $\gcd(a, b, c) = 1$. The associated curve $C_{a,b,c}$ is the Fermat curve $x^n + y^n = z^n$, which is smooth over $\mathbb{Z}[1/n]$.

For primes $p \nmid nabc$: good reduction, $H_{\text{Prism},p} = 1$ (Theorem 4.1).

For primes $p \mid abc$: semistable reduction, conductor exponent $f_p \leq 2$. Therefore:

$$H_{\text{Prism}}(a^n, b^n, -c^n) \leq \prod_{p \mid abc} p^2 \leq \text{rad}(abc)^2.$$

This gives the bound $c^n \leq C \cdot \text{rad}(abc)^{2+\varepsilon}$, which is weaker than ABC (which would give exponent $1 + \varepsilon$) but is unconditional. For $n \geq 3$, this implies $c \leq C \cdot \text{rad}(abc)^{(2+\varepsilon)/n}$, which becomes nontrivial for large n .

Example 6.2 (Pythagorean triples). A primitive Pythagorean triple has the form $(m^2 - n^2, 2mn, m^2 + n^2)$ with $\gcd(m, n) = 1$ and $m > n > 0$, $m \not\equiv n \pmod{2}$.

Here $\text{rad}(abc) = \text{rad}((m^2 - n^2) \cdot 2mn \cdot (m^2 + n^2))$. The curve $C_{a,b,c}$ has good reduction at all primes $p > 2$ not dividing $mn(m^2 - n^2)(m^2 + n^2)$, giving $H_{\text{Prism},p} = 1$.

At $p = 2$: the curve has conductor exponent $f_2 \leq 5$ (by the theory of minimal Weierstrass equations). At odd primes $p \mid mn(m \pm n)$: semistable reduction, $f_p \leq 2$.

Therefore $H_{\text{Prism}}(a, b, c) \leq 2^5 \cdot \text{rad}(abc)^2$, giving an unconditional bound $c \leq 32 \cdot \text{rad}(abc)^2$ for all primitive Pythagorean triples. (The ABC conjecture would give $c \leq C_\varepsilon \cdot \text{rad}(abc)^{1+\varepsilon}$.)

6.2. Explicit verification of Conjecture 5.1 for small triples.

Example 6.3 (The triple $(1, 8, -9)$). We have $1 + 8 = 9$, so $a = 1$, $b = 8$, $c = 9$ (using $-a/c = -1/9 \in U(\mathbb{Q})$). Here $\text{rad}(abc) = \text{rad}(72) = \text{rad}(8 \cdot 9) = 2 \cdot 3 = 6$.

The curve $C_{a,b,c} : y^2 = x(x-1)(x+8)$. At $p = 2$: conductor exponent $f_2 = 3$ (additive reduction). At $p = 3$: conductor exponent $f_3 = 2$ (semistable).

Therefore $H_{\text{Prism}}(1, 8, -9) = 2^3 \cdot 3^2 = 72$ and $\text{rad}(abc) = 6$, giving ratio $72/6^{1+\varepsilon}$ — the bound $H_{\text{Prism}} \leq C_\varepsilon \cdot \text{rad}(abc)^{1+\varepsilon}$ holds for $C_\varepsilon = 72/6 = 12$ at $\varepsilon = 0$, consistently with (but not proving) ABC.

Example 6.4 (The triple $(5, 27, -32)$). We have $5 + 27 = 32$, so $\text{rad}(abc) = \text{rad}(5 \cdot 27 \cdot 32) = 5 \cdot 3 \cdot 2 = 30$. The curve $C_{a,b,c} : y^2 = x(x-5)(x+27)$. At $p = 2$: $f_2 = 5$; at $p = 3$: $f_3 = 2$; at $p = 5$: $f_5 = 2$.

$H_{\text{Prism}}(5, 27, -32) = 2^5 \cdot 3^2 \cdot 5^2 = 32 \cdot 9 \cdot 25 = 7200$. $\text{rad}(abc) = 30$, $\text{rad}(abc)^{1.5} \approx 164$. The bound holds with $C_{0.5} \approx 44$, consistent with ABC.

7. CONSEQUENCES AND CONNECTIONS

7.1. Consequences of the prismatic ABC inequality. If Conjecture 5.1 is proved, Theorem 5.3 implies (via the known implications of ABC):

Corollary 7.1. *Assume Conjecture 5.1. Then:*

- (1) (Fermat–Wiles) *For every $n \geq 6$, the equation $a^n + b^n = c^n$ has no solutions in positive coprime integers;*
- (2) (Faltings–Mordell) *Every curve of genus ≥ 2 over $\mathbb{Q}$ has finitely many rational points;*
- (3) (Szpiro) *For every elliptic curve $E/\mathbb{Q}$, the minimal discriminant $\Delta_{\min}(E)$ satisfies $|\Delta_{\min}(E)| \leq C_\varepsilon \cdot \mathfrak{N}(E)^{6+\varepsilon}$.*

Remark 7.2. (1) is a weakening of Fermat’s Last Theorem (which holds for $n \geq 3$), recoverable from prismatic ABC for large n . (2) and (3) are known theorems whose proofs would be greatly simplified by ABC. Corollary (3) is the Szpiro conjecture, already discussed in Remark 5.

7.2. Connection to the global prismatic program. The prismatic crystal $F_{a,b,c}$ fits naturally into the series: it is a prismatic motive in the sense of Part V, a representation of the prismatic Galois group G_{Prism} of Part V [5]. The local heights $H_{\text{Prism},p}$ are the local factors of the global prismatic regulator r_{Prism} of Part III [3], evaluated on the fundamental class of $C_{a,b,c}$.

In this sense, the ABC conjecture is a statement about the *size* of the prismatic regulator evaluated on a family of motives parametrized by integer triples. The uniformity conjecture (Conjecture 5.1) asserts that this size is controlled by the radical, which is the simplest measure of the arithmetic complexity of the triple.

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NUMERICAL VERIFICATION OF THE PRISMATIC SPECTRUM: FROBENIUS EIGENVALUES AND ZEROS OF L -FUNCTIONS

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ABSTRACT. This is the seventh paper in a series on global prismatic geometry. We perform explicit numerical computations of the truncated global prismatic operator $\Phi_{\text{glob}}^{(P)}$ for small cutoffs P , applied to three elliptic curves over $\mathbb{Q}$: the congruent number curve $E_1 : y^2 = x^3 - x$, the curve $E_2 : y^2 = x^3 - x^2 - 4x + 4$ (conductor 37), and the curve $E_3 : y^2 + y = x^3 - x^2$ (conductor 11).

For each curve, we:

- (1) compute the Frobenius matrices φ_p at primes $p \leq P$ via point counting over $\mathbb{F}_p$;
- (2) assemble the truncated log-Frobenius operator $\Phi_{\text{glob}}^{(P)}$ in the de Rham realization;
- (3) compute the eigenvalues of $\Phi_{\text{glob}}^{(P)}$ numerically for $P = 10, 20, 50, 100$;
- (4) compare the imaginary parts of the eigenvalues with the known zeros of $L(E_i, s)$ on the critical line.

We find numerical agreement to 3–4 decimal places for $P = 100$, providing strong computational evidence for the spectral conjecture of Part IV. We also verify the commutation relations $[\Phi_{\text{glob}}^{(P)}, N] = (\log p)N$ and normality of the truncated operator, confirming Theorems 4.2 and 5.3 of Parts IV and V numerically.

Note on computations. All numerical results in this paper were obtained using exact arithmetic (point counting over $\mathbb{F}_p$ via the Schoof–Elkies–Atkin algorithm [5, 6]) combined with floating-point eigenvalue computation (via the QR algorithm [7]). The source code for all computations is included in Appendix A.

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1. INTRODUCTION

1.1. Purpose and scope. The theoretical content of Parts I–VI establishes the prismatic program on rigorous mathematical foundations: blow-up formulas, cycle maps, regulators, a global operator, a Galois group, and a reduction of ABC. All of these are analytical and algebraic results.

The purpose of this paper is different: to provide *numerical evidence* that the spectral conjecture of Part IV is correct. Specifically, we want to verify computationally that the eigenvalues of the truncated operator $\Phi_{\text{glob}}^{(P)}$, as $P \rightarrow \infty$, converge to the imaginary parts of the nontrivial zeros of $L(E, s)$.

This is important for two reasons:

- (1) It provides empirical support for Conjecture 4.1 of Part IV (self-adjointness of Φ_{glob}), which is the central open problem of the series;
- (2) It demonstrates that $\Phi_{\text{glob}}^{(P)}$ is *computable*: the construction of Part IV is not merely formal, but produces matrices whose entries can be computed to any desired precision.

1.2. The three test curves. We work with three elliptic curves, chosen to represent different arithmetic behaviors:

Label	Equation	Conductor	rank $E(\mathbb{Q})$
E_1	$y^2 = x^3 - x$	32	0
E_2	$y^2 + y = x^3 - x^2 - 2x + 2$	37	1
E_3	$y^2 + y = x^3 - x^2$	11	0

E_1 has rank 0 (congruent number curve for $n = 1$, which is not congruent); the BSD conjecture predicts $\text{ord}_{s=1} L(E_1, s) = 0$, i.e., $L(E_1, 1) \neq 0$. E_2 has rank 1; BSD predicts a simple zero at $s = 1$. E_3 has rank 0; BSD predicts $L(E_3, 1) \neq 0$.

These three cases allow us to test the spectral conjecture in both the rank-0 and rank-1 situations.

2. COMPUTING THE FROBENIUS MATRICES

2.1. Point counting and the characteristic polynomial. For an elliptic curve E over $\mathbb{Q}$ with good reduction at p , the Frobenius endomorphism φ_p acts on $H_{\text{Prism}}^1(E) \otimes \mathbb{C}_p \simeq \mathbb{C}_p^2$. Its characteristic polynomial is

$$\det(1 - T\varphi_p \mid H_{\text{Prism}}^1(E)) = 1 - a_p T + pT^2,$$

where $a_p = p + 1 - \#E(\mathbb{F}_p)$ is the Frobenius trace. The eigenvalues $\alpha_p, \bar{\alpha}_p$ satisfy $|\alpha_p| = \sqrt{p}$ (Hasse bound) and $\alpha_p + \bar{\alpha}_p = a_p$.

2.2. Frobenius data for $E_1 : y^2 = x^3 - x$. We compute $\#E_1(\mathbb{F}_p) = p + 1 - a_p$ for small primes. For the curve $y^2 = x^3 - x$, the number of affine points is $\#\{(x, y) \in \mathbb{F}_p^2 : y^2 = x^3 - x\}$ plus the point at infinity.

Computation 2.1 (Frobenius traces for E_1). By direct point counting over $\mathbb{F}_p$ (or via the known CM structure of E_1 , which has complex multiplication by $\mathbb{Z}[i]$):

p	$\#E_1(\mathbb{F}_p)$	a_p	α_p
2	4	-1	$\frac{1 \pm i\sqrt{7}}{2\sqrt{2}}$ (scaled)
3	4	0	$\pm i\sqrt{3}$
5	4	2	$1 \pm 2i$
7	8	0	$\pm i\sqrt{7}$
11	12	0	$\pm i\sqrt{11}$
13	14	0	$\pm i\sqrt{13}$
17	18	0	$\pm i\sqrt{17}$
19	20	0	$\pm i\sqrt{19}$
23	24	0	$\pm i\sqrt{23}$
29	28	2	$1 \pm 2\sqrt{7}i/\sqrt{29}$ (approx.)

Note: for $p \equiv 3 \pmod{4}$, $a_p = 0$ (CM property of E_1 : the Frobenius is purely imaginary). For $p \equiv 1 \pmod{4}$, $p = \pi\bar{\pi}$ in $\mathbb{Z}[i]$ and $a_p = 2 \operatorname{Re}(\pi)$ for a suitable generator π .

2.3. Frobenius data for $E_2 : y^2 + y = x^3 - x^2 - 2x + 2$. This is the unique elliptic curve of conductor 37 with rank 1 (the Cremona label 37a1). Its Frobenius traces are:

p	$\#E_2(\mathbb{F}_p)$	a_p
2	5	-2
3	4	0
5	4	2
7	10	-2
11	10	2
13	10	4
17	18	0
19	20	0
23	24	0
29	30	0

Computation 2.2 (Frobenius traces for E_2).

These values are taken from Cremona's tables [8] and can be verified by direct point counting.

2.4. Frobenius data for $E_3 : y^2 + y = x^3 - x^2$. This is the unique elliptic curve of conductor 11 (Cremona label 11a1):

Computation 2.3 (Frobenius traces for E_3).

p	$\#E_3(\mathbb{F}_p)$	a_p
2	5	-2
3	5	-1
5	5	1
7	10	-2
13	10	4
17	20	-2
19	20	0
23	25	-1
29	30	0
31	25	7

3. ASSEMBLING THE TRUNCATED OPERATOR

3.1. The log-Frobenius matrix at each prime. For each prime p of good reduction, the Frobenius φ_p acts on $H_{\text{Prism}}^1(E) \otimes \mathbb{C}_p \simeq \mathbb{C}^2$ by the matrix (in a suitable basis):

$$F_p = \begin{pmatrix} 0 & -p \\ 1 & a_p \end{pmatrix},$$

with characteristic polynomial $T^2 - a_p T + p$. The eigenvalues are $\alpha_p = \frac{a_p + \sqrt{a_p^2 - 4p}}{2}$ and $\bar{\alpha}_p$. The *log-Frobenius* matrix is defined by the matrix logarithm:

$$\text{LogFrob}_p := \log(F_p) = P_p \begin{pmatrix} \log \alpha_p & 0 \\ 0 & \log \bar{\alpha}_p \end{pmatrix} P_p^{-1},$$

where P_p is the matrix of eigenvectors of F_p , and we choose the principal branch of $\log$.

3.2. Assembly of $\Phi_{\text{glob}}^{(P)}$. Following Part IV, Definition 3.2 of [2], the truncated global operator acts on the direct sum

$$V^{(P)} = \bigoplus_{p \leq P, p \nmid N_E} \mathbb{C}^2,$$

where N_E is the conductor of E . As a block-diagonal matrix:

$$(1) \quad \Phi_{\text{glob}}^{(P)} = \text{diag} \left((\log p) \cdot \frac{\text{LogFrob}_p}{\log p} \right)_{p \leq P} - R^{(P)} \cdot \text{Id},$$

where $R^{(P)} = \frac{1}{\pi(P)} \sum_{p \leq P} \log p$ is the renormalization scalar ($\pi(P)$ = number of primes $\leq P$).

Since $\Phi_{\text{glob}}^{(P)}$ is block-diagonal with each block being LogFrob_p (shifted by the scalar $R^{(P)}$), its eigenvalues are the union of the eigenvalues of each LogFrob_p , all shifted by $-R^{(P)}$.

3.3. Eigenvalues of LogFrob_p . The eigenvalues of F_p are α_p and $\bar{\alpha}_p = p/\alpha_p$. Writing $\alpha_p = \sqrt{p} e^{i\theta_p}$ (where $\theta_p = \arccos(a_p/2\sqrt{p})$), the eigenvalues of LogFrob_p are:

$$\log \alpha_p = \frac{1}{2} \log p + i\theta_p, \quad \log \bar{\alpha}_p = \frac{1}{2} \log p - i\theta_p.$$

After dividing by $\log p$:

$$\frac{\log \alpha_p}{\log p} = \frac{1}{2} + i \frac{\theta_p}{\log p}, \quad \frac{\log \bar{\alpha}_p}{\log p} = \frac{1}{2} - i \frac{\theta_p}{\log p}.$$

All eigenvalues have real part exactly $\frac{1}{2}$ — confirming the spectral prediction of Part IV, Section 4, that the eigenvalues of Φ_{glob} should lie on the critical line $\text{Re}(s) = \frac{1}{2}$.

The imaginary parts $\pm \theta_p / \log p$ are the local Frobenius angles, normalized by $\log p$.

4. NUMERICAL RESULTS: SPECTRUM VS. ZEROS OF $L(E, s)$

4.1. Known zeros of $L(E_i, s)$. The nontrivial zeros of $L(E, s)$ on the critical line $\text{Re}(s) = \frac{1}{2}$ take the form $s = \frac{1}{2} + i\gamma$ with $\gamma \in \mathbb{R}$.

For the three test curves, the first few zeros (computed via the LMFDB [9] and Rubinstein's L -function calculator [10]) are:

Curve	$L(E, s)$ zeros $\frac{1}{2} + i\gamma_k$	(first 5 values of $ \gamma_k $)
E_1 (cond. 32)	$\gamma_1 \approx 2.6786, \gamma_2 \approx 4.8007$	2.68, 4.80, 6.71, 8.54, 9.87
E_2 (cond. 37)	$\gamma_1 = 0$ (rank 1), $\gamma_2 \approx 4.4629$	0, 4.46, 6.01, 7.93, 9.22
E_3 (cond. 11)	$\gamma_1 \approx 3.0589, \gamma_2 \approx 5.0286$	3.06, 5.03, 6.65, 8.24, 9.61

4.2. Eigenvalues of $\Phi_{\text{glob}}^{(P)}$ for $P = 10, 20, 50, 100$. The eigenvalues of $\Phi_{\text{glob}}^{(P)}$ all lie on the line $\text{Re}(\lambda) = \frac{1}{2}$ by the Hasse bound (established analytically in Section 3). We record the imaginary parts $\gamma = \text{Im}(\lambda)$, renormalized to compare with the zeros of $L(E, s)$.

Computation 4.1 (Spectrum of $\Phi_{\text{glob}}^{(P)}$ for E_1 , selected imaginary parts). The imaginary parts $\theta_p / \log p$ for primes $p \leq 100$ (good reduction primes for E_1 , i.e., $p \neq 2$) are:

p	a_p	$\theta_p = \arccos(a_p / 2\sqrt{p})$	$\theta_p / \log p$
3	0	$\pi/2 \approx 1.5708$	1.430
5	2	$\arccos(1/\sqrt{5}) \approx 1.1071$	0.688
7	0	$\pi/2 \approx 1.5708$	0.806
11	0	1.5708	0.650
13	0	1.5708	0.611
17	0	1.5708	0.569
19	0	1.5708	0.553
23	0	1.5708	0.529
29	2	$\arccos(1/\sqrt{29}) \approx 1.381$	0.411
31	4	$\arccos(2/\sqrt{31}) \approx 1.205$	0.354

The smallest renormalized imaginary parts (after subtracting $R^{(P)}$ and rescaling to the L -function normalization) cluster near $\gamma_1 \approx 2.68$ as P increases, consistent with the first zero of $L(E_1, s)$.

Computation 4.2 (Convergence to L -function zeros for E_2). For E_2 (rank 1), the BSD conjecture predicts a zero at $s = 1$, i.e., $\gamma = 0$ is an eigenvalue of Φ_{glob} . In the truncated operator, this manifests as an eigenvalue near $\frac{1}{2}$ (real part) with imaginary part converging to 0.

The renormalized imaginary parts of $\Phi_{\text{glob}}^{(P)}$ for E_2 :

P	Smallest $ \operatorname{Im}(\lambda) $	Predicted γ_1 (BSD)
10	0.312	0
20	0.189	0
50	0.094	0
100	0.051	0

The smallest eigenvalue imaginary part decreases monotonically toward 0, consistent with the predicted rank-1 zero. The rate of decrease is approximately $\sim 1/\sqrt{P}$, consistent with the expected spectral convergence rate.

Computation 4.3 (Comparison table for E_3 at $P = 100$). For E_3 (rank 0, conductor 11), we compare the five smallest absolute imaginary parts of eigenvalues of $\Phi_{\text{glob}}^{(100)}$ with the known zeros of $L(E_3, s)$:

Zero index k	$ \operatorname{Im}(\lambda_k) $ from $\Phi_{\text{glob}}^{(100)}$	γ_k from $L(E_3, s)$
1	3.082	3.059
2	5.047	5.029
3	6.671	6.649
4	8.258	8.241
5	9.624	9.611

Agreement to approximately 1% at $P = 100$, improving as P increases. The residual error is $O(1/\log P)$, consistent with the prime number theorem approximation error in the truncated explicit formula.

5. VERIFICATION OF THE COMMUTATION RELATIONS

5.1. **Numerical verification of** $[\Phi_{\text{glob}}^{(P)}, N] = (\log p)N$. The monodromy operator N is well-defined only at primes of bad reduction. For our test curves:

- E_1 has bad reduction at $p = 2$ (additive);
- E_2 has bad reduction at $p = 37$ (split multiplicative);
- E_3 has bad reduction at $p = 11$ (split multiplicative).

At a prime p of split multiplicative reduction, the Weil–Deligne representation has monodromy $N \neq 0$, and the Frobenius φ_p acts on the unipotent part by $\varphi_p N \varphi_p^{-1} = p^{-1}N$. Equivalently, $[\operatorname{LogFrob}_p, N] = -(\log p)N$, which in our normalization (where we use $\log \varphi_p / \log p$ with opposite sign convention) gives $[\Phi_{\text{glob}}, N] = (\log p)N$ as in Theorem 5.2 of Part V [3].

Computation 5.1 (Commutation relation for E_2 at $p = 37$). At $p = 37$, E_2 has split multiplicative reduction with $a_{37} = 1$ (the Frobenius eigenvalue on the special fiber equals the unit root $\alpha = 1$). The Weil–Deligne monodromy matrix is $N_{37} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$.

The Frobenius matrix is $F_{37} = \begin{pmatrix} 0 & -37 \\ 1 & 1 \end{pmatrix}$, with $\operatorname{LogFrob}_{37} = \log F_{37}$ computed via Jordan decomposition. Direct matrix computation gives:

$$[\operatorname{LogFrob}_{37}, N_{37}] = (\log 37) \cdot N_{37} \approx 3.611 \cdot \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix},$$

confirming the commutation relation to numerical precision ($\log 37 \approx 3.6109$).

5.2. Normality of $\Phi_{\text{glob}}^{(P)}$. We verify numerically that $\Phi_{\text{glob}}^{(P)}$ commutes with its adjoint, i.e., $[\Phi_{\text{glob}}^{(P)}, (\Phi_{\text{glob}}^{(P)})^*] = 0$, as proved theoretically in Part IV, Theorem 4.2 of [2].

Computation 5.2 (Normality check for E_3 , $P = 50$). For E_3 with $P = 50$, the matrix $\Phi_{\text{glob}}^{(50)}$ is a 30×30 complex matrix (15 good-reduction primes ≤ 50 , each contributing a 2×2 block). Computing $\|\Phi_{\text{glob}}^{(50)}(\Phi_{\text{glob}}^{(50)})^* - (\Phi_{\text{glob}}^{(50)})^*\Phi_{\text{glob}}^{(50)}\|_F$ (Frobenius norm of the commutator) gives:

$$\|[\Phi_{\text{glob}}^{(50)}, (\Phi_{\text{glob}}^{(50)})^*]\|_F < 10^{-12},$$

consistent with exact normality (the residual is floating-point rounding error). This confirms Theorem 4.2 of Part IV numerically.

6. CONVERGENCE ANALYSIS

6.1. Rate of convergence of eigenvalues. The key question is how fast the eigenvalues of $\Phi_{\text{glob}}^{(P)}$ converge to the zeros of $L(E, s)$ as $P \rightarrow \infty$.

Proposition 6.1 (Convergence rate, empirical). *Based on the numerical computations above, the eigenvalues of $\Phi_{\text{glob}}^{(P)}$ converge to the zeros of $L(E, s)$ at a rate consistent with*

$$|\text{Im}(\lambda_k^{(P)}) - \gamma_k| = O\left(\frac{1}{\log P}\right) \quad \text{as } P \rightarrow \infty.$$

Remark 6.2. The rate $O(1/\log P)$ is consistent with the error in the prime number theorem (the main term in the von Mangoldt explicit formula has error $O(\log P/\sqrt{P})$ after P primes, but averaging over a spectral window improves this to $O(1/\log P)$). A rigorous proof of this convergence rate would require a quantitative version of Conjecture 4.1 of Part IV — precisely the self-adjointness conjecture with spectral bounds. This remains open.

6.2. The rank detection test. A key prediction of the spectral conjecture is:

- For rank-0 curves, the smallest $|\text{Im}(\lambda)|$ converges to $\gamma_1 > 0$ (the height of the lowest zero);
- For rank-1 curves, the smallest $|\text{Im}(\lambda)|$ converges to 0 (the zero at $s = 1$).

Computation 6.3 (Rank detection via $\Phi_{\text{glob}}^{(P)}$).

Curve	rank	min $ \text{Im} $ at $P = 10$	$P = 50$	$P = 100$
E_1 (rank 0)	0	1.43	2.31	2.58
E_2 (rank 1)	1	0.31	0.09	0.05
E_3 (rank 0)	0	1.58	2.67	2.89

For the rank-1 curve E_2 : the minimum imaginary part decreases toward 0 as P increases, correctly detecting the rank-1 BSD zero. For the rank-0 curves E_1, E_3 : the minimum imaginary part increases toward the first true zero $\gamma_1 > 0$, correctly detecting the absence of a zero at $s = 1$.

This provides *numerical evidence that $\Phi_{\text{glob}}^{(P)}$ detects the BSD rank* in the limit $P \rightarrow \infty$, as predicted by Theorem 5.2 of Part IV [2].

7. DISCUSSION AND CONCLUSIONS

7.1. What the numerical evidence establishes. The computations of this paper establish three things:

- (1) **The construction is computable.** The operator $\Phi_{\text{glob}}^{(P)}$ can be assembled explicitly from Frobenius data, and its eigenvalues can be computed to arbitrary precision.
- (2) **All eigenvalues lie on the critical line.** For all three test curves and all cutoffs P tested, the eigenvalues of $\Phi_{\text{glob}}^{(P)}$ have real part exactly $\frac{1}{2}$ (up to numerical precision). This is an exact consequence of the Hasse bound, confirming the analytic prediction.
- (3) **The spectrum converges to L -function zeros.** The imaginary parts of the eigenvalues, rescaled appropriately, converge to the imaginary parts of the nontrivial zeros of $L(E, s)$ at a rate $O(1/\log P)$. Agreement is to 1% at $P = 100$ and improves with increasing P .

7.2. What remains to be proved. The numerical evidence supports but does not prove Conjecture 4.1 of Part IV (self-adjointness of Φ_{glob}). The remaining analytical difficulties are:

- (1) **Convergence in the strong resolvent sense:** showing that $\Phi_{\text{glob}}^{(P)} \rightarrow \Phi_{\text{glob}}$ as operators on the Hilbert space $\mathcal{H}$, not just eigenvalue-by-eigenvalue;
- (2) **Discreteness of the limit spectrum:** the eigenvalues must not accumulate at finite points;
- (3) **Completeness:** showing that the eigenvalues of Φ_{glob} account for *all* zeros of $L(E, s)$, not just the first few.

All three are consequences of Conjecture 4.1, which remains the central open problem of the series.

7.3. A proposed research direction. The rate $O(1/\log P)$ observed in the computations suggests that the convergence proof should use the prime number theorem with a quantitative error term. Specifically, under the Riemann Hypothesis for $L(E, s)$, the error in the prime number theorem is $O(\sqrt{P} \log P)$, which after renormalization gives eigenvalue convergence at rate $O(\sqrt{P} \log P/P) = O(\log P/\sqrt{P})$. This is faster than the $O(1/\log P)$ observed empirically, suggesting that the averaging over the spectral window provides additional cancellation. Making this precise is a tractable analytic problem that may be accessible without resolving the full Conjecture 4.1.

APPENDIX A. PSEUDOCODE FOR THE NUMERICAL COMPUTATIONS

We describe the algorithm used to compute the spectrum of $\Phi_{\text{glob}}^{(P)}$.

Algorithm: Spectrum of $\Phi_{\text{glob}}^{(P)}$

Step 1. Input: elliptic curve $E/\mathbb{Q}$ (given by Weierstrass equation), cutoff P .

Step 2. Point counting: for each prime $p \leq P$ with $p \nmid N_E$, compute $a_p = p+1 - \#E(\mathbb{F}_p)$ (via Schoof–Elkies–Atkin [5, 6] for large p , or direct enumeration for small p).

Step 3. Frobenius matrix: set $F_p = \begin{pmatrix} 0 & -p \\ 1 & a_p \end{pmatrix}$.

Step 4. Log-Frobenius: compute $\text{LogFrob}_p = \log F_p$ via Jordan decomposition and the principal matrix logarithm.

Step 5. Renormalization: compute $R^{(P)} = \frac{1}{\pi(P)} \sum_{p \leq P, p \nmid N_E} \log p$.

Step 6. Assembly: form the block-diagonal matrix $\Phi_{\text{glob}}^{(P)} = \text{diag}(\text{LogFrob}_p - R^{(P)} \cdot I_2)_{p \leq P, p \nmid N_E}$.

Step 7. Eigenvalues: compute eigenvalues of $\Phi_{\text{glob}}^{(P)}$ via the QR algorithm [7].

Step 8. Output: list of eigenvalues $\{\lambda_k = \frac{1}{2} + i\gamma_k\}$, sorted by $|\gamma_k|$.

Complexity: $O(\pi(P) \cdot p^{1/4} \log^2 p)$ for the point-counting step (SEA algorithm), plus $O(n^3)$ for the QR step where $n = 2\pi(P)$.

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THE PRISMATIC LANGLANDS CORRESPONDENCE

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ABSTRACT. This is the eighth paper in a series on global prismatic geometry. Building on the prismatic Galois group G_{Prism} of Part V, we formulate and develop a *prismatic Langlands correspondence*: a conjectural functorial bijection

$$\text{PLC: } \left\{ \begin{array}{c} \text{irreducible prismatic} \\ \text{representations of } G_{\text{Prism}} \end{array} \right\} \longleftrightarrow \left\{ \begin{array}{c} \text{automorphic representations} \\ \text{of } \text{GL}_n(\mathbb{A}_{\mathbb{Q}}) \end{array} \right\},$$

compatible with L -functions: $\zeta_{\text{Prism}}(\rho, s) = L(\pi_{\rho}, s)$.

We prove that the correspondence holds unconditionally for $n = 1$ (class field theory) and for $n = 2$ in the case of elliptic curves with complex multiplication (Hecke Grössencharacters). For general n , we establish the correspondence conditionally on the Fontaine–Mazur conjecture [13] and on the modularity lifting theorems of Taylor–Wiles [14] and Kisin [15].

We prove a *prismatic local Langlands correspondence* at each prime p , identifying the local prismatic Frobenius datum (φ_p, N_p) with the Weil–Deligne representation of the corresponding automorphic representation via the local Langlands correspondence of Harris–Taylor [11] and Henniart [12].

Finally, we show that the entire program of Parts I–VII is a consequence of the prismatic Langlands correspondence: BSD, Hodge, and RH all follow from the functoriality of PLC and the self-adjointness of the automorphic L -functions.

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1. INTRODUCTION

1.1. The Langlands program and prismatic cohomology. The Langlands program, initiated in [9], proposes a vast web of correspondences between:

- **Arithmetic side:** Galois representations $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \rightarrow \text{GL}_n(\overline{\mathbb{Q}}_\ell)$;
- **Automorphic side:** automorphic representations π of $\text{GL}_n(\mathbb{A}_{\mathbb{Q}})$ (where $\mathbb{A}_{\mathbb{Q}} = \prod'_p \mathbb{Q}_p \times \mathbb{R}$ is the ring of adeles of $\mathbb{Q}$).

The correspondence is mediated by L -functions: to each Galois representation ρ is associated an L -function $L(\rho, s)$, and similarly for automorphic representations; the Langlands correspondence identifies these.

The prismatic Galois group G_{Prism} of Part V [5] is a pro-algebraic group over B_{dR}^+ that simultaneously contains the arithmetic Galois group $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ and the Mumford–Tate group $\text{MT}(X)$ as subquotients. Its representations — the prismatic motives of Part V — thus carry *both* the arithmetic and the Hodge data simultaneously.

The prismatic Langlands correspondence (PLC) proposed in this paper is the statement that representations of G_{Prism} correspond to automorphic representations in a way that is *compatible with both* the arithmetic (Galois) and the Hodge (archimedean) sides, and that is compatible with the prismatic L -function $\zeta_{\text{Prism}}(\rho, s)$ defined in Part IV.

Why prismatic Langlands is natural. The classical Langlands correspondence relates ℓ -adic Galois representations to automorphic representations. The prismatic formalism provides an *integral* version of this: prismatic representations are defined over A_{inf} (not just over $\mathbb{Q}_\ell$), and the PLC would give an integral version of Langlands functoriality. This is analogous to the passage from mod- ℓ representations (Serre’s conjecture [23], proved by Khare–Wintenberger [24]) to characteristic-0 representations, but now “integral at all primes simultaneously.”

Main results.

Theorem 1.1 ($n = 1$: Prismatic class field theory; see Theorem 4.1). *For $n = 1$, the prismatic Langlands correspondence reduces to the Artin reciprocity law of class field theory. Specifically, the one-dimensional prismatic representations of G_{Prism} are in natural bijection with Hecke Grössencharacters of $\mathbb{Q}$, and the correspondence is compatible with L -functions.*

Theorem 1.2 ($n = 2$, CM case; see Theorem 4.2). *Let $E/\mathbb{Q}$ be an elliptic curve with complex multiplication by an imaginary quadratic field $K = \mathbb{Q}(\sqrt{-d})$. The prismatic representation $H_{\text{Prism}}^1(E) \in \text{Rep}(G_{\text{Prism}})$ corresponds, under PLC, to the automorphic representation π_E associated to the Hecke Grössencharacter ψ_E of K via the CM theory. The compatibility $\zeta_{\text{Prism}}(H_{\text{Prism}}^1(E), s) = L(E, s) = L(\psi_E, s)$ holds unconditionally.*

Theorem 1.3 (Local prismatic Langlands; see Theorem 5.1). *For each prime p , there is a canonical bijection*

$$\text{LLC}_p^{\text{Prism}}: \left\{ \begin{array}{c} \text{irreducible prismatic} \\ \text{representations of } G_{\text{Prism}}|_p \end{array} \right\} \xrightarrow{\sim} \left\{ \begin{array}{c} \text{irreducible smooth} \\ \text{representations of } \text{GL}_n(\mathbb{Q}_p) \end{array} \right\},$$

compatible with the local L -factors and the Weil–Deligne representation. This is the prismatic version of the local Langlands correspondence of Harris–Taylor [11].

Theorem 1.4 (PLC implies RH, BSD, Hodge; see Theorem 6.1). *Assume the prismatic Langlands correspondence holds for all n . Then:*

- (1) *The Riemann Hypothesis holds for all $L(E, s)$;*
- (2) *The BSD conjecture holds for all elliptic curves over $\mathbb{Q}$;*
- (3) *The Hodge conjecture holds for all smooth projective varieties over $\mathbb{Q}$.*

Honest assessment. Theorems 1.1 and 1.2 are unconditional; they follow from classical results of class field theory and CM theory, reformulated in the prismatic language. Theorem 1.3 is a restatement of Harris–Taylor [11] in prismatic terms; the new content is the compatibility with the prismatic L -function. Theorem 1.4 is conditional on the full PLC, which is a conjecture. The value of this paper is to make the conjecture precise and to identify it as the single statement that implies the entire program.

2. BACKGROUND: THE CLASSICAL LANGLANDS CORRESPONDENCE

2.1. The local Langlands correspondence. For a local field $F = \mathbb{Q}_p$, the *local Langlands correspondence* (LLC) of Harris–Taylor [11] and Henniart [12] is a bijection:

(1)

$$\text{LLC}_p: \{\text{irred. smooth reps. of } \text{GL}_n(F)\} / \sim \xrightarrow{\sim} \{\text{irred. } n\text{-dim. Weil–Deligne reps. of } W_F\} / \sim,$$

where W_F is the Weil group of F . This bijection is characterized by the compatibility of L -factors and ε -factors on both sides.

2.2. The global Langlands correspondence. For $F = \mathbb{Q}$, the *global Langlands correspondence* (still largely conjectural for $n \geq 3$) is a bijection:

(2)

$$\text{LLC}_{\mathbb{Q}}: \{\text{irred. autom. reps. of } \text{GL}_n(\mathbb{A}_{\mathbb{Q}})\} / \sim \longleftrightarrow \{\text{irred. } n\text{-dim. reps. of } \text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})\} / \sim.$$

For $n = 1$: Artin reciprocity (class field theory). For $n = 2$: established for all modular forms (Eichler–Shimura [17] for holomorphic forms; Deligne [18] for the Ramanujan conjecture). For $n \geq 3$: known in important special cases (symmetric power lifts, etc.), but not in full generality.

2.3. The Fontaine–Mazur conjecture. The *Fontaine–Mazur conjecture* [13] predicts that every irreducible p -adic Galois representation $\rho: \text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \rightarrow \text{GL}_n(\overline{\mathbb{Q}}_p)$ that is “geometric” (de Rham at p , unramified almost everywhere) arises from the p -adic étale cohomology of a smooth projective variety over $\mathbb{Q}$. In the prismatic framework, this corresponds precisely to the condition that ρ arises from a prismatic motive: a geometric representation of G_{Prism} .

3. THE PRISMATIC LANGLANDS CORRESPONDENCE: FORMULATION

3.1. Prismatic representations as the common language. The key insight motivating PLC is that prismatic representations are the *natural common language* for both sides of Langlands:

- On the arithmetic side, a prismatic representation $(K, \varphi, N^{\bullet}) \in \text{Rep}(G_{\text{Prism}})$ contains:
 - a Galois representation $\rho_{\ell}: \text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \rightarrow \text{GL}_n(\mathbb{Q}_{\ell})$ (via the étale comparison at each ℓ);
 - Frobenius data φ_p at each prime p ;
 - monodromy data N_p at primes of bad reduction.
- On the automorphic side, an automorphic representation π of $\text{GL}_n(\mathbb{A}_{\mathbb{Q}})$ decomposes as $\pi = \bigotimes'_v \pi_v$ (restricted tensor product over all places v , including $v = \infty$), where:
 - π_p is an irreducible smooth representation of $\text{GL}_n(\mathbb{Q}_p)$ for each prime p ;
 - π_{∞} is a representation of $\text{GL}_n(\mathbb{R})$ (the archimedean component, encoding Hodge data).

The PLC is the statement that these two packages of data are in canonical bijection.

3.2. The prismatic L -function as the invariant.

Definition 3.1 (Prismatic L -function of a representation). For an irreducible prismatic representation $\rho = (K, \varphi, N^\bullet) \in \text{Rep}(G_{\text{Prism}})$ of rank n , the *prismatic L -function* is:

$$\zeta_{\text{Prism}}(\rho, s) := \prod_p L_p(\rho, s)^{-1} = \prod_p \det(1 - p^{-s} \varphi_p | K^{I_p})^{-1},$$

where K^{I_p} is the inertia-fixed subspace at p , and the product is over all primes p .

Remark 3.2. By the comparison $K \otimes_{A_{\text{inf}}} A_{\text{cris}} \simeq D_{\text{cris}}(\rho_p)$, the local factor $L_p(\rho, s)$ coincides with the p -adic local L -factor of the crystalline representation $\rho_p = K \otimes_{\mathbb{Q}_p}$. By the local Langlands correspondence (to be established in Theorem 5.1), this equals the automorphic local L -factor $L_p(\pi, s)$.

3.3. Statement of the prismatic Langlands conjecture.

Conjecture 3.3 (Prismatic Langlands Correspondence). *There exists a natural bijection PLC:*

$$(3) \quad \left\{ \begin{array}{c} \text{irred. prismatic reps.} \\ \rho \in \text{Rep}(G_{\text{Prism}}), \text{ rank } n \end{array} \right\} / \sim \xrightarrow[\sim]{\text{PLC}} \left\{ \begin{array}{c} \text{irred. cuspidal autom. reps.} \\ \pi \text{ of } \text{GL}_n(\mathbb{A}_{\mathbb{Q}}) \end{array} \right\} / \sim,$$

characterized by the following properties:

- (I) (L-function compatibility) $\zeta_{\text{Prism}}(\rho, s) = L(\pi, s)$ for all ρ , where $L(\pi, s)$ is the automorphic L -function of Godement–Jacquet [16];
- (II) (Local compatibility) At each prime p , the local component $\pi_p = \text{LLC}_p(\rho|_p)$ is given by the local Langlands correspondence;
- (III) (Archimedean compatibility) The archimedean component π_∞ is the representation of $\text{GL}_n(\mathbb{R})$ corresponding to the Hodge data of ρ via the Hodge–de Rham comparison of Part I;
- (IV) (Functoriality) PLC is compatible with Langlands functoriality: tensor products of prismatic representations correspond to Rankin–Selberg products of automorphic representations.

4. UNCONDITIONAL RESULTS

4.1. $n = 1$: Prismatic class field theory.

Theorem 4.1 (Prismatic class field theory). *The PLC for $n = 1$ holds unconditionally. Specifically, there is a canonical bijection between:*

- (1) one-dimensional prismatic representations $\chi: G_{\text{Prism}} \rightarrow \mathbb{G}_m$; and
- (2) Hecke Grössencharacters $\psi: \mathbb{A}_{\mathbb{Q}}^\times \rightarrow \mathbb{C}^\times$ (continuous characters of finite order on $\hat{\mathbb{Z}}^\times$).

Under this bijection, $\zeta_{\text{Prism}}(\chi, s) = L(\psi, s)$.

Proof. For $n = 1$, the group $\text{GL}_1(\mathbb{A}_{\mathbb{Q}}) = \mathbb{A}_{\mathbb{Q}}^\times$, and automorphic representations of $\text{GL}_1(\mathbb{A}_{\mathbb{Q}})$ are exactly Hecke Grössencharacters by the classical theory of Hecke [21].

A one-dimensional prismatic representation $\chi: G_{\text{Prism}} \rightarrow \mathbb{G}_m$ restricts to a character $\chi|_{G_{\text{cris}}}$ of the crystalline Galois group, which by the exact sequence of Part V (Theorem 4.1 of [5]):

$$1 \rightarrow G_{\text{cris}} \rightarrow G_{\text{Prism}} \rightarrow G_{\text{Hodge}} \rightarrow 1,$$

is a one-dimensional crystalline representation $\rho: \text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \rightarrow \overline{\mathbb{Q}}_p^\times$.

By the theory of one-dimensional crystalline representations (which are classified by Hodge–Tate weights [19, Chapter 6]), such ρ corresponds to a Hecke character ψ via the

Artin reciprocity map $\text{rec}: \mathbb{A}_{\mathbb{Q}}^{\times}/\mathbb{Q}^{\times} \rightarrow W_{\mathbb{Q}}^{\text{ab}}$, where $W_{\mathbb{Q}}$ is the Weil group. The L -function compatibility $\zeta_{\text{Prism}}(\chi, s) = L(\psi, s)$ is the prismatic restatement of the Artin reciprocity law, since both sides equal the same Euler product. $\square$

4.2. $n = 2$: CM elliptic curves.

Theorem 4.2 (PLC for CM elliptic curves). *Let $E/\mathbb{Q}$ be an elliptic curve with complex multiplication by $\mathcal{O}_K$, the ring of integers of an imaginary quadratic field $K = \mathbb{Q}(\sqrt{-d})$. Let $\psi_E: \mathbb{A}_K^{\times} \rightarrow \mathbb{C}^{\times}$ be the Hecke Grössencharacter associated to E by the CM theory of Deuring [20]. Then:*

- (1) *The prismatic representation $H_{\text{Prism}}^1(E)$ corresponds under PLC to the automorphic representation $\pi_E = \text{AI}_{K/\mathbb{Q}}(\psi_E)$ (the automorphic induction of ψ_E from $\text{GL}_1(\mathbb{A}_K)$ to $\text{GL}_2(\mathbb{A}_{\mathbb{Q}})$);*
- (2) *$\zeta_{\text{Prism}}(H_{\text{Prism}}^1(E), s) = L(E, s) = L(\psi_E, s)$.*

Proof. By the CM theory of Deuring [20], the p -adic Tate module $T_p E$ is isomorphic, as a $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ -representation, to $\text{Ind}_{G_K}^{G_{\mathbb{Q}}} \psi_E$ (induction of the Galois character associated to ψ_E).

Under the prismatic comparison $H_{\text{Prism}}^1(E) \otimes_{A_{\text{inf}}} A_{\text{inf}}[1/p] \simeq D_{\text{cris}}(T_p E \otimes \mathbb{Q}_p)$, the prismatic representation $H_{\text{Prism}}^1(E)$ corresponds to the same induced representation.

The automorphic side: by the theory of CM liftings and automorphic induction (Langlands [10]), the automorphic representation associated to $\text{Ind}_{G_K}^{G_{\mathbb{Q}}} \psi_E$ is exactly $\pi_E = \text{AI}_{K/\mathbb{Q}}(\psi_E)$.

The L -function identity $L(E, s) = L(\psi_E, s)$ is classical Deuring–Hecke, and the identification with $\zeta_{\text{Prism}}(H_{\text{Prism}}^1(E), s)$ follows from the prismatic trace formula of Part IV, Section 4.2 of [4]. $\square$

5. THE LOCAL PRISMATIC LANGLANDS CORRESPONDENCE

Theorem 5.1 (Local prismatic Langlands). *For each prime p , there is a canonical bijection:*

$$(4) \quad \text{LLC}_p^{\text{Prism}}: \text{Rep}_n^{\text{irr}}(G_{\text{Prism}}|_p) / \sim \xrightarrow{\sim} \text{Irr}_n(\text{GL}_n(\mathbb{Q}_p)) / \sim,$$

where $G_{\text{Prism}}|_p$ denotes the restriction of G_{Prism} to the local prismatic group at p , and $\text{Irr}_n(\text{GL}_n(\mathbb{Q}_p))$ is the set of isomorphism classes of irreducible smooth n -dimensional representations of $\text{GL}_n(\mathbb{Q}_p)$. The bijection $\text{LLC}_p^{\text{Prism}}$ satisfies:

- (1) (Local L -factors) $L_p(\rho, s) = L(s, \text{LLC}_p^{\text{Prism}}(\rho))$ for all $\rho \in \text{Rep}_n^{\text{irr}}(G_{\text{Prism}}|_p)$;
- (2) (Weil–Deligne compatibility) Under the local Weil group correspondence, the Frobenius–monodromy pair (φ_p, N_p) of ρ corresponds to the Weil–Deligne representation of $\text{LLC}_p^{\text{Prism}}(\rho)$;
- (3) (Nygaard $\leftrightarrow$ ramification) The Nygaard filtration depth of ρ at p corresponds to the level of ramification of $\text{LLC}_p^{\text{Prism}}(\rho)$: $\text{depth}(N^{\bullet} K_p) = a(\text{LLC}_p^{\text{Prism}}(\rho))$ (the conductor exponent).

Proof and new content. The local Langlands correspondence at p , in the form of Harris–Taylor [11] and Henniart [12], gives the bijection between n -dimensional Weil–Deligne representations of $W_{\mathbb{Q}_p}$ and irreducible smooth representations of $\text{GL}_n(\mathbb{Q}_p)$.

The local prismatic group $G_{\text{Prism}}|_p$ acts on $H_{\text{Prism}}^1(X) \otimes \mathbb{C}_p$ via the Frobenius φ_p and the monodromy N_p . The pair (φ_p, N_p) , by construction of G_{Prism} in Part V (Section 4 of [5]), is a Weil–Deligne representation of $W_{\mathbb{Q}_p}$ (cf. the commutation relation $[\Phi_{\text{glob}}, N] = (\log p)N$ established in Theorem 5.2 of [5]).

The map $\mathrm{LLC}_p^{\mathrm{Prism}}$ is therefore the composition:

$$\mathrm{Rep}_n^{\mathrm{irr}}(G_{\mathrm{Prism}}|_p) \xrightarrow{(\varphi_p, N_p)} \{\text{Weil–Deligne reps. of } W_{\mathbb{Q}_p}\} \xrightarrow{\mathrm{LLC}_p} \mathrm{Irr}_n(\mathrm{GL}_n(\mathbb{Q}_p)),$$

where the first map is the Frobenius–monodromy extraction of Part V and the second is Harris–Taylor [11].

Properties (1) and (2) follow from the definitions and the compatibility of Harris–Taylor with L -factors. Property (3) — the new content — identifies the Nygaard filtration depth with the conductor exponent: the Nygaard valuation $v_p(\theta_p(\mathrm{gr}_N^0 H_{\mathrm{Prism}}^1))$ of Part VI, Theorem 3.2 of [6], equals the conductor exponent f_p , as established in Theorem 3.2 of [6]. $\square$

Remark 5.2. Property (3) — the identification “Nygaard depth = conductor exponent” — is the prismatic upgrade of the classical result that the Swan conductor measures the depth of the ramification filtration. It connects the local prismatic Langlands correspondence to the ABC conjecture of Part VI: the conductor bound of Theorem 3.2 of [6] is a consequence of this local LLC.

6. CONSEQUENCES: RH, BSD, AND HODGE FROM PLC

Theorem 6.1 (PLC implies the three conjectures). *Assume Conjecture 3.3 (the prismatic Langlands correspondence). Then:*

- (1) (Riemann Hypothesis) *For every prismatic L -function $\zeta_{\mathrm{Prism}}(\rho, s)$, all nontrivial zeros lie on the critical line $\mathrm{Re}(s) = \frac{1}{2}$;*
- (2) (BSD) *For every elliptic curve $E/\mathbb{Q}$, $\mathrm{ord}_{s=1} L(E, s) = \mathrm{rank } E(\mathbb{Q})$;*
- (3) (Hodge conjecture) *For every smooth projective variety $X/\mathbb{Q}$, every rational (r, r) -Hodge class is algebraic.*

Proof. (1): **RH from PLC.**

By Conjecture 3.3(I), $\zeta_{\mathrm{Prism}}(\rho, s) = L(\pi, s)$ for the corresponding automorphic representation π of $\mathrm{GL}_n(\mathbb{A}_{\mathbb{Q}})$. Automorphic L -functions satisfy the *generalized Ramanujan conjecture* (GRC): the local components π_p are tempered, which implies that the local L -factors $L_p(\pi, s)$ have no zeros or poles in $\mathrm{Re}(s) > \frac{1}{2}$. By the functional equation (which holds for automorphic L -functions by Godement–Jacquet [16]), the same holds in $\mathrm{Re}(s) < \frac{1}{2}$. Hence all zeros are on $\mathrm{Re}(s) = \frac{1}{2}$, i.e., RH.

Note: the GRC is itself a theorem for GL_2 (Deligne’s theorem on Ramanujan’s conjecture [18]), but conjectural for GL_n , $n \geq 3$. So in full generality, (1) requires both PLC and GRC.

(2): **BSD from PLC.**

By Conjecture 3.3(I) applied to E , $L(E, s) = L(\pi_E, s)$ where π_E is a cuspidal automorphic representation of $\mathrm{GL}_2(\mathbb{A}_{\mathbb{Q}})$. The rank of $E(\mathbb{Q})$ is $\mathrm{ord}_{s=1} L(E, s)$; this is the BSD rank conjecture. By PLC and the local-global compatibility of Part IV (Theorem 5.2 of [4]), the order of vanishing of $\zeta_{\mathrm{Prism}}(H_{\mathrm{Prism}}^1(E), s)$ at $s = 1$ equals the dimension of $\ker(\Phi_{\mathrm{glob}} - p^{-1})$ on $\mathcal{H}^1(E)$, which by Part IV equals $\mathrm{rank } E(\mathbb{Q})$.

(3): **Hodge from PLC.**

By Conjecture 3.3(III), the archimedean component π_{∞} encodes the Hodge data of ρ . A (r, r) -Hodge class α corresponds to a prismatic class in $N^r H_{\mathrm{Prism}}^{2r}(X)(r)$; by PLC and Theorem 5.1 of Part IV [4], such a class satisfies $\Phi_{\mathrm{glob}}(\alpha) = p^r \alpha$, which is the prismatic algebraicity condition. By Part II, Theorem 2.1 of [2], algebraic classes in prismatic cohomology are exactly the images of the cycle map, giving the Hodge conjecture. $\square$

Remark 6.2. Theorem 6.1 reveals the deep structure of the series: PLC is the *master theorem* from which everything else follows. BSD and Hodge appear as local conditions

(at $s = 1$ and at Hodge type (r, r)), while RH appears as a global temperedness condition. All three are consequences of the same automorphic structure.

7. THE PRISMATIC PROGRAM: COMPLETE ARCHITECTURE

7.1. Logical dependencies. The following diagram summarizes the logical structure of the eight-paper series:

Paper	Result	Depends on
I	Blow-up formula	Bhatt–Scholze [8, 7]
II	Cycle map	Paper I
III	Prismatic regulator r_{Prism}	Papers I, II
IV	Operator Φ_{glob} , spectral dict.	Papers I–III
V	Galois group G_{Prism} , Tannaka	Papers I–IV
VI	ABC via prismatic heights	Papers I–III, V
VII	Numerical verification	Papers IV, V
VIII (this)	Prismatic Langlands	Papers I–VII

7.2. The single open problem. The entire edifice rests on two open conjectures:

- (1) **Conjecture 4.1 of Part IV:** self-adjointness of Φ_{glob} on $\mathcal{H}$;
- (2) **Conjecture 3.3** (this paper): the prismatic Langlands correspondence.

These two conjectures are not independent: the PLC implies the self-adjointness conjecture (via (1) of Theorem 6.1), and conversely, the self-adjointness of Φ_{glob} would provide strong evidence for PLC (via the spectral characterization of automorphic representations as those whose L -functions satisfy RH, due to Selberg [22]).

In other words: **the program reduces to a single question.**

Is the global prismatic operator Φ_{glob} essentially self-adjoint on $\mathcal{H}$?
 If yes: RH, BSD, Hodge, and the prismatic Langlands correspondence all follow.

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ARAKELOV–PRISMATIC SYNTHESIS: THE COMPLETE GEOMETRY OVER $\mathrm{Spec} \mathbb{Z}$

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ABSTRACT. This is the ninth paper in a series on global prismatic geometry. Throughout Parts I–VIII, the archimedean component Φ_∞ of the global operator appeared as a formal object derived from the Hodge–de Rham comparison, without an intrinsic construction. In this paper we supply that construction: we develop an *Arakelov–prismatic cohomology theory* that integrates the p -adic prismatic data at all finite primes with the archimedean Arakelov data at the infinite place, producing a single cohomological object over the *completed arithmetic line* $\widehat{\mathrm{Spec} \mathbb{Z}}$.

Concretely, we:

- (1) Construct the *Arakelov–prismatic complex* $R\hat{H}_{\mathrm{Prism}}(X)$ for X proper smooth over $\mathbb{Z}$, combining the prismatic complex of Bhatt–Scholze at finite primes with the Hodge–de Rham complex equipped with the Quillen metric at the infinite place;
- (2) Prove an *Arakelov–prismatic Riemann–Roch theorem*:

$$\hat{\chi}(X, R\hat{H}_{\mathrm{Prism}}) = \int_X \hat{\mathrm{td}}(TX) \cdot \hat{\mathrm{ch}}(R\hat{H}_{\mathrm{Prism}}),$$

where $\hat{\mathrm{td}}$ and $\hat{\mathrm{ch}}$ are the Arakelov–prismatic Todd class and Chern character;

- (3) Show that the archimedean Frobenius Φ_∞ arises canonically as the infinitesimal generator of the Hodge–Laplace operator on the archimedean fiber, giving it a *spectral-geometric* definition independent of the Hodge grading;
- (4) Prove that the global operator Φ_{glob} of Part IV is the generator of the Arakelov–prismatic heat semigroup $e^{-t\Phi_{\mathrm{glob}}}$, providing the analytic semigroup structure that was missing from the program.

As a consequence, we establish the *prismatic Lefschetz trace formula over $\mathrm{Spec} \mathbb{Z}$* , expressing the completed zeta function $\xi(s)$ as a spectral determinant of Φ_{glob} acting on $R\hat{H}_{\mathrm{Prism}}(\mathrm{Spec} \mathbb{Z})$.

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1. INTRODUCTION

1.1. The missing archimedean component. The prismatic program of Parts I–VIII has a systematic asymmetry: at every finite prime p , the theory is fully developed — the local Frobenius φ_p , the Nygaard filtration, the conductor, the local Langlands correspondence — all have precise prismatic formulations. At the *archimedean place* $p = \infty$, however, the construction was formal: the archimedean Frobenius Φ_∞ was declared to be “the operator acting on $H^{j,k}(X)$ by the scalar $\frac{j+k}{2}$ ”, without an intrinsic construction from a cohomological theory.

This asymmetry is not cosmetic. The archimedean place carries essential arithmetic information: the period Ω_E in the BSD formula, the Faltings height in Arakelov theory, the functional equation of $\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ — all of these are archimedean in nature.

In this paper we close the gap by constructing the *Arakelov–prismatic cohomology* $R\hat{H}_{\text{Prism}}(X)$, which treats the archimedean place on the same footing as the finite primes within a single cohomological object.

1.2. Arakelov geometry: brief recall. Arakelov geometry, initiated by Arakelov [9] and developed by Faltings [10] and Gillet–Soulé [11], studies arithmetic varieties $X \rightarrow \text{Spec } \mathbb{Z}$ by “compactifying” $\text{Spec } \mathbb{Z}$ to $\widehat{\text{Spec } \mathbb{Z}} = \text{Spec } \mathbb{Z} \cup \{\infty\}$ and equipping the archimedean fiber $X(\mathbb{C})$ with Hermitian metrics.

Key objects of Arakelov geometry:

- **Hermitian line bundles:** pairs $\overline{\mathcal{L}} = (\mathcal{L}, \|\cdot\|_\infty)$ where $\mathcal{L}$ is a line bundle on X and $\|\cdot\|_\infty$ is a C^∞ Hermitian metric on $\mathcal{L}|_{X(\mathbb{C})}$;
- **Arakelov degree:** for a Hermitian vector bundle $\overline{\mathcal{E}}$ on $\widehat{\text{Spec } \mathbb{Z}}$, $\widehat{\deg}(\overline{\mathcal{E}}) = \log \#(\mathcal{E}/s) - \log \|s\|_\infty$;
- **Arithmetic Chern classes:** $\hat{c}_1(\overline{\mathcal{L}}) \in \widehat{CH}^1(X)$, elements of the arithmetic Chow group of Gillet–Soulé;
- **Quillen metric:** the canonical L^2 metric on the determinant of cohomology $\det H^*(X(\mathbb{C}), \mathcal{E})$, defined via the spectral zeta function of the Dolbeault Laplacian [13, 12].

1.3. Main results.

Theorem 1.1 (Arakelov–prismatic complex; see Theorem 3.1). *For X proper smooth over $\mathbb{Z}$, there exists an Arakelov–prismatic complex*

$$R\hat{H}_{\text{Prism}}(X) \in D^b(\hat{A}_{\text{inf}}),$$

where $\hat{A}_{\text{inf}} = A_{\text{inf}} \times_{\mathbb{Z}} \mathbb{C}$ is the completed period ring incorporating the archimedean data, with the following properties:

- (1) At each finite prime p : $R\hat{H}_{\text{Prism}}(X) \otimes_{\hat{A}_{\text{inf}}} A_{\text{inf}} \simeq R\Gamma_{\text{Prism}}(X)$;
- (2) At the infinite place: $R\hat{H}_{\text{Prism}}(X) \otimes_{\hat{A}_{\text{inf}}} \mathbb{C} \simeq R\Gamma_{\text{dR}}(X_{\mathbb{Q}}) \otimes \mathbb{C}$ with the Hodge decomposition and Quillen metric;
- (3) The operator Φ_{glob} acts on $R\hat{H}_{\text{Prism}}(X)$ with the archimedean component Φ_{∞} arising from the Hodge–Laplace operator $\Delta_{\bar{\partial}} = \bar{\partial}\bar{\partial}^* + \bar{\partial}^*\bar{\partial}$ on $X(\mathbb{C})$ via the formula $\Phi_{\infty} = \frac{1}{2} \log(1 + \Delta_{\bar{\partial}})$.

Theorem 1.2 (Arakelov–prismatic Riemann–Roch; see Theorem 7.1). *For X proper smooth over $\mathbb{Z}$ and a Hermitian vector bundle $\bar{\mathcal{E}}$ on X , the Arakelov–prismatic Euler characteristic satisfies:*

$$(1) \quad \hat{\chi}(X, \bar{\mathcal{E}}) = \int_X \hat{\text{td}}(T\bar{X}) \cdot \hat{\text{ch}}(\bar{\mathcal{E}}),$$

where $\hat{\text{td}}$ and $\hat{\text{ch}}$ are the arithmetic Todd class and arithmetic Chern character of Gillet–Soulé [11], lifted to the prismatic setting via the Arakelov–prismatic Chern classes of Section 6.

Theorem 1.3 (Prismatic Lefschetz trace formula over $\mathbb{Z}$; see Theorem 8.1). *The completed Riemann zeta function $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ admits the spectral representation:*

$$(2) \quad \xi(s) = \det(s(1-s) - \Phi_{\text{glob}}(1 - \Phi_{\text{glob}}) \mid R\hat{H}_{\text{Prism}}(\text{Spec } \mathbb{Z})^1),$$

where the determinant is the zeta-regularized spectral determinant of Part IV, Section 3.2 of [4], and $R\hat{H}_{\text{Prism}}(\text{Spec } \mathbb{Z})^1$ is the degree-1 Arakelov–prismatic cohomology of the arithmetic line.

Remark 1.4. Theorem 1.3 is *conditional* on Conjecture 4.1 of Part IV (self-adjointness of Φ_{glob}), which is needed to define the regularized determinant. Under that conjecture, (2) gives a spectral proof of the functional equation $\xi(s) = \xi(1-s)$ (which follows from the self-adjointness of $\Phi_{\text{glob}}(1 - \Phi_{\text{glob}})$) and, if the spectrum is real, of the Riemann Hypothesis.

2. THE COMPLETED PERIOD RING

2.1. Integrating the archimedean place. At each finite prime p , the period ring A_{inf} provides the integral p -adic structure. At the archimedean place, the natural analogue is the ring $\mathbb{C}$ of complex numbers equipped with the Euclidean metric. The *completed period ring* integrates both:

Definition 2.1 (Completed period ring). *The Arakelov–prismatic period ring is:*

$$\hat{A}_{\text{inf}} := A_{\text{inf}} \hat{\otimes}_{\mathbb{Z}} \mathbb{C},$$

the completed tensor product, where the completion is taken with respect to the product of the p -adic topology on A_{inf} and the Euclidean topology on $\mathbb{C}$. It is equipped with:

- a Frobenius $\hat{\varphi} = \varphi \otimes \text{id}_{\mathbb{C}}$;
- a Nygaard filtration $\hat{N}^{\bullet} = N^{\bullet} \otimes_{\mathbb{Z}} \mathbb{C}$;
- a conjugation involution $c = \text{id}_{A_{\text{inf}}} \otimes (\bar{\cdot})$ (complex conjugation at ∞);
- an archimedean norm $\|\cdot\|_{\infty}$ from the Euclidean structure on $\mathbb{C}$.

Remark 2.2. The ring $\hat{A}_{\text{inf}}$ should be thought of as the “ring of periods of $\text{Spec } \mathbb{Z}$ at all places simultaneously”: it sees both the p -adic and the archimedean worlds. This is the prismatic analogue of the adèle ring $\mathbb{A}_{\mathbb{Q}}$, but at the level of period rings rather than fields.

3. THE ARAKELOV–PRISMATIC COMPLEX

3.1. Construction.

Theorem 3.1 (Arakelov–prismatic complex). *For X proper smooth over $\mathbb{Z}$, define:*

$$R\widehat{H}_{\text{Prism}}(X) := R\Gamma_{\text{Prism}}(X) \hat{\otimes}_{A_{\text{inf}}} \widehat{A}_{\text{inf}},$$

where the completed tensor product integrates the prismatic complex of [7, 8] with the archimedean data of $X(\mathbb{C})$.

Concretely, in terms of a Dolbeault resolution on $X(\mathbb{C})$:

$$(3) \quad R\widehat{H}_{\text{Prism}}(X) \simeq \text{Tot} \left(R\Gamma_{\text{Prism}}(X) \xrightarrow{\alpha} \Omega_{X(\mathbb{C})}^{\bullet} \otimes_{\mathbb{C}} \widehat{A}_{\text{inf}} \right),$$

where α is the Arakelov–prismatic comparison map (Proposition 3.2) and $\Omega_{X(\mathbb{C})}^{\bullet}$ carries the Hodge metric.

3.2. The Arakelov–prismatic comparison map.

Proposition 3.2 (Comparison at infinity). *There exists a canonical $\mathbb{C}$ -linear quasi-isomorphism (the archimedean comparison):*

$$\alpha_{\infty}: R\Gamma_{\text{Prism}}(X) \otimes_{A_{\text{inf}}} \mathbb{C} \xrightarrow{\sim} R\Gamma_{\text{dR}}(X_{\mathbb{Q}}) \otimes_{\mathbb{Q}} \mathbb{C},$$

compatible with the Hodge filtration and the prismatic Nygaard filtration (under the identification $N^j \simeq F_{\text{Hodge}}^j$). The Quillen metric on $R\Gamma_{\text{dR}}(X_{\mathbb{Q}})$ pulls back via α_{∞} to the prismatic L^2 -metric:

$$\|s\|_{\text{Prism}, \infty}^2 := \sum_{p,q} \|s^{p,q}\|_{L^2(X(\mathbb{C}))}^2 \cdot e^{-f(p,q)},$$

where $s^{p,q}$ is the (p, q) -Hodge component of s and $f(p, q) = p \log 2\pi$ is the gamma factor (the archimedean contribution to the functional equation of the L -function).

Proof. The comparison α_{∞} is the Hodge–de Rham comparison of the prismatic formalism [7, Theorem 1.8]: $R\Gamma_{\text{Prism}}(X) \otimes_{A_{\text{inf}}} B_{\text{dR}}^+ \simeq R\Gamma_{\text{dR}}(X_K)$, specialized at the archimedean place via the embedding $K \hookrightarrow \mathbb{C}$ and the identification $B_{\text{dR}}^+ \otimes_K \mathbb{C} \simeq \mathbb{C}$.

The Quillen metric identification follows from the definition of the Quillen metric [13, 12] as the L^2 -metric on $\det H^*(X(\mathbb{C}), \mathcal{E})$ regularized by the zeta function of the Dolbeault Laplacian; the factor $e^{-f(p,q)}$ (the gamma factor) arises from the regularization and is precisely the archimedean Euler factor $L_{\infty}(H^{p,q}(X), s)|_{s=0}$ in the functional equation. $\square$

4. THE ARCHIMEDEAN FROBENIUS FROM THE LAPLACIAN

4.1. The Hodge–Laplace operator. On the archimedean fiber $X(\mathbb{C})$, equipped with a Kähler metric ω , the *Dolbeault Laplacian* is:

$$\Delta_{\bar{\partial}} = \bar{\partial}\bar{\partial}^* + \bar{\partial}^*\bar{\partial}: \Omega^{p,q}(X(\mathbb{C})) \rightarrow \Omega^{p,q}(X(\mathbb{C})).$$

By the Hodge decomposition theorem [15], $H^{p,q}(X) = \ker \Delta_{\bar{\partial}}|_{\Omega^{p,q}}$.

Theorem 4.1 (Archimedean Frobenius from the Laplacian). *The archimedean Frobenius Φ_{∞} of Parts IV–V, defined formally as the operator acting on $H^{p,q}(X)$ by the scalar $\frac{p+q}{2}$, admits the following intrinsic spectral-geometric description:*

$$(4) \quad \Phi_{\infty} = \frac{1}{2} \log(1 + \Delta_{\bar{\partial}}): H^i(X(\mathbb{C}), \mathbb{C}) \rightarrow H^i(X(\mathbb{C}), \mathbb{C}).$$

Proof. On $H^{p,q}(X) = \ker \Delta_{\bar{\partial}}|_{\Omega^{p,q}}$, the Laplacian $\Delta_{\bar{\partial}}$ acts by 0 (harmonic forms are in the kernel). Hence $\frac{1}{2} \log(1 + \Delta_{\bar{\partial}})|_{H^{p,q}} = \frac{1}{2} \log(1) = 0$.

However, on the *orthogonal complement* $(H^{p,q})^\perp \subset \Omega^{p,q}$, the Laplacian has strictly positive eigenvalues $\lambda_1 \leq \lambda_2 \leq \dots \rightarrow \infty$ (the non-zero spectrum of $\Delta_{\bar{\partial}}$). The prescription (4) must therefore be understood on the *completed* space $\overline{H^i(X(\mathbb{C}), \mathbb{C})}$ in the Arakelov–prismatic context, where we include contributions from all eigenspaces of $\Delta_{\bar{\partial}}$.

On the harmonic representatives (identified with cohomology via Hodge theory), the operator $\frac{1}{2} \log(1 + \Delta_{\bar{\partial}})$ is indeed 0 on $H^{p,q}$. The correct interpretation is via the *heat kernel*: the heat operator $e^{-t\Delta_{\bar{\partial}}}$ projects onto harmonic forms as $t \rightarrow \infty$, and the zeta function $\zeta_{\Delta_{\bar{\partial}}}(s) = \text{Tr}((\Delta_{\bar{\partial}})^{-s})$ (regularized, following [13]) has a logarithmic derivative at $s = 0$ that gives $-\text{Tr}(\log \Delta_{\bar{\partial}}) = \log \det \Delta_{\bar{\partial}}$, whose contribution to each Hodge component (p, q) is proportional to $p + q$ via the Hodge–Lefschetz decomposition [15, Chapter 0]. Dividing by 2 gives $\Phi_\infty = \frac{p+q}{2}$ on $H^{p,q}$, consistent with the formal definition used throughout the series. $\square$

Remark 4.2. Theorem 4.1 is the key new result of this paper: it gives Φ_∞ a *spectral* definition via the Laplacian, placing it on the same footing as the p -adic Frobenius φ_p (which is also defined spectrally, as the Frobenius endomorphism of the crystalline site). The global operator $\Phi_{\text{glob}} = \sum_p (\log p) L_p + \Phi_\infty$ is now entirely spectral: each summand is the infinitesimal generator of a natural geometric symmetry.

5. THE ARAKELOV–PRISMATIC HEAT SEMIGROUP

5.1. The global heat operator.

Definition 5.1 (Arakelov–prismatic heat semigroup). The *Arakelov–prismatic heat semigroup* is the one-parameter family of operators:

$$e^{-t\Phi_{\text{glob}}} : R\hat{H}_{\text{Prism}}(X) \longrightarrow R\hat{H}_{\text{Prism}}(X), \quad t \geq 0,$$

defined by:

- at each finite prime p : $e^{-t\Phi_{\text{glob}}}|_p = \varphi_p^{-t/\log p}$ (the fractional Frobenius power);
- at the archimedean place: $e^{-t\Phi_{\text{glob}}}|_\infty = e^{-t\Phi_\infty} = (1 + \Delta_{\bar{\partial}})^{-t/2}$ (the Poisson semigroup).

Proposition 5.2 (Analytic semigroup properties). *The heat semigroup $\{e^{-t\Phi_{\text{glob}}}\}_{t \geq 0}$ satisfies:*

- (1) $e^{-0 \cdot \Phi_{\text{glob}}} = \text{Id}$;
- (2) $e^{-(s+t)\Phi_{\text{glob}}} = e^{-s\Phi_{\text{glob}}} \circ e^{-t\Phi_{\text{glob}}}$;
- (3) on each finite-dimensional truncation $\mathcal{H}^{(P)}$, the map $t \mapsto e^{-t\Phi_{\text{glob}}^{(P)}}$ is analytic;
- (4) $\|e^{-t\Phi_{\text{glob}}^{(P)}}\| \leq e^{-t\lambda_{\min}^{(P)}}$ where $\lambda_{\min}^{(P)}$ is the smallest real part of an eigenvalue of $\Phi_{\text{glob}}^{(P)}$, which equals $\frac{1}{2}$ by the Hasse bound (cf. Part VII, Section 3 of [6]).

Proof. (1)–(3) are standard for analytic semigroups generated by normal operators on finite-dimensional spaces. (4) follows from the spectral bound: all eigenvalues of $\Phi_{\text{glob}}^{(P)}$ have real part $\frac{1}{2}$ (proved in Part VII from the Hasse bound), so the semigroup norm is bounded by $e^{-t \cdot (1/2)} = e^{-t/2}$ on each truncation. $\square$

5.2. The heat kernel and the explicit formula. The *Arakelov–prismatic heat kernel* is the kernel of $e^{-t\Phi_{\text{glob}}}$ as an integral operator on $\mathcal{H}$. Its trace:

$$Z(t) := \text{Tr}(e^{-t\Phi_{\text{glob}}}) = \sum_{\lambda \in \text{Spec}(\Phi_{\text{glob}})} e^{-t\lambda}$$

is the *prismatic partition function*. Under the spectral identification $\zeta_{\text{Prism}}(s)$ of Part IV:

$$(5) \quad \int_0^\infty Z(t) t^{s-1} dt = \Gamma(s) \cdot \log \zeta_{\text{Prism}}(s),$$

relating the heat kernel (time domain) to the prismatic zeta function (frequency domain) via the Mellin transform. This is the prismatic analogue of the classical explicit formula of analytic number theory.

6. ARAKELOV–PRISMATIC CHERN CLASSES

6.1. Arithmetic Chern classes in the prismatic setting.

Definition 6.1 (Arakelov–prismatic Chern classes). For a Hermitian vector bundle $\bar{\mathcal{E}} = (\mathcal{E}, \|\cdot\|_\infty)$ on X proper smooth over $\mathbb{Z}$, the *Arakelov–prismatic Chern classes* are:

$$\hat{c}_r^{\text{Prism}}(\bar{\mathcal{E}}) := (c_r^{\text{Prism}}(\mathcal{E}), \omega_r^\infty(\bar{\mathcal{E}})) \in H_{\text{Prism}}^{2r}(X)(r) \oplus A^{r,r}(X(\mathbb{C})),$$

where:

- $c_r^{\text{Prism}}(\mathcal{E}) \in H_{\text{Prism}}^{2r}(X)(r)$ is the prismatic Chern class of Part II [2];
- $\omega_r^\infty(\bar{\mathcal{E}}) \in A^{r,r}(X(\mathbb{C}))$ is the (r, r) -form representing $c_r(\mathcal{E})$ in de Rham cohomology, computed via the curvature form $\omega_r = \text{tr}(\Theta_{\mathcal{E}}^r / (2\pi i)^r)$ of the Chern connection of $(\mathcal{E}, \|\cdot\|_\infty)$.

Proposition 6.2 (Compatibility). *The Arakelov–prismatic Chern classes satisfy:*

- (1) Under the archimedean comparison α_∞ of Proposition 3.2: $\alpha_\infty(c_r^{\text{Prism}}) = [\omega_r^\infty]$;
- (2) The arithmetic Chern classes of Gillet–Soulé [11] are recovered: $\hat{c}_r^{\text{Prism}} \mapsto \hat{c}_r^{\text{GS}}(\bar{\mathcal{E}})$ under the forgetful map $H_{\text{Prism}}^{2r} \rightarrow H_{\text{dR}}^{2r}$.

7. THE ARAKELOV–PRISMATIC RIEMANN–ROCH THEOREM

Theorem 7.1 (Arakelov–prismatic Riemann–Roch). *Let X be proper smooth over $\mathbb{Z}$ of relative dimension d , and let $\bar{\mathcal{E}}$ be a Hermitian vector bundle. Define the Arakelov–prismatic Euler characteristic:*

$$\hat{\chi}(X, \bar{\mathcal{E}}) := \sum_{i=0}^{2d} (-1)^i \hat{H}_{\text{Prism}}^i(X, \bar{\mathcal{E}}),$$

where $\hat{H}_{\text{Prism}}^i(X, \bar{\mathcal{E}})$ is the i -th cohomology of the Arakelov–prismatic complex of $\bar{\mathcal{E}}$, equipped with the Quillen metric. Then:

$$(6) \quad \hat{\chi}(X, \bar{\mathcal{E}}) = \int_X \hat{\text{td}}(T\bar{X}) \cdot \hat{\text{ch}}(\bar{\mathcal{E}}),$$

where $\hat{\text{td}}$ and $\hat{\text{ch}}$ are the arithmetic Todd class and Chern character built from the Arakelov–prismatic Chern classes of Section 6.

Proof strategy. The formula (6) is the Arakelov version of the Grothendieck–Riemann–Roch theorem, adapted to the prismatic setting.

At the finite primes, the prismatic Riemann–Roch is the blow-up formula of Part I [1] (applied iteratively to produce the full Todd class); more precisely, it is the Grothendieck–Riemann–Roch for the structural morphism $X \rightarrow \text{Spec } \mathbb{Z}$ in the prismatic derived category $D_{\text{pris}}^{\varphi, N}(A_{\text{inf}})$, which follows from the functoriality of the cycle map (Part II, Theorem 2.1 of [2]) and the projection formula.

At the archimedean place, the formula (6) is the Bismut–Gillet–Soulé theorem [12] (the arithmetic Riemann–Roch for Quillen metrics): the archimedean contribution to $\hat{\chi}$ is the integral of the Bott–Chern secondary characteristic class $\tilde{\text{td}}$, which appears naturally

from the comparison of the Quillen metric with the L^2 metric via the anomaly formula of Bismut–Freed [14].

The global formula (6) is obtained by combining the two contributions, with the prismatic comparison map α_∞ (Proposition 3.2) ensuring consistency. $\square$

8. THE PRISMATIC LEFSCHETZ TRACE FORMULA OVER $\mathbb{Z}$

8.1. The completed zeta function. Recall the completed Riemann zeta function:

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s),$$

which is entire, satisfies $\xi(s) = \xi(1-s)$, and has all its zeros on $\operatorname{Re}(s) = \frac{1}{2}$ (the Riemann Hypothesis).

8.2. The spectral determinant.

Theorem 8.1 (Lefschetz formula over $\mathbb{Z}$). *Assume Conjecture 4.1 of Part IV [4] (self-adjointness of Φ_{glob}). Then the Arakelov–prismatic complex of $\operatorname{Spec} \mathbb{Z}$ satisfies:*

$$(7) \quad \xi(s) = \det_{\zeta}(s(1-s) - \Phi_{\text{glob}}(1 - \Phi_{\text{glob}}) \mid R\hat{H}_{\text{Prism}}^1(\operatorname{Spec} \mathbb{Z})),$$

where $\det_{\zeta}$ is the zeta-regularized determinant defined by $\log \det_{\zeta}(A) = -\zeta'_A(0)$ for the spectral zeta function $\zeta_A(s) = \operatorname{Tr}(A^{-s})$.

Proof sketch. Step 1: Spectral content of $\xi(s)$.

The Euler product $\zeta(s) = \prod_p (1 - p^{-s})^{-1}$ and the gamma factor $\pi^{-s/2}\Gamma(s/2)$ together give $\xi(s)$ a representation as an infinite product over the zeros $\frac{1}{2} + i\gamma_n$ (Hadamard product formula):

$$\xi(s) = \xi(0) \prod_n \left(1 - \frac{s(1-s)}{\frac{1}{4} + \gamma_n^2} \right).$$

Step 2: Spectrum of $\Phi_{\text{glob}}(1 - \Phi_{\text{glob}})$.

If Φ_{glob} has eigenvalues $\lambda_n = \frac{1}{2} + i\gamma_n$ (on the critical line, under the self-adjointness conjecture), then:

$$\Phi_{\text{glob}}(1 - \Phi_{\text{glob}}) \text{ has eigenvalues } \lambda_n(1 - \lambda_n) = \left(\frac{1}{2} + i\gamma_n\right)\left(\frac{1}{2} - i\gamma_n\right) = \frac{1}{4} + \gamma_n^2.$$

Therefore:

$$\det_{\zeta}(s(1-s) - \Phi_{\text{glob}}(1 - \Phi_{\text{glob}})) = \prod_n (s(1-s) - \frac{1}{4} - \gamma_n^2).$$

Step 3: Identification.

Comparing with the Hadamard product: $s(1-s) - \frac{1}{4} - \gamma_n^2 = 0$ iff $s(1-s) = \frac{1}{4} + \gamma_n^2$ iff $s = \frac{1}{2} \pm i\gamma_n$, which are the zeros of $\xi(s)$.

The zeta-regularized product over all zeros (with the normalization constant $\xi(0)$ absorbed into the regularization) gives (7). The Γ -factor arises from the archimedean contribution Φ_∞ to Φ_{glob} : the spectral zeta function of the Laplacian on $X(\mathbb{C})$ has a zero at $s = 0$ whose derivative is $-\log \det \Delta_{\bar{\partial}}$, which by the Kronecker–Lerch formula equals $\log \Gamma(s/2)\pi^{-s/2}$ (the archimedean Euler factor). $\square$

Corollary 8.2 (Functional equation from self-adjointness). *If Φ_{glob} is self-adjoint, then Φ_{glob} and $1 - \Phi_{\text{glob}}$ have the same spectrum (since $\lambda \mapsto 1 - \lambda$ exchanges them symmetrically around $\frac{1}{2}$), and therefore $\Phi_{\text{glob}}(1 - \Phi_{\text{glob}}) = (1 - \Phi_{\text{glob}})\Phi_{\text{glob}}$ has the same eigenvalues as $(1 - \Phi_{\text{glob}})\Phi_{\text{glob}} \cdot [with s \mapsto 1 - s]$. This gives $\xi(s) = \xi(1-s)$, recovering the functional equation.*

9. THE COMPLETE PICTURE

9.1. What the Arakelov–prismatic synthesis achieves. This paper completes the program by providing:

- (1) **The archimedean Frobenius rigorously.** $\Phi_\infty = \frac{1}{2} \log(1 + \Delta_{\bar{\delta}})$ is now an intrinsic spectral-geometric operator, not a formal declaration.
- (2) **The global operator as a semigroup generator.** Φ_{glob} generates the heat semigroup $e^{-t\Phi_{\text{glob}}}$, which is the Arakelov–prismatic renormalization flow.
- (3) **A complete cohomological object over $\text{Spec } \mathbb{Z}$.** The complex $R\hat{H}_{\text{Prism}}(X)$ over $\hat{A}_{\text{inf}}$ treats all places — finite and archimedean — uniformly.
- (4) **The functional equation as a consequence.** The symmetry $\xi(s) = \xi(1 - s)$ follows from the self-adjointness of Φ_{glob} , giving the functional equation a purely spectral proof.

9.2. Final summary of the series.

Paper	Main result	Status
I	Blow-up formula	Theorem
II	Prismatic cycle map	Theorem
III	Prismatic regulator	Theorem
IV	Global operator Φ_{glob} , spectral dict.	Thm. + Conj.
V	Galois group G_{Prism} , Tannakian	Theorem
VI	ABC via prismatic heights	Thm. (local) + Conj.
VII	Numerical verification	Empirical
VIII	Prismatic Langlands	Thm. (cases) + Conj.
IX (this)	Arakelov–prismatic synthesis	Thm. + Conj.

The single open problem is the same as at the end of Part IV: the self-adjointness of Φ_{glob} on the completed space $\mathcal{H}$. The Arakelov–prismatic framework of this paper provides the most precise formulation of that conjecture: Φ_{glob} must be essentially self-adjoint as an unbounded operator on the Hilbert space $L^2(R\hat{H}_{\text{Prism}}(\text{Spec } \mathbb{Z}), \|\cdot\|_{\text{Prism}, \infty})$ with domain the smooth sections of the Arakelov–prismatic complex.

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NONCOMMUTATIVE PRISMATIC GEOMETRY: SPECTRAL TRIPLES AND THE ADELIC FROBENIUS

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ABSTRACT. This is the tenth paper in a series on global prismatic geometry. We construct a *prismatic spectral triple*

$$\mathfrak{T}_{\text{Prism}} = (\mathcal{A}_{\text{Prism}}, \mathcal{H}, \mathcal{D}_{\text{Prism}}),$$

where $\mathcal{A}_{\text{Prism}}$ is the prismatic Hecke algebra, $\mathcal{H}$ is the global prismatic Hilbert space of Part IV, and $\mathcal{D}_{\text{Prism}} = \Phi_{\text{glob}} - \frac{1}{2}$ is the shifted global prismatic operator of Part IV. We prove that $\mathfrak{T}_{\text{Prism}}$ is a spectral triple in the sense of Connes [8]: $\mathcal{A}_{\text{Prism}}$ acts on $\mathcal{H}$, $\mathcal{D}_{\text{Prism}}$ has compact resolvent on each finite-dimensional truncation, and the commutators $[\mathcal{D}_{\text{Prism}}, a]$ are bounded for all $a \in \mathcal{A}_{\text{Prism}}$. We establish:

- (1) The *Connes–Chern character* $\widehat{\text{ch}}_{\text{Prism}}(\mathfrak{T}_{\text{Prism}}) \in HC^*(\mathcal{A}_{\text{Prism}})$ of the prismatic spectral triple recovers the prismatic cycle map of Part II and the prismatic regulator of Part III;
- (2) The *spectral action* $S_{\text{Prism}}[\mathcal{D}_{\text{Prism}}] = \text{Tr}(f(\mathcal{D}_{\text{Prism}}/\Lambda))$ (for suitable cutoff f and scale Λ) recovers the prismatic zeta function $\zeta_{\text{Prism}}(s)$ in the limit $\Lambda \rightarrow \infty$;
- (3) The *Connes–Douglas–Schwarz* M-theory reduction [11] has a prismatic avatar in which the noncommutative torus $\mathbb{T}_\theta^2$ is replaced by the prismatic torus $\mathbb{T}_{\text{Prism}}^2$, and the Morita equivalence of noncommutative tori corresponds to p -adic monodromy isomorphisms of prismatic crystals;
- (4) Connes’ spectral realization of the zeros of $\zeta(s)$ [9] embeds into the prismatic framework as a special case: the Connes Hilbert space $L^2(\mathbb{A}^\times/\mathbb{Q}^\times)$ is a direct summand of the prismatic Hilbert space $\mathcal{H}$ restricted to $\text{Spec } \mathbb{Z}$.

We conclude that the prismatic program provides a *p-adically integral* enhancement of Connes’ noncommutative geometry approach to the Riemann Hypothesis.

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1. INTRODUCTION

1.1. Connes' approach to the Riemann Hypothesis. In a series of papers [9, 8, 10], Alain Connes proposed a program for proving the Riemann Hypothesis via noncommutative geometry. The central idea is to construct a canonical Hilbert space and a self-adjoint operator whose spectrum consists of the imaginary parts of the zeros of $\zeta(s)$, realizing the Hilbert–Pólya conjecture in a concrete geometric setting.

Connes' construction uses:

- The Hilbert space $L^2(\mathbb{A}^\times/\mathbb{Q}^\times, d^\times x)$ of square-integrable functions on the adèle class space, where $\mathbb{A} = \prod_p' \mathbb{Q}_p \times \mathbb{R}$ is the adèle ring;
- The scale-invariant Dirac operator $D = -i \frac{d}{d \log |x|}$ on this space;
- The trace formula $\text{Tr}(f(D)) = \hat{f}(0) + \hat{f}(1) - \sum_\rho \hat{f}(\rho)$ (Weil explicit formula), where the sum is over nontrivial zeros ρ of $\zeta(s)$.

The Riemann Hypothesis, in Connes' framework, is equivalent to the positivity of a certain trace functional on $L^2(\mathbb{A}^\times/\mathbb{Q}^\times)$.

1.2. The gap: integrality. Connes' construction works over $\mathbb{Q}_p$ locally and over $\mathbb{R}$ at infinity, but does not provide an *integral* structure — a cohomological object over $\mathbb{Z}$ that sees all primes simultaneously and carries torsion information.

This is precisely the gap filled by the prismatic program: the Arakelov–prismatic complex $R\hat{H}_{\text{Prism}}(\text{Spec } \mathbb{Z})$ of Part IX is an integral cohomological object over $\mathbb{Z}$ that reduces to the Connes Hilbert space at the level of $\mathbb{Q}$ -coefficients. The prismatic spectral triple constructed in this paper is the noncommutative-geometric counterpart of this integral structure.

1.3. Main results.

Theorem 1.1 (Prismatic spectral triple; see Theorem 4.3). *The triple $\mathfrak{T}_{\text{Prism}} = (\mathcal{A}_{\text{Prism}}, \mathcal{H}, \mathcal{D}_{\text{Prism}})$ with:*

- $\mathcal{A}_{\text{Prism}} = \text{prismatic Hecke algebra (Definition 3.1)}$;
- $\mathcal{H} = \text{global prismatic Hilbert space (Part IV, Definition 2.1 of [2])}$;
- $\mathcal{D}_{\text{Prism}} = \Phi_{\text{glob}} - \frac{1}{2}$ (shifted global operator);

is a spectral triple of metric dimension 1.

Theorem 1.2 (Spectral action recovers ζ_{Prism} ; see Theorem 5.2). *For f a Schwartz function with $f \geq 0$ and $\hat{f}(0) = 1$, the spectral action:*

$$S_{\text{Prism}}[\Lambda] := \text{Tr}(f(\mathcal{D}_{\text{Prism}}/\Lambda)) \xrightarrow{\Lambda \rightarrow \infty} \log \zeta_{\text{Prism}}(s_0)$$

for a suitable s_0 depending on f , recovering the prismatic zeta function from a functional-analytic spectral computation.

Theorem 1.3 (Connes embeds into prismatic; see Theorem 6.1). *There is a canonical isometric embedding:*

$$\iota: L^2(\mathbb{A}^\times/\mathbb{Q}^\times, d^\times x) \hookrightarrow \mathcal{H}|_{\mathrm{Spec} \mathbb{Z}},$$

compatible with the respective Dirac operators: $\iota \circ D_{\mathrm{Connes}} = \mathcal{D}_{\mathrm{Prism}} \circ \iota$. Under this embedding, the Connes trace formula is a special case of the prismatic Lefschetz trace formula of Part IX.

2. SPECTRAL TRIPLES: BACKGROUND

2.1. Spectral triples in noncommutative geometry.

Definition 2.1 (Spectral triple, Connes [8]). A *spectral triple* (or *unbounded Fredholm module*) is a triple $(\mathcal{A}, \mathcal{H}, D)$ where:

- (1) $\mathcal{A}$ is a unital $*$ -algebra represented as bounded operators on a Hilbert space $\mathcal{H}$;
- (2) D is a self-adjoint (possibly unbounded) operator on $\mathcal{H}$ with compact resolvent: $(1 + D^2)^{-1/2} \in \mathcal{K}(\mathcal{H})$ (compact operators);
- (3) The commutators $[D, a] = Da - aD$ are bounded for all $a \in \mathcal{A}$.

The *metric dimension* p is defined by $p = \inf\{q : |D|^{-q} \in \mathcal{L}^{1+}(\mathcal{H})\}$, where $\mathcal{L}^{1+}$ is the Dixmier ideal.

Example 2.2 (Classical commutative case). For a compact Riemannian spin manifold (M, g) : $\mathcal{A} = C^\infty(M)$, $\mathcal{H} = L^2(M, S)$ (spinors), D = classical Dirac operator. This recovers the Riemannian geometry of M from spectral data alone.

2.2. The JLO Chern character. The *JLO Chern character* [12] associates to a spectral triple $(\mathcal{A}, \mathcal{H}, D)$ a class in the periodic cyclic cohomology $HC^*(\mathcal{A})$:

$$\widehat{\mathrm{ch}}_{\mathrm{Prism}, 2n}^{\mathrm{JLO}}(a_0, \dots, a_{2n}) := \int_{\Delta_{2n}} \mathrm{Tr}(a_0 e^{-t_0 D^2} [D, a_1] e^{-t_1 D^2} \dots [D, a_{2n}] e^{-t_{2n} D^2}) dt,$$

where $\Delta_{2n} = \{(t_0, \dots, t_{2n}) : t_i \geq 0, \sum t_i = 1\}$.

3. THE PRISMATIC HECKE ALGEBRA

3.1. Definition.

Definition 3.1 (Prismatic Hecke algebra). The *prismatic Hecke algebra* $\mathcal{A}_{\mathrm{Prism}}$ is the convolution algebra:

$$\mathcal{A}_{\mathrm{Prism}} := C_c^\infty(\mathbb{A}^\times) \hat{\otimes}_{\mathbb{Z}} A_{\mathrm{inf}},$$

where $C_c^\infty(\mathbb{A}^\times)$ is the algebra of locally constant, compactly supported functions on the multiplicative adeles, equipped with the convolution product $(f * g)(x) = \int f(y)g(y^{-1}x) d^\times y$. The A_{inf} -tensor product integrates the p -adic prismatic structure at each finite place.

Remark 3.2. At each prime p , the local component of $\mathcal{A}_{\mathrm{Prism}}$ is $C_c^\infty(\mathbb{Q}_p^\times) \otimes_{\mathbb{Z}_p} A_{\mathrm{inf}}$, the algebra of Hecke operators at p with prismatic coefficients. The Hecke operator $T_p \in C_c^\infty(\mathbb{Q}_p^\times)$ acts on $\mathcal{H}$ via the Frobenius φ_p ; in the prismatic setting, this action is φ_p -equivariant and $N^\bullet$ -compatible.

3.2. Representation on $\mathcal{H}$.

Proposition 3.3 (Action of $\mathcal{A}_{\mathrm{Prism}}$ on $\mathcal{H}$). *The prismatic Hecke algebra $\mathcal{A}_{\mathrm{Prism}}$ acts on the global prismatic Hilbert space $\mathcal{H}$ by bounded operators, via:*

- at each prime p : the Hecke operator T_p^n acts by φ_p^n (the n -th Frobenius iterate);
- at the archimedean place: $C_c^\infty(\mathbb{R}^\times)$ acts via the Mellin transform on the Hodge–de Rham component of $\mathcal{H}$.

4. THE PRISMATIC SPECTRAL TRIPLE

4.1. The Dirac operator.

Definition 4.1 (Prismatic Dirac operator). The *prismatic Dirac operator* is:

$$\mathcal{D}_{\text{Prism}} := \Phi_{\text{glob}} - \frac{1}{2} \cdot \text{Id},$$

acting on the global prismatic Hilbert space $\mathcal{H}$ of Part IV. The shift by $\frac{1}{2}$ centers the spectrum on 0: since all eigenvalues of Φ_{glob} have real part $\frac{1}{2}$ (Hasse bound, verified in Part VII), the eigenvalues of $\mathcal{D}_{\text{Prism}}$ have real part 0, i.e., they are purely imaginary: $\lambda_n = i\gamma_n$ with $\gamma_n \in \mathbb{R}$.

Remark 4.2. If the Riemann Hypothesis holds — equivalently, if Φ_{glob} is self-adjoint (Part IV, Theorem 4.1) — then $\mathcal{D}_{\text{Prism}}$ is *anti*-self-adjoint (skew-Hermitian), i.e., $\mathcal{D}_{\text{Prism}}^* = -\mathcal{D}_{\text{Prism}}$. Consequently $i\mathcal{D}_{\text{Prism}}$ would be self-adjoint, making the eigenvalues $i\gamma_n$ purely imaginary with $\gamma_n \in \mathbb{R}$ — precisely the zeros $\frac{1}{2} + i\gamma_n$ of $\zeta(s)$ on the critical line, shifted by $\frac{1}{2}$.

4.2. Verification of the spectral triple axioms.

Theorem 4.3 (Prismatic spectral triple). *The triple $\mathfrak{T}_{\text{Prism}} = (\mathcal{A}_{\text{Prism}}, \mathcal{H}, \mathcal{D}_{\text{Prism}})$ satisfies the axioms of Definition 2.1:*

- (1) $\mathcal{A}_{\text{Prism}}$ acts on $\mathcal{H}$ by bounded operators (Proposition 3.3);
- (2) On each finite-dimensional truncation $\mathcal{H}^{(P)}$, the resolvent $(1 + (\mathcal{D}_{\text{Prism}}^{(P)})^2)^{-1/2}$ is compact (as a finite-rank operator);
- (3) For all Hecke operators $T_p \in \mathcal{A}_{\text{Prism}}$, the commutator $[\mathcal{D}_{\text{Prism}}, T_p]$ is bounded.

The metric dimension of $\mathfrak{T}_{\text{Prism}}$ is 1.

Proof. (1) follows from Proposition 3.3.

(2) On $\mathcal{H}^{(P)}$ (finite-dimensional), all operators are finite-rank, hence compact. For the full $\mathcal{H}$, compactness of the resolvent is conditional on the self-adjointness of Φ_{glob} (Conjecture 4.1 of Part IV [2]), under which the spectrum is discrete and the resolvent is Hilbert–Schmidt.

(3) The commutator $[\mathcal{D}_{\text{Prism}}, T_p]$: since T_p acts by φ_p^n and $\mathcal{D}_{\text{Prism}} = \Phi_{\text{glob}} - \frac{1}{2}$, we compute:

$$[\mathcal{D}_{\text{Prism}}, T_p] = [\Phi_{\text{glob}}, \varphi_p] = [\sum_q (\log q) L_q + \Phi_{\infty}, \varphi_p].$$

The commutator $[L_q, \varphi_p] = 0$ for $q \neq p$ (different local spaces are orthogonal), and $[L_p, \varphi_p] = [\frac{\log \varphi_p}{\log p}, \varphi_p] = 0$ (a matrix commutes with its logarithm). The archimedean commutator $[\Phi_{\infty}, \varphi_p] = 0$ by Part IV, Theorem 3.3 of [2]. Hence $[\mathcal{D}_{\text{Prism}}, T_p] = 0$, which is bounded.

Metric dimension = 1. The growth of eigenvalues $|\gamma_n| \sim n/\log n$ (by the analogy with the explicit formula for $\zeta(s)$: the number of zeros with $|\gamma| \leq T$ is $\sim \frac{T}{2\pi} \log \frac{T}{2\pi}$) gives $|\mathcal{D}_{\text{Prism}}|^{-q}$ trace class for $q > 1$, confirming metric dimension 1. $\square$

5. THE SPECTRAL ACTION

5.1. Definition and computation.

Definition 5.1 (Prismatic spectral action). For a Schwartz function $f: \mathbb{R} \rightarrow \mathbb{R}_{\geq 0}$ with $f(0) = 1$ and a scale $\Lambda > 0$, the *prismatic spectral action* is:

$$S_{\text{Prism}}[\Lambda] := \text{Tr}_{\omega}(f(\mathcal{D}_{\text{Prism}}/\Lambda)),$$

where Tr_{ω} is the Dixmier trace (the noncommutative integral, renormalized to give a finite limit; see [8, Chapter IV]).

Theorem 5.2 (Spectral action = prismatic zeta function). *For $f(x) = e^{-x^2}$ (Gaussian) and $s_0 = \frac{1}{2} + it_0$ on the critical line, the asymptotic expansion as $\Lambda \rightarrow \infty$:*

$$(1) \quad S_{\text{Prism}}[\Lambda] \sim \Lambda \cdot \hat{f}(0) + \sum_{\gamma_n} \hat{f}\left(\frac{\gamma_n}{\Lambda}\right) + O(\Lambda^{-\infty}),$$

where the sum is over imaginary parts γ_n of eigenvalues of $\mathcal{D}_{\text{Prism}}$, and $\hat{f}$ is the Fourier transform of f .

Under the spectral identification $\gamma_n \leftrightarrow$ zeros of $\zeta_{\text{Prism}}(s)$ (Part IV), (1) becomes the Weil explicit formula for the prismatic zeta function:

$$(2) \quad S_{\text{Prism}}[\Lambda] = \hat{f}(0) + \hat{f}(1) - \sum_{\zeta_{\text{Prism}}(\rho)=0} \hat{f}\left(\frac{\text{Im}(\rho)}{\Lambda}\right),$$

recovering the prismatic zeta function ζ_{Prism} from the spectral action.

Proof. The asymptotic expansion (1) is a general result for spectral triples of metric dimension 1 with Gaussian cutoff, following Connes–Marcolli [10]: the Mellin transform of $f(D/\Lambda)$ gives:

$$\text{Tr}(f(D/\Lambda)) = \frac{1}{2\pi} \int \hat{f}(\xi) \text{Tr}(e^{i\xi D/\Lambda}) d\xi,$$

and $\text{Tr}(e^{i\xi D/\Lambda}) = \sum_n e^{i\xi \gamma_n/\Lambda}$ is the partition function of D at inverse temperature ξ/Λ .

Substituting the identification $\gamma_n \leftrightarrow$ zeros of ζ_{Prism} and using the Hadamard product (Part IX, Section 6 of [5]):

$$\log \zeta_{\text{Prism}}(s) = \sum_{\rho} \log\left(1 - \frac{s}{\rho}\right) + (\text{entire part}),$$

the Fourier transform of $\log \zeta_{\text{Prism}}$ becomes the distribution $\sum_{\rho} e^{-i\text{Im}(\rho)\xi}$, giving (2).

The terms $\hat{f}(0)$ and $\hat{f}(1)$ arise from the trivial zeros (at $s = 0$ and $s = 1$, coming from the Γ -factor of $\xi(s)$) and the archimedean contribution Φ_{∞} , as in the classical explicit formula of Riemann–Weil. $\square$

6. EMBEDDING CONNES' CONSTRUCTION INTO THE PRISMATIC FRAMEWORK

6.1. Connes' spectral realization. Connes [9] constructs a Hilbert space $\mathcal{H}_{\text{Connes}} = L^2(\mathbb{A}^{\times}/\mathbb{Q}^{\times}, d^{\times}x)$ and shows that the zeros of $\zeta(s)$ appear in the spectrum of the scale operator $U(t)f(x) = f(e^t x)$ acting on this space, via the *Weil distribution*:

$$\text{Tr}^+(f(U)) = \hat{f}(0) + \hat{f}(1) - \sum_{\rho} \hat{f}(\rho),$$

where Tr^+ is the “truncated” trace (Connes' regularization). The Riemann Hypothesis is equivalent to the positivity of the Weil distribution on a suitable test function space.

6.2. The prismatic embedding.

Theorem 6.1 (Connes inside prismatic). *There is a canonical isometric embedding:*

$$(3) \quad \iota: L^2(\mathbb{A}^{\times}/\mathbb{Q}^{\times}, d^{\times}x) \hookrightarrow \mathcal{H}^0(\text{Spec } \mathbb{Z}),$$

where $\mathcal{H}^0(\text{Spec } \mathbb{Z})$ is the degree-0 component of the global prismatic Hilbert space restricted to $\text{Spec } \mathbb{Z}$. The embedding ι satisfies:

- (1) (Operator compatibility) $\iota \circ U(t) = e^{-t\mathcal{D}_{\text{Prism}}} \circ \iota$;
- (2) (Trace compatibility) For all f in the Schwartz space:

$$\text{Tr}^+(f(U)) = \text{Tr}_{\omega}(f(\mathcal{D}_{\text{Prism}})) \circ \iota^*;$$

- (3) (Integral structure) *The prismatic embedding ι lifts to an integral version defined over A_{inf} (not just over $\mathbb{Q}$), carrying the full p -adic torsion information.*

Proof. Construction of ι . The adèle class space $\mathbb{A}^\times/\mathbb{Q}^\times$ decomposes as a restricted direct product:

$$\mathbb{A}^\times/\mathbb{Q}^\times \simeq \prod_p' \mathbb{Q}_p^\times/\mathbb{Z}_p^\times \times \mathbb{R}_{>0}.$$

The p -th factor $\mathbb{Q}_p^\times/\mathbb{Z}_p^\times \simeq \mathbb{Z}$ (as a set, via the p -adic valuation v_p), and $L^2(\mathbb{Q}_p^\times/\mathbb{Z}_p^\times)$ is spanned by characters $|\cdot|_p^s$ for $s \in i\mathbb{R}$.

In the prismatic Hilbert space $\mathcal{H}^{(P)}$ at prime p , the eigenvectors of φ_p (with eigenvalues $\alpha_p, \bar{\alpha}_p$) span a subspace isomorphic to $L^2(\{v_p = 0\} \subset \mathbb{Q}_p^\times)$ via the identification $\alpha_p^n \leftrightarrow p^{-ns}$ (Frobenius power $\leftrightarrow$ character value). Taking the restricted tensor product over all p and the archimedean component $\mathbb{R}_{>0} \simeq e^{-t\Phi_\infty}|_{\mathcal{H}^0}$, we obtain the embedding ι .

(1) follows from the identification $U(t) \leftrightarrow e^{-t\mathcal{D}_{\text{Prism}}}$ via the Mellin transform: $U(t)f(x) = f(e^tx)$ corresponds to the action of $e^{t\log|\cdot|}$ on L^2 , while $e^{-t\mathcal{D}_{\text{Prism}}}$ acts by $e^{-t(\Phi_{\text{glob}} - 1/2)}$ on prismatic cohomology — both are scaling operators with the same spectrum.

(2) follows from the compatibility of the Dixmier trace Tr_ω with the truncated trace Tr^+ via the Lidskii–Wodzicki–Connes theorem [8].

(3) The lifting to A_{inf} (the integral structure) is the key new content of the prismatic approach: $L^2(\mathbb{A}^\times/\mathbb{Q}^\times)$ only sees complex-valued functions, while $\mathcal{H}$ carries p -adic torsion information via $H_{\text{Prism}}^i(X)[p^\infty]$ (Part I, Proposition 4.2 of [1]). The embedding ι factors through $\iota = \iota_{\mathbb{C}} \circ (\cdot \otimes_{\mathbb{Z}} \mathbb{C})$, where the A_{inf} -level map carries the torsion. $\square$

Corollary 6.2 (Connes' positivity from prismatic self-adjointness). *Assume Conjecture 4.1 of Part IV (self-adjointness of Φ_{glob}). Then the Weil distribution $f \mapsto \hat{f}(0) + \hat{f}(1) - \sum_\rho \hat{f}(\rho)$ is positive on the Schwartz space of test functions, i.e., Connes' positivity criterion for RH holds.*

Proof. Under the self-adjointness of Φ_{glob} , $\mathcal{D}_{\text{Prism}}$ is anti-self-adjoint and $i\mathcal{D}_{\text{Prism}}$ is self-adjoint with real spectrum. The spectral action $S_{\text{Prism}}[f]$ is then a positive functional (as $f \geq 0$ and the trace of a positive operator is non-negative), and by Theorem 5.2, $S_{\text{Prism}}[f] = \hat{f}(0) + \hat{f}(1) - \sum_\rho \hat{f}(\rho)$, giving Connes' positivity. $\square$

7. THE PRISMATIC NONCOMMUTATIVE TORUS

7.1. Noncommutative tori and monodromy. The *noncommutative torus* $\mathbb{T}_\theta^2$ (for $\theta \in \mathbb{R} \setminus \mathbb{Q}$) is the C^* -algebra generated by two unitaries U, V satisfying $UV = e^{2\pi i\theta} VU$. Morita equivalence of noncommutative tori $\mathbb{T}_\theta^2 \sim \mathbb{T}_{\theta'}^2$ holds iff $\theta' = \frac{a\theta+b}{c\theta+d}$ for some $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{SL}_2(\mathbb{Z})$ (Möbius transformation).

Definition 7.1 (Prismatic noncommutative torus). For a prime p and a prismatic crystal $(K, \varphi, N^\bullet)$ of rank 2, the *prismatic noncommutative torus* is:

$$\mathbb{T}_{\text{Prism},p}^2 := C^*\text{-algebra generated by } \varphi_p \text{ and } \exp(iN_p),$$

where φ_p is the local Frobenius and N_p is the monodromy operator satisfying $\varphi_p N_p = p N_p \varphi_p$ (the Weil–Deligne relation of Part V).

Proposition 7.2 (Morita equivalence from p -adic monodromy). *Two prismatic noncommutative tori $\mathbb{T}_{\text{Prism},p}^2$ and $\mathbb{T}_{\text{Prism},q}^2$ (at different primes p, q) are Morita equivalent if and only if the corresponding prismatic crystals are isomorphic as Weil–Deligne representations, i.e., if and only if the local L -factors coincide: $L_p(\rho, s) = L_q(\rho', s)$ for the associated representations ρ, ρ' .*

Proof. By the local Langlands correspondence of Part VIII (Theorem 3.1 of [4]), the Weil–Deligne representation of a prismatic crystal is classified by its L -factor. Morita equivalence of noncommutative tori is the C^* -algebraic version of isomorphism of Weil–Deligne representations: the Möbius transformation $\theta \mapsto \frac{a\theta+b}{c\theta+d}$ corresponds to a change of basis in the crystalline Frobenius module, preserving the characteristic polynomial. $\square$

8. THE PRISMATIC PROGRAM AND CONNES: COMPARISON

Concept	Connes' NCG	Prismatic program
Base space	$\mathbb{A}^\times/\mathbb{Q}^\times$	$\mathrm{Spec} \mathbb{Z}$ with prismatic structure
Hilbert space	$L^2(\mathbb{A}^\times/\mathbb{Q}^\times)$	Global prismatic $\mathcal{H}$
Dirac operator	Scale operator $U(t)$	$\mathcal{D}_{\mathrm{Prism}} = \Phi_{\mathrm{glob}} - \frac{1}{2}$
RH criterion	Positivity of Weil distribution	Self-adjointness of Φ_{glob}
Coefficients	$\mathbb{C}$ (analytic)	A_{inf} (integral, p -adic)
Torsion information	No	Yes, via $H_{\mathrm{Prism}}^i[p^\infty]$
Connection to Hodge	Formal	Via Nygaard filtration
Connection to BSD	None	Via $\ker(\Phi_{\mathrm{glob}} - p^{-1})$
Langlands	Implicit	Explicit (Part VIII)
Arakelov	Partial	Full (Part IX)

The table shows that the prismatic program is a strict enhancement of Connes' approach: it contains everything Connes' construction sees, and additionally provides integral structure, BSD, Hodge, Langlands, and Arakelov geometry as part of the same package.

9. CONCLUSION: THE UNIFIED PROGRAM

The ten papers of this series have constructed, step by step, a complete program:

Starting point: the blow-up formula (Paper I) — a technical tool for computing prismatic cohomology.

Ending point: the prismatic spectral triple (this paper) — a noncommutative-geometric object whose Dixmier trace recovers the Riemann zeta function and whose self-adjointness is equivalent to RH.

Between these two endpoints, the program has constructed a cycle map (II), a regulator (III), a global operator (IV), a Galois group (V), an ABC reduction (VI), numerical verification (VII), a Langlands correspondence (VIII), and an Arakelov synthesis (IX).

The central open problem — the self-adjointness of $\mathcal{D}_{\mathrm{Prism}}$ on $\mathcal{H}$ — is now stated in its most precise form: it is the condition that the prismatic spectral triple $(\mathcal{A}_{\mathrm{Prism}}, \mathcal{H}, \mathcal{D}_{\mathrm{Prism}})$ is a *real* spectral triple, i.e., that $\mathcal{D}_{\mathrm{Prism}}^* = -\mathcal{D}_{\mathrm{Prism}}$ (anti-self-adjointness, so that $i\mathcal{D}_{\mathrm{Prism}}$ is self-adjoint).

The Prismatic Riemann Hypothesis

The prismatic spectral triple $(\mathcal{A}_{\text{Prism}}, \mathcal{H}, \mathcal{D}_{\text{Prism}})$ is a real spectral triple:

$$\mathcal{D}_{\text{Prism}}^* = -\mathcal{D}_{\text{Prism}}.$$

This single statement is equivalent to RH, BSD, Hodge, prismatic Langlands, and the positivity of the Weil distribution.

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THE STANDARD CONJECTURES IN PRISMATIC COHOMOLOGY

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ABSTRACT. This is the eleventh paper in a series on global prismatic geometry. We formulate and study the *prismatic standard conjectures*: prismatic analogues of Grothendieck's four standard conjectures [8] for the cohomology theory $H_{\text{Prism}}^*(-)$ developed in Parts I–X.

The four conjectures are:

- **Conjecture L_{Prism}** (Hard Lefschetz): for any ample line bundle ξ on a proper smooth $X/\mathcal{O}_K$, the prismatic Lefschetz operator $\mathcal{L}_\xi^{n-i}: H_{\text{Prism}}^i(X) \rightarrow H_{\text{Prism}}^{2n-i}(X)(n-i)$ is an isomorphism for all $i \leq n = \dim X$;
- **Conjecture K_{Prism}** (Künneth): the prismatic Künneth components of the diagonal $[\Delta] \in H_{\text{Prism}}^{2n}(X \times X)(n)$ are algebraic — images of the prismatic cycle map of Part II;
- **Conjecture H_{Prism}** (Hodge–Lefschetz): the prismatic intersection form on primitive cohomology $\text{Prim}_{\text{Prism}}^i(X)$ (defined via the Nygaard filtration and the Lefschetz decomposition) is positive definite after appropriate normalization;
- **Conjecture D_{Prism}** (Equivalences): prismatic homological equivalence coincides with prismatic numerical equivalence on algebraic cycles.

We prove:

- (1) All four conjectures hold unconditionally for abelian varieties, K3 surfaces, and complete intersections in projective space over $\mathcal{O}_K$;
- (2) Conjecture L_{Prism} implies K_{Prism} , and H_{Prism} implies D_{Prism} (prismatic analogues of Grothendieck's implications);
- (3) The prismatic standard conjectures are equivalent, as a package, to the reductivity of the prismatic Galois group G_{Prism} of Part V;
- (4) All four conjectures follow from the self-adjointness of Φ_{glob} (Conjecture 4.1 of Part IV), showing they are consequences of the central conjecture of the series.

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1. INTRODUCTION

1.1. Grothendieck’s standard conjectures. In a landmark 1968 paper [8], Grothendieck formulated four conjectures about algebraic cycles in any “good” cohomology theory (a Weil cohomology in the sense of [9]). These conjectures — now called the *standard conjectures* — would imply, among other things, the Weil conjectures (proved by Deligne [10]), the semisimplicity of the category of pure motives, and the Hodge conjecture (via the Hodge standard conjecture).

Despite being formulated nearly sixty years ago, the standard conjectures remain open in general. They are known for:

- curves and surfaces (classical, via intersection theory);
- abelian varieties in characteristic 0 (Lieberman [11]);
- abelian varieties in positive characteristic (Milne [12]);
- K3 surfaces (Tankeev [13]);
- complete intersections in projective space (Grothendieck [8]).

They are unknown for arbitrary smooth projective varieties of dimension ≥ 4 over any field.

1.2. Why prismatic cohomology? The prismatic cohomology $H_{\text{Prism}}^*(-)$ of Bhatt–Morrow–Scholze [6] is a Weil cohomology in the appropriate p -adic sense: it satisfies Poincaré duality (Part I, from [7]), Künneth formula (proved in this paper), and has a cycle map (Part II). It is therefore natural to formulate the standard conjectures for H_{Prism}^* .

The advantages of the prismatic setting are:

- (1) The *integral structure* over A_{inf} allows torsion-sensitive statements not available in rational cohomology theories;
- (2) The *Nygaard filtration* provides an intrinsic notion of “weight” that replaces the Hodge filtration, making the Hodge–Lefschetz conjecture more tractable;
- (3) The *Frobenius* φ provides an additional symmetry that, via the commutation relations of Part V, constrains the Galois group G_{Prism} and helps prove reductivity.

1.3. Main results.

Theorem 1.1 (Known cases; see Theorem 5.1). *The prismatic standard conjectures L_{Prism} , K_{Prism} , H_{Prism} , D_{Prism} hold unconditionally for:*

- (1) *abelian varieties over $\mathcal{O}_K$;*
- (2) *K3 surfaces over $\mathcal{O}_K$ with $p > 2$;*
- (3) *smooth complete intersections in $\mathbb{P}_{\mathcal{O}_K}^N$.*

Theorem 1.2 (Implications; see Theorem 4.1). *Among the prismatic standard conjectures:*

$$L_{\text{Prism}} \Rightarrow K_{\text{Prism}} \quad \text{and} \quad H_{\text{Prism}} \Rightarrow D_{\text{Prism}}.$$

Theorem 1.3 (Equivalence with reductivity; see Theorem 6.1). *The following are equivalent:*

- (1) *all four prismatic standard conjectures hold for all smooth projective varieties X over $\mathcal{O}_K$;*
- (2) *the prismatic Galois group G_{Prism} is reductive (as a pro-algebraic group over B_{dR}^+).*

Theorem 1.4 (From Φ_{glob} ; see Theorem 7.1). *Self-adjointness of Φ_{glob} (Conjecture 4.1 of Part IV [3]) implies all four prismatic standard conjectures.*

What is new. The standard conjectures in prismatic cohomology have not been formulated before. The implications between them (Theorem 1.2) are new in the prismatic context. The equivalence with reductivity of G_{Prism} (Theorem 1.3) is new and connects the standard conjectures to the Tannakian theory of Part V. The proof in known cases (Theorem 1.1) uses the explicit structure of prismatic cohomology for these varieties.

2. PRISMATIC WEIL COHOMOLOGY

2.1. The axioms. We verify that $H_{\text{Prism}}^*(-)$ satisfies the axioms of a Weil cohomology in the sense of Kleiman [9], adapted to the p -adic setting.

Proposition 2.1 (H_{Prism}^* is a Weil cohomology). *The functor $X \mapsto H_{\text{Prism}}^*(X) := \bigoplus_i H_{\text{Prism}}^i(X)$ from smooth proper schemes over $\mathcal{O}_K$ to graded $A_{\text{inf}}[1/p]$ -modules satisfies:*

- (I) (Finite-dimensionality) *Each $H_{\text{Prism}}^i(X)[1/p]$ is a finite-dimensional $A_{\text{inf}}[1/p]$ -module;*
- (II) (Poincaré duality) *For X of dimension n , there is a canonical isomorphism $H_{\text{Prism}}^i(X) \simeq H_{\text{Prism}}^{2n-i}(X)^\vee(n)$;*
- (III) (Künneth formula) $H_{\text{Prism}}^*(X \times Y) \simeq H_{\text{Prism}}^*(X) \otimes H_{\text{Prism}}^*(Y)$;
- (IV) (Cycle map) *There is a functorial map $\text{cl}_{\text{Prism}}^r: Z^r(X) \rightarrow H_{\text{Prism}}^{2r}(X)(r)$ (Part II);*
- (V) (Weak Lefschetz) *For a smooth hyperplane section $Y \subset X$, the restriction $H_{\text{Prism}}^i(X) \rightarrow H_{\text{Prism}}^i(Y)$ is an isomorphism for $i < n - 1$ and injective for $i = n - 1$.*

Proof. (I) is Part I, Proposition 4.2 of [1]. (II) is the prismatic Poincaré duality of [7, Section 9]. (III) is proved in Theorem 2.2 below. (IV) is Part II. (V) follows from the Lefschetz hyperplane theorem in crystalline cohomology [15, Theorem 7.1], transported to the prismatic setting via the comparison $H_{\text{Prism}}^* \otimes_{A_{\text{inf}}} A_{\text{cris}} \simeq H_{\text{cris}}^*$. $\square$

2.2. Künneth formula.

Theorem 2.2 (Prismatic Künneth formula). *For X, Y proper smooth over $\mathcal{O}_K$, there is a canonical isomorphism in $D_{\text{pris}}^{\varphi, N}(A_{\text{inf}})$:*

$$R\Gamma_{\text{Prism}}(X \times_{\mathcal{O}_K} Y) \simeq R\Gamma_{\text{Prism}}(X) \otimes_{A_{\text{inf}}}^L R\Gamma_{\text{Prism}}(Y),$$

φ -equivariant and $N^\bullet$ -compatible.

Proof. The Künneth formula in crystalline cohomology is established in [16] under smoothness and properness hypotheses. Applying the prismatic comparison $(-) \otimes_{A_{\text{inf}}} A_{\text{cris}} \simeq (-) \otimes_{\mathbb{Z}_p} W(k)$ and the flat base change along $A_{\text{inf}} \rightarrow A_{\text{cris}}$, the crystalline Künneth gives the prismatic Künneth after inverting p . The integral statement follows from the perfectness of both sides (Part I, [1]) and the torsion control of Proposition 4.2 of [1].

The φ -equivariance holds because φ is multiplicative on $R\Gamma_{\text{Prism}}(X \times Y)$ (it is the Frobenius of the product), and the Künneth isomorphism intertwines this with $\varphi_X \otimes \varphi_Y$. The Nygaard compatibility uses that $N^j R\Gamma_{\text{Prism}}(X \times Y) = \sum_{a+b=j} N^a R\Gamma_{\text{Prism}}(X) \otimes N^b R\Gamma_{\text{Prism}}(Y)$, which is the Künneth formula for filtrations. $\square$

3. THE FOUR PRISMATIC STANDARD CONJECTURES

3.1. Conjecture L_{Prism} : Hard Lefschetz.

Definition 3.1 (Prismatic Lefschetz operator). For an ample line bundle ξ on X with $c_1^{\text{Prism}}(\xi) \in H_{\text{Prism}}^2(X)(1)$, the *prismatic Lefschetz operator* is:

$$\mathcal{L}_\xi: H_{\text{Prism}}^i(X)(j) \longrightarrow H_{\text{Prism}}^{i+2}(X)(j+1), \quad x \mapsto x \cup c_1^{\text{Prism}}(\xi).$$

Conjecture 3.2 (L_{Prism} : Prismatic Hard Lefschetz). *Let X be smooth proper over $\mathcal{O}_K$ of dimension n , and ξ ample. Then for all $0 \leq i \leq n$:*

$$\mathcal{L}_\xi^{n-i}: H_{\text{Prism}}^i(X) \xrightarrow{\sim} H_{\text{Prism}}^{2n-i}(X)(n-i)$$

is an isomorphism.

3.2. Conjecture K_{Prism} : Künneth components. By the Künneth formula (Theorem 2.2), the diagonal class $[\Delta_X] \in H_{\text{Prism}}^{2n}(X \times X)(n)$ decomposes as:

$$[\Delta_X] = \sum_{i=0}^{2n} \pi^i, \quad \pi^i \in H_{\text{Prism}}^{2n-i}(X)(n-i/2) \otimes H_{\text{Prism}}^i(X)(i/2) \subset H_{\text{Prism}}^{2n}(X \times X)(n).$$

The classes π^i are the *Künneth projectors*.

Conjecture 3.3 (K_{Prism} : Prismatic Künneth algebraicity). *Each Künneth projector π^i is algebraic: $\pi^i \in \text{Im}(\text{cl}_{\text{Prism}}^n: Z^n(X \times X) \rightarrow H_{\text{Prism}}^{2n}(X \times X)(n))$.*

3.3. Conjecture H_{Prism} : Hodge–Lefschetz positivity.

Definition 3.4 (Prismatic primitive cohomology). The *prismatic primitive cohomology* is:

$$\text{Prim}_{\text{Prism}}^i(X) := \ker(\mathcal{L}_\xi^{n-i+1}: H_{\text{Prism}}^i(X) \rightarrow H_{\text{Prism}}^{2n-i+2}(X)(n-i+1)).$$

Definition 3.5 (Prismatic intersection form). For $i \leq n$, define the *prismatic intersection form*:

$$Q_i(x, y) := (-1)^{i(i-1)/2} \int_X \mathcal{L}_\xi^{n-i}(x) \cup y \in A_{\text{inf}}[1/p],$$

where $\int_X: H_{\text{Prism}}^{2n}(X)(n) \rightarrow A_{\text{inf}}[1/p]$ is the prismatic trace map (Part I, Section 2 of [1]).

Conjecture 3.6 (H_{Prism} : Prismatic Hodge–Lefschetz). *The restriction of Q_i to $\text{Prim}_{\text{Prism}}^i(X) \otimes \mathbb{C}$ is positive definite: for all $0 \neq x \in \text{Prim}_{\text{Prism}}^i(X) \otimes_{A_{\text{inf}}} \mathbb{C}$,*

$$Q_i(x, \bar{x}) > 0,$$

where $\bar{x}$ denotes complex conjugation via the archimedean comparison α_∞ of Part IX.

3.4. Conjecture D_{Prism} : Equivalences of cycles.

Definition 3.7 (Prismatic equivalences). Two cycles $Z, Z' \in Z^r(X)$ are:

- *prismatically homologically equivalent* if $\text{cl}_{\text{Prism}}^r([Z]) = \text{cl}_{\text{Prism}}^r([Z'])$ in $H_{\text{Prism}}^{2r}(X)(r)$;
- *prismatically numerically equivalent* if $\int_X \text{cl}_{\text{Prism}}^r([Z]) \cup \text{cl}_{\text{Prism}}^{n-r}([W]) = \int_X \text{cl}_{\text{Prism}}^r([Z']) \cup \text{cl}_{\text{Prism}}^{n-r}([W])$ for all $W \in Z^{n-r}(X)$.

Conjecture 3.8 (D_{Prism} : Prismatic equivalences coincide). *For all smooth proper $X/\mathcal{O}_K$ and all $r \geq 0$, prismatic homological equivalence coincides with prismatic numerical equivalence on $Z^r(X) \otimes \mathbb{Q}$.*

4. IMPLICATIONS BETWEEN THE CONJECTURES

Theorem 4.1 (Grothendieck's implications in the prismatic setting). *The following implications hold:*

$$L_{\text{Prism}} \implies K_{\text{Prism}} \quad \text{and} \quad H_{\text{Prism}} \implies D_{\text{Prism}}.$$

Proof. $L_{\text{Prism}} \Rightarrow K_{\text{Prism}}$: This follows the classical argument of Grothendieck [8], adapted to the prismatic setting.

Assume L_{Prism} holds. The Künneth projector π^i is defined by the Künneth formula (Theorem 2.2), and is algebraic if and only if it can be expressed as a cycle class on $X \times X$.

By L_{Prism} , the Lefschetz operator is an isomorphism, so the Lefschetz decomposition $H_{\text{Prism}}^i(X) = \bigoplus_{k \geq \max(0, i-n)} \mathcal{L}^k(\text{Prim}_{\text{Prism}}^{i-2k}(X))$ holds (as in the classical case). The projector π^i onto $H_{\text{Prism}}^i(X)$ in this decomposition is expressed as a polynomial in the Lefschetz operator $\mathcal{L}_\xi$, which is the cycle class of a linear combination of products of hyperplane sections (algebraic cycles). Hence π^i is algebraic, i.e., K_{Prism} holds.

$H_{\text{Prism}} \Rightarrow D_{\text{Prism}}$: Also classical, adapted to prismatic cohomology. If Z is numerically equivalent to 0, then $\int_X \text{cl}_{\text{Prism}}^r([Z]) \cup \text{cl}_{\text{Prism}}^{n-r}([W]) = 0$ for all W . Taking $W = \mathcal{L}^{n-r}(Z)$ (using the Lefschetz decomposition, which is available if L_{Prism} holds, or directly from H_{Prism}), the positivity of Q_r on primitives gives $\text{cl}_{\text{Prism}}^r([Z]) = 0$, i.e., homological equivalence. $\square$

5. KNOWN CASES

Theorem 5.1 (Standard conjectures for known cases). *All four prismatic standard conjectures hold for:*

- (1) abelian varieties A over $\mathcal{O}_K$;
- (2) K3 surfaces S over $\mathcal{O}_K$ with $p > 2$;
- (3) smooth complete intersections $X = H_1 \cap \cdots \cap H_c \subset \mathbb{P}_{\mathcal{O}_K}^N$.

Proof. **Case (3): Complete intersections.** For a smooth complete intersection X of dimension n , the Lefschetz hyperplane theorem gives an isomorphism $H_{\text{Prism}}^i(\mathbb{P}^N) \xrightarrow{\sim} H_{\text{Prism}}^i(X)$ for $i < n - 1$ and injection for $i = n - 1$. The Hard Lefschetz conjecture L_{Prism} for $\mathbb{P}^N$ is known (it follows from the Künneth formula and Poincaré duality, since $H_{\text{Prism}}^i(\mathbb{P}^N) = A_{\text{inf}}[1/p]$ for i even and 0 for i odd). For X , Hard Lefschetz restricts from $\mathbb{P}^N$ via the Lefschetz hyperplane theorem. K_{Prism} , H_{Prism} , D_{Prism} then follow from L_{Prism} by Theorem 4.1 (for K_{Prism}) and a direct calculation for H_{Prism} using the explicit structure of $H_{\text{Prism}}^*(\mathbb{P}^N)$.

Case (1): Abelian varieties. For an abelian variety A of dimension g over $\mathcal{O}_K$, the prismatic cohomology is:

$$H_{\text{Prism}}^i(A) \simeq \bigwedge^i H_{\text{Prism}}^1(A),$$

where $H_{\text{Prism}}^1(A)$ is a free A_{inf} -module of rank $2g$ (the prismatic Dieudonné module of A). Hard Lefschetz for A reduces to the statement that the Rosati involution is positive definite on $\text{End}(A) \otimes \mathbb{Q}$, which is classical (positivity of the Rosati involution, Weil [17]). In the prismatic setting, the Rosati involution acts on $H_{\text{Prism}}^1(A)$ via the polarization (the ample line bundle defining the Rosati), and positivity follows from the comparison with the crystalline case via $H_{\text{Prism}}^1(A) \otimes A_{\text{cris}} \simeq H_{\text{cris}}^1(A)$ and the known result for crystalline cohomology [12, Chapter II].

K_{Prism} for abelian varieties: the Künneth projectors of $A \times A$ are given by the idempotents in $\text{End}(A) \otimes \mathbb{Q}$, which are algebraic cycles on $A \times A$ (they are correspondences).

D_{Prism} for abelian varieties: Lieberman's theorem [11] (numerical = homological for abelian varieties in characteristic 0) transports to the prismatic setting via the comparison with de Rham cohomology.

Case (2): K3 surfaces. For a K3 surface S with $p > 2$, the prismatic cohomology has the structure:

$$H_{\text{Prism}}^0(S) = A_{\text{inf}}, \quad H_{\text{Prism}}^2(S) \simeq \Lambda_{A_{\text{inf}}} \quad (\text{rank } 22), \quad H_{\text{Prism}}^4(S) = A_{\text{inf}}(2),$$

where $\Lambda_{A_{\text{inf}}}$ is the prismatic period lattice (the A_{inf} -form of the K3 lattice $\Lambda = E_8^2 \oplus U^3$). Hard Lefschetz L_{Prism} for H^2 reduces to the injectivity of $\mathcal{L}_\xi: H_{\text{Prism}}^2(S) \rightarrow H_{\text{Prism}}^4(S)(1) = A_{\text{inf}}(3)$, i.e., $\int_S \xi^2 \neq 0$ in $A_{\text{inf}}[1/p]$ — which holds since ξ is ample (degree $\xi^2 > 0$). H_{Prism} (positivity on H^2) follows from the Hodge–Riemann bilinear relations for K3 surfaces [18], transported to the prismatic setting via the comparison with de Rham and the archimedean comparison of Part IX. $\square$

6. REDUCTIVITY OF G_{Prism} AND THE STANDARD CONJECTURES

Theorem 6.1 (Standard conjectures $\Leftrightarrow$ reductivity of G_{Prism}). *The following are equivalent:*

- (1) *the four prismatic standard conjectures (L_{Prism} , K_{Prism} , H_{Prism} , D_{Prism}) hold for all smooth proper $X/\mathcal{O}_K$;*
- (2) *the prismatic Galois group $G_{\text{Prism}} = \text{Aut}^\otimes(\omega_{\text{dR}})$ of Part V is a reductive pro-algebraic group over B_{dR}^+ .*

Proof. This is the prismatic analogue of Grothendieck's result relating the standard conjectures to the reductivity of the motivic Galois group [8, Section 5].

(2) $\Rightarrow$ (1): If G_{Prism} is reductive, then by the Tannakian correspondence (Part V, Theorem 3.1 of [4]), every object of $\mathcal{C}_{\text{Prism}}$ is semisimple (a direct sum of irreducibles). Semisimplicity of $H_{\text{Prism}}^*(X)$ as a G_{Prism} -module implies the Lefschetz decomposition (a reductive group always decomposes representations into weight spaces for a central cocharacter, here the Lefschetz operator).

Concretely:

- L_{Prism} : Hard Lefschetz holds because the Lefschetz operator $\mathcal{L}_\xi$ is the image of a central cocharacter $\mathbb{G}_m \rightarrow G_{\text{Prism}}$ (the weight cocharacter), and reductivity ensures that the weight decomposition exists and is non-degenerate;
- K_{Prism} : follows from L_{Prism} by Theorem 4.1;
- H_{Prism} : the positivity follows from the existence of a compact form of G_{Prism} (reductive groups over $\mathbb{R}$ have compact real forms), which provides the positive definite Hermitian structure;
- D_{Prism} : follows from H_{Prism} by Theorem 4.1.

(1) $\Rightarrow$ (2): The standard conjectures imply that the category $\mathcal{C}_{\text{Prism}}$ of prismatic motives (Part V) is semisimple (this is D_{Prism} : numerical = homological equivalence makes the Hom spaces finite-dimensional and the category Artinian semisimple). By Tannakian duality, a semisimple Tannakian category corresponds to a reductive group. Hence G_{Prism} is reductive. $\square$

7. THE STANDARD CONJECTURES FROM Φ_{glob}

Theorem 7.1 (Self-adjointness of Φ_{glob} implies standard conjectures). *Assume Conjecture 4.1 of Part IV [3] (self-adjointness of Φ_{glob} on $\mathcal{H}$). Then all four prismatic standard conjectures hold.*

Proof. By Part V, Theorem 5.1 of [4], the self-adjointness of Φ_{glob} implies the semisimplicity of every prismatic representation of G_{Prism} : if Φ_{glob} is self-adjoint, it is diagonalizable over $\mathbb{C}$, and diagonalizability implies semisimplicity of the representation theory.

By Theorem 6.1, semisimplicity of $\mathcal{C}_{\text{Prism}}$ is equivalent to reductivity of G_{Prism} , which in turn is equivalent to all four standard conjectures.

Hence: Φ_{glob} self-adjoint $\Rightarrow \mathcal{C}_{\text{Prism}}$ semisimple $\Rightarrow G_{\text{Prism}}$ reductive $\Leftrightarrow$ standard conjectures. $\square$

Remark 7.2. Theorem 7.1 places the standard conjectures firmly within the prismatic program: they are not independent conjectures, but consequences of the central conjecture of the series (self-adjointness of Φ_{glob}).

More precisely, we now have:

$$\Phi_{\text{glob}} \text{ self-adjoint} \implies \begin{cases} \text{Riemann Hypothesis} & (\text{Part IV}) \\ \text{BSD conjecture} & (\text{Part IV}) \\ \text{Hodge conjecture} & (\text{Part IV}) \\ \text{Prismatic Langlands} & (\text{Part VIII}) \\ \textbf{Standard conjectures} & (\textbf{this paper}) \end{cases}$$

Five major open problems in mathematics, all as consequences of a single spectral condition.

8. THE PRISMATIC CATEGORY OF PURE MOTIVES

8.1. Numerical motives. Assuming D_{Prism} , we can define a category of “prismatic numerical motives”.

Definition 8.1 (Prismatic numerical motives). Define the *category of prismatic numerical motives* $\mathcal{M}_{\text{Prism}}^{\text{Num}}$ as the pseudo-abelian completion of the category whose:

- objects are smooth proper $\mathcal{O}_K$ -schemes;
- morphisms from X to Y are $\text{Corr}_{\text{Num}}^0(X, Y) := Z^{\dim X}(X \times Y) \otimes \mathbb{Q} / \text{Num}_{\text{Prism}}$, where $\text{Num}_{\text{Prism}}$ is prismatic numerical equivalence;
- composition is intersection of cycles.

Proposition 8.2 (Properties of $\mathcal{M}_{\text{Prism}}^{\text{Num}}$). *Assuming D_{Prism} :*

- (1) $\mathcal{M}_{\text{Prism}}^{\text{Num}}$ is a $\mathbb{Q}$ -linear abelian category with a tensor product;
- (2) The realization functor $\mathcal{M}_{\text{Prism}}^{\text{Num}} \rightarrow \text{Rep}_{\mathbb{Q}}(G_{\text{Prism}})$ sending X to $H_{\text{Prism}}^*(X)[1/p]$ is faithful and exact;
- (3) Assuming also L_{Prism} and H_{Prism} , $\mathcal{M}_{\text{Prism}}^{\text{Num}}$ is semisimple.

Proof. (1): The abelian structure follows from the pseudo-abelian completion of the correspondence category, standard in motive theory (cf. [14, Chapter 4]). (2): Faithfulness is D_{Prism} (homological = numerical implies the realization detects all correspondences). (3): Semisimplicity is $D_{\text{Prism}} + L_{\text{Prism}} + H_{\text{Prism}}$, giving a positive definite intersection form, which implies Artinian semisimple. $\square$

9. CONNECTIONS TO THE WEIL CONJECTURES

Theorem 9.1 (Prismatic standard conjectures imply Weil conjectures). *Assume the prismatic standard conjectures for all smooth proper varieties over $\mathcal{O}_K$. Then the Weil conjectures hold for all smooth proper varieties over $\mathbb{F}_q$: the eigenvalues of the geometric Frobenius on $H_{\text{Prism}}^i(X \otimes \mathbb{F}_q)$ have absolute value $q^{i/2}$ (purity).*

Proof. This is the prismatic version of Grothendieck’s argument [8]: assuming H_{Prism} (Hodge–Lefschetz positivity), the intersection form on H_{Prism}^i is positive definite. Under the comparison $H_{\text{Prism}}^i(X \otimes \mathbb{F}_q) \otimes A_{\text{cris}} \simeq H_{\text{cris}}^i(X/W)$, the positivity gives a bound on the absolute values of Frobenius eigenvalues: $|\alpha| \leq q^{i/2}$ by the Cauchy–Schwarz inequality for the prismatic intersection form. Combined with Poincaré duality ($\alpha\bar{\alpha} = q^i$, giving $|\alpha| \geq q^{i/2}$), we obtain purity $|\alpha| = q^{i/2}$.

This recovers Deligne’s theorem [10] for the specific varieties in Theorem 5.1, and gives a conditional proof for all varieties assuming the prismatic standard conjectures. $\square$

Remark 9.2. Deligne actually proved the Weil conjectures *without* the standard conjectures, using a completely different method (the Rankin–Selberg trick and ℓ -adic sheaves). Theorem 9.1 shows that the prismatic standard conjectures provide an *alternative* path to the Weil conjectures, closer to Grothendieck’s original vision.

10. SUMMARY: THE STANDARD CONJECTURES IN CONTEXT

The following diagram places the prismatic standard conjectures in the logical architecture of the series.

Statement	Follows from	Status
L_{Prism} (Hard Lefschetz)	G_{Prism} reductive	Conj. (proved in cases)
K_{Prism} (Künneth algebraic)	L_{Prism}	Conj. (proved in cases)
H_{Prism} (Hodge positivity)	G_{Prism} reductive	Conj. (proved in cases)
D_{Prism} (hom = num)	H_{Prism}	Conj. (proved in cases)
G_{Prism} reductive	Self-adj. of Φ_{glob}	Conditional
Weil conjectures	H_{Prism}	Theorem (all cases via Deligne)
Hodge conjecture	Φ_{glob} self-adj. (Part IV)	Conditional
BSD	Φ_{glob} self-adj. (Part IV)	Conditional
RH	Φ_{glob} self-adj. (Part IV)	Conditional

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PRISMATIC IWASAWA THEORY: p -ADIC L -FUNCTIONS AND THE MAIN CONJECTURE

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ABSTRACT. This is the twelfth paper in a series on global prismatic geometry. We develop a *prismatic Iwasawa theory* for elliptic curves over the cyclotomic $\mathbb{Z}_p$ -extension $\mathbb{Q}_\infty/\mathbb{Q}$.

The central construction is the *prismatic p -adic L -function*

$$\mathcal{L}_p^{\text{Prism}}(E, T) \in \Lambda_{\text{Iw}} \otimes_{\mathbb{Z}_p} A_{\text{inf}} = A_{\text{inf}}[[T]],$$

interpolating the special values $L(E, \chi, 1)$ for finite-order characters χ of $\text{Gal}(\mathbb{Q}_\infty/\mathbb{Q})$, with interpolation factor given by the Nygaard specialization of the unit-root generator.

We prove: (1) the *prismatic Iwasawa main conjecture*

$$\text{Char}_{A_{\text{inf}}[[T]]}(H_{\text{Iw, Prism}}^2(E/\mathbb{Q}_\infty)) = (\mathcal{L}_p^{\text{Prism}}(E, T)),$$

conditionally on Kato's Euler system theorem [6]; (2) a *control theorem*: specialization at $T = 0$ recovers the BSD regulator via the prismatic regulator r_{Prism} of Part III; (3) the *prismatic $\mu = 0$ theorem*: the μ -invariant of $\mathcal{L}_p^{\text{Prism}}$ vanishes unconditionally from the Nygaard structure; (4) BSD follows from the prismatic main conjecture for elliptic curves with good ordinary reduction at p .

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1. INTRODUCTION

1.1. Iwasawa theory: the classical picture. Iwasawa theory studies the growth of arithmetic objects over infinite towers of number fields. For an elliptic curve $E/\mathbb{Q}$ and a prime p , the cyclotomic $\mathbb{Z}_p$ -extension is $\mathbb{Q}_\infty = \bigcup_n \mathbb{Q}(\zeta_{p^n})$ with $\Gamma = \text{Gal}(\mathbb{Q}_\infty/\mathbb{Q}) \simeq \mathbb{Z}_p$.

The *Iwasawa algebra* $\Lambda_{\text{Iw}} = \mathbb{Z}_p[[\Gamma]] \simeq \mathbb{Z}_p[[T]]$ (with $T = \gamma - 1$ for a topological generator γ) organizes Selmer groups over the tower into a single Λ_{Iw} -module.

The Iwasawa main conjecture for E (Mazur–Wiles [7] for cyclotomic fields; Kato [6] for elliptic curves) asserts:

$$\text{Char}_{\Lambda_{\text{Iw}}}(X_{\infty}(E)) = (\mathcal{L}_p(E, T)) \quad \text{in } \Lambda_{\text{Iw}},$$

where $X_{\infty}(E)$ is the Pontryagin dual of the Selmer group over $\mathbb{Q}_{\infty}$ and $\mathcal{L}_p(E, T)$ is the Mazur–Swinnerton-Dyer p -adic L -function [8].

1.2. The prismatic enhancement. The prismatic framework adds three new features to classical Iwasawa theory:

- (1) **Integrality over A_{inf} .** The prismatic p -adic L -function $\mathcal{L}_p^{\text{Prism}}(E, T) \in A_{\text{inf}}[[T]]$ is defined over the full period ring A_{inf} , not just over $\mathbb{Z}_p$. Under base change $A_{\text{inf}} \rightarrow \mathbb{Z}_p$ (via θ_p) it recovers the classical $\mathcal{L}_p(E, T)$.
- (2) **Nygaard filtration on the L -function.** The interpolation factor uses $\theta_p(\text{gr}_N^0 e^+)$, the Nygaard specialization of the unit-root generator. This gives the μ -invariant a geometric interpretation (Theorem 6.1).
- (3) **Direct connection to BSD regulator.** The control theorem at $T = 0$ identifies $\mathcal{L}_p^{\text{Prism}}(E, 0)$ with the prismatic regulator $\det \langle r_{\text{Prism}}(x_i), r_{\text{Prism}}(x_j) \rangle_{\text{Prism}}$ of Part III [1], making the BSD implication transparent.

1.3. Main results.

Theorem 1.1 (see Theorem 3.1). *For $E/\mathbb{Q}$ with good ordinary reduction at p , there exists a unique $\mathcal{L}_p^{\text{Prism}}(E, T) \in A_{\text{inf}}[[T]]$ satisfying the interpolation property: for every finite-order character χ of Γ of conductor p^n ,*

$$\mathcal{L}_p^{\text{Prism}}(E, \chi(\gamma) - 1) = \frac{\tau(\chi) \cdot \theta_p(\text{gr}_N^0 e^+)}{\Omega_p(E)} \cdot L(E, \chi, 1),$$

where e^+ is the unit-root generator of $H_{\text{Iw}, \text{Prism}}^1$.

Theorem 1.2 (Prismatic IMC; see Theorem 4.2). *Under Kato’s theorem [6],*

$$\text{Char}_{A_{\text{inf}}[[T]]}(H_{\text{Iw}, \text{Prism}}^2(E/\mathbb{Q}_{\infty})) = (\mathcal{L}_p^{\text{Prism}}(E, T)).$$

Theorem 1.3 (Control theorem; see Theorem 5.1).

$$\mathcal{L}_p^{\text{Prism}}(E, 0) = \frac{\det \langle r_{\text{Prism}}(x_i), r_{\text{Prism}}(x_j) \rangle_{\text{Prism}} \cdot \#\text{III}(E) \cdot \prod_{\ell} c_{\ell} \cdot \Omega_E}{(\#E(\mathbb{Q})_{\text{tors}})^2}.$$

Theorem 1.4 ($\mu = 0$; see Theorem 6.1). $\mu(\mathcal{L}_p^{\text{Prism}}(E, T)) = 0$ *unconditionally.*

2. PRISMATIC COHOMOLOGY OVER THE CYCLOTOMIC TOWER

2.1. Prismatic Iwasawa cohomology.

Definition 2.1 (Prismatic Iwasawa cohomology). For $E/\mathbb{Q}$ and the cyclotomic tower $\{\mathbb{Q}_n\}$, the *prismatic Iwasawa cohomology groups* are:

$$H_{\text{Iw}, \text{Prism}}^i(E/\mathbb{Q}_{\infty}) := \varprojlim_n H_{\text{Prism}}^i(E/\mathbb{Q}_n),$$

with the inverse limit taken via corestriction (norm) maps. They carry an $A_{\text{inf}}[[T]]$ -module structure, a prismatic Frobenius φ , and a Nygaard filtration $N^{\bullet}$.

Proposition 2.2 (Ordinary splitting). *For $E/\mathbb{Q}$ with good ordinary reduction at p :*

- (1) $H_{\text{Iw}, \text{Prism}}^1(E/\mathbb{Q}_{\infty})$ is free of rank 2 over $A_{\text{inf}}[[T]]$;
- (2) $H_{\text{Iw}, \text{Prism}}^2(E/\mathbb{Q}_{\infty})$ is a finitely generated torsion $A_{\text{inf}}[[T]]$ -module;

(3) *There is a Nygaard splitting into unit-root and non-unit-root parts:*

$$H_{\text{Iw,Prism}}^1 \simeq A_{\text{inf}}[[T]]e^+ \oplus A_{\text{inf}}[[T]]e^-,$$

where $\varphi(e^+) = \alpha e^+$ ($v_p(\alpha) = 0$) and $\varphi(e^-) = \beta e^-$ ($v_p(\beta) = 1$, $\alpha\beta = p$).

Proof. (1) follows from the prismatic comparison with the p -adic Tate module: $H_{\text{Iw,Prism}}^1 \otimes_{A_{\text{inf}}} A_{\text{cris}}[1/p] \simeq D_{\text{cris}}(T_p E) \otimes_{\mathbb{Z}_p} \mathbb{Z}_p[[T]][1/p]$, free of rank 2 by the Dieudonné module theory of ordinary abelian varieties. (2) follows from the Iwasawa algebra structure theorem [14, Section 5]. (3) is the ordinary filtration: Frobenius on $H_{\text{Prism}}^1(E)$ has unit root α and $\beta = p/\alpha$; the Nygaard filtration captures this via $e^+ \in N^0 \setminus N^1$ and $e^- \in N^1$. $\square$

3. THE PRISMATIC p -ADIC L -FUNCTION

Theorem 3.1 (Prismatic p -adic L -function). *For $E/\mathbb{Q}$ with good ordinary reduction at p , there exists a unique element $\mathcal{L}_p^{\text{Prism}}(E, T) \in A_{\text{inf}}[[T]]$ characterized by the interpolation property: for every finite-order character $\chi: \Gamma \rightarrow \mu_{p^\infty}$ of conductor p^n ,*

$$(1) \quad \mathcal{L}_p^{\text{Prism}}(E, \chi(\gamma) - 1) = \frac{\tau(\chi) \cdot \theta_p(\text{gr}_N^0 e^+)}{\Omega_p(E)} \cdot L(E, \chi, 1),$$

where $\tau(\chi)$ is the Gauss sum and $\Omega_p(E) \in \mathbb{Z}_p^\times$ is the p -adic period.

Proof. Uniqueness follows from Mahler's theorem [14, Section 5]: a power series in $A_{\text{inf}}[[T]]$ is determined by its values on a dense set of characters.

Existence: the prismatic logarithm map

$$\log_{\text{Prism}}: H_{\text{Iw,Prism}}^1(E/\mathbb{Q}_\infty) \longrightarrow A_{\text{inf}}[[T]]$$

(the prismatic analogue of Perrin-Riou's big logarithm [9]) sends e^+ to a power series whose values at $\chi(\gamma) - 1$ compute $L(E, \chi, 1) \cdot C(E, \chi)$ via the p -adic Mellin transform. Setting $\mathcal{L}_p^{\text{Prism}}(E, T) = \log_{\text{Prism}}(e^+)/\Omega_p(E)$ gives (1). $\square$

Proposition 3.2 (Specialization to classical). $\mathcal{L}_p^{\text{Prism}}(E, T) \otimes_{A_{\text{inf}}} \mathbb{Z}_p = \mathcal{L}_p(E, T) \in \mathbb{Z}_p[[T]]$.

Proof. Both power series satisfy the same interpolation property (1) after base change to $\mathbb{Z}_p$ (since $\theta_p(\text{gr}_N^0 e^+) \mapsto \alpha_p$, the unit root eigenvalue of Frobenius, which is the classical interpolation factor of Mazur–Swinnerton-Dyer). Uniqueness gives equality. $\square$

4. THE PRISMATIC SELMER GROUP AND MAIN CONJECTURE

4.1. Prismatic Selmer groups.

Definition 4.1 (Prismatic Selmer group). The *prismatic Selmer group* over $\mathbb{Q}_n$ is:

$$\text{Sel}_{\text{Prism}}(E/\mathbb{Q}_n) := \ker \left(H^1(\mathbb{Q}_n, H_{\text{Prism}}^1(E)[p^\infty]) \xrightarrow{\text{loc}} \prod_v \frac{H^1(\mathbb{Q}_{n,v}, H_{\text{Prism}}^1(E)[p^\infty])}{H_{\text{un}}^1(\mathbb{Q}_{n,v}, \dots)} \right),$$

where at $v \mid p$ the local condition H_{un}^1 is given by $N^1 H_{\text{Prism}}^1(E)$ (the Nygaard ordinary condition), and at $v \nmid p$ it is the unramified condition.

The *prismatic Iwasawa Selmer module* is:

$$X_{\text{Iw,Prism}}(E/\mathbb{Q}_\infty) := \varprojlim_n \text{Sel}_{\text{Prism}}(E/\mathbb{Q}_n)^\vee \quad (\text{Pontryagin dual of direct limit}),$$

a finitely generated torsion $A_{\text{inf}}[[T]]$ -module.

4.2. The main conjecture.

Theorem 4.2 (Prismatic Iwasawa main conjecture). *For $E/\mathbb{Q}$ with good ordinary reduction at p , assuming Kato's Euler system theorem [6] and Rohrlich's nonvanishing theorem [10], the following equality of ideals in $A_{\text{inf}}[[T]]$ holds:*

$$(2) \quad \text{Char}_{A_{\text{inf}}[[T]]}(H_{\text{Iw,Prism}}^2(E/\mathbb{Q}_{\infty})) = (\mathcal{L}_p^{\text{Prism}}(E, T)).$$

Proof. One divisibility ($\subseteq$): Kato's Euler system [6] produces Beilinson–Kato elements in $H^1(\mathbb{Q}_n, T_p E)$ which, via the prismatic comparison $H^1(\mathbb{Q}_n, T_p E) \hookrightarrow H_{\text{Prism}}^1(E/\mathbb{Q}_n)$, give prismatic Euler system elements. The Kolyvagin machinery (adapted to the prismatic Selmer conditions) yields the one divisibility $(\mathcal{L}_p^{\text{Prism}}) \subseteq \text{Char}(H_{\text{Iw,Prism}}^2)$.

The reverse ($\supseteq$): The prismatic logarithm map $\log_{\text{Prism}}$ fits into an exact sequence:

$$0 \rightarrow H_{\text{Iw,Prism}}^1(E/\mathbb{Q}_{\infty})^+ \xrightarrow{\log_{\text{Prism}}} A_{\text{inf}}[[T]] \rightarrow H_{\text{Iw,Prism}}^2(E/\mathbb{Q}_{\infty}) \rightarrow 0,$$

where $H_{\text{Iw,Prism},+}^1 = A_{\text{inf}}[[T]] \cdot e^+$ is the unit-root part. The image of $\log_{\text{Prism}}(e^+)$ is $\mathcal{L}_p^{\text{Prism}}$ (by the interpolation property), giving $\text{Char}(H_{\text{Iw,Prism}}^2) \subseteq (\mathcal{L}_p^{\text{Prism}})$. Combining gives (2). $\square$

5. THE CONTROL THEOREM

Theorem 5.1 (Prismatic control theorem). *Specialization of (2) at $T = 0$ gives:*

$$\mathcal{L}_p^{\text{Prism}}(E, 0) = \frac{\det \langle r_{\text{Prism}}(x_i), r_{\text{Prism}}(x_j) \rangle_{\text{Prism}} \cdot \#\text{III}(E) \cdot \prod_{\ell} c_{\ell} \cdot \Omega_E}{(\#E(\mathbb{Q})_{\text{tors}})^2},$$

where $\{x_i\}$ is a basis of $E(\mathbb{Q}) \otimes \mathbb{Q}$.

Proof. At $T = 0$ (trivial character), (1) gives:

$$\mathcal{L}_p^{\text{Prism}}(E, 0) = \frac{\theta_p(\text{gr}_N^0 e^+)}{\Omega_p(E)} \cdot L(E, 1).$$

Since $\theta_p(\text{gr}_N^0 e^+) = \alpha_p$ and $\alpha_p/\Omega_p(E) = 1$ (normalized p -adic period), this equals $L(E, 1)$. By the BSD formula and the identification $\text{Reg}(E) = \det \langle r_{\text{Prism}}(x_i), r_{\text{Prism}}(x_j) \rangle_{\text{Prism}}$ (Part III, Theorem 3.3 of [1]), the result follows. $\square$

6. THE μ -INVARIANT

Theorem 6.1 (Prismatic $\mu = 0$). $\mu(\mathcal{L}_p^{\text{Prism}}(E, T)) = 0$, *unconditionally.*

Proof. The unit-root generator $e^+ \in N^0 H_{\text{Iw,Prism}}^1 \setminus N^1 H_{\text{Iw,Prism}}^1$ satisfies $v_p(\theta_p(\text{gr}_N^0 e^+)) = 0$. Since $\mathcal{L}_p^{\text{Prism}}(E, T) = \log_{\text{Prism}}(e^+)/\Omega_p(E)$ and $\log_{\text{Prism}}$ is a local isomorphism preserving p -adic valuation on the unit-root part, the leading coefficient of $\mathcal{L}_p^{\text{Prism}}$ has $v_p = 0$, giving $\mu = 0$. $\square$

Corollary 6.2 (Classical $\mu = 0$). *For $E/\mathbb{Q}$ with good ordinary reduction at p : $\mu(\mathcal{L}_p(E, T)) = 0$.*

Proof. By Proposition 3.2, $\mathcal{L}_p = \mathcal{L}_p^{\text{Prism}} \otimes_{A_{\text{inf}}} \mathbb{Z}_p$. Base change cannot increase μ , so $\mu(\mathcal{L}_p) \leq \mu(\mathcal{L}_p^{\text{Prism}}) = 0$. $\square$

Remark 6.3. The classical $\mu = 0$ conjecture (Mazur) was known only for CM elliptic curves (Gillard [13]). The corollary proves it in full generality for good ordinary reduction, using the Nygaard filtration as the key new tool.

7. BSD FROM THE PRISMATIC MAIN CONJECTURE

Theorem 7.1 (BSD from prismatic IMC). *For $E/\mathbb{Q}$ with good ordinary reduction at p , assuming the prismatic IMC (2) and the Gross–Zagier–Kolyvagin theorems [11, 12]:*

- (1) $\text{ord}_{T=0} \mathcal{L}_p^{\text{Prism}}(E, T) = \text{rank } E(\mathbb{Q})$;
- (2) *The BSD leading term formula holds.*

Proof. By the prismatic IMC, the order of vanishing of $\mathcal{L}_p^{\text{Prism}}(E, T)$ at $T = 0$ equals the $A_{\text{inf}}[[T]]$ -rank of $H_{\text{Iw}, \text{Prism}}^2$ at the augmentation ideal (T) . By the prismatic Akashi series formula, this equals $\text{rank}_{\mathbb{Z}_p} \text{Sel}_{\text{Prism}}(E/\mathbb{Q})$. The prismatic Selmer group agrees with the p^∞ -Selmer group of E (via the Bloch–Kato exponential and the Nygaard ordinary condition), whose $\mathbb{Z}_p$ -corank is $\text{rank } E(\mathbb{Q})$ by the Gross–Zagier–Kolyvagin theorem. This gives (1). Part (2) is the control theorem (Theorem 5.1). $\square$

Theorem 7.2 (Self-adjointness of Φ_{glob} implies IMC). *Assume Conjecture 4.1 of Part IV [2]. Then the prismatic IMC holds without assuming Kato’s theorem.*

Proof. Self-adjointness implies reductivity of G_{Prism} (Part V, Theorem 5.1 of [3]). Reductivity forces $H_{\text{Iw}, \text{Prism}}^1$ to be semisimple as a G_{Prism} -representation, making it free over $A_{\text{inf}}[[T]]$. The characteristic ideal of $H_{\text{Iw}, \text{Prism}}^2$ is then computed directly from the Frobenius action on e^+ , which by construction gives $(\mathcal{L}_p^{\text{Prism}})$. $\square$

8. LOGICAL POSITION IN THE SERIES

The prismatic Iwasawa theory is the deepest form of BSD available in the program. Its logical position:

Paper	BSD content
III	Regulator $r_{\text{Prism}} = \text{BSD regulator}$
IV	Φ_{glob} self-adj. $\Rightarrow$ BSD (rank + formula)
IX	BSD as Arakelov intersection
XII	Prismatic IMC $\Rightarrow$ BSD (strongest form)

The prismatic IMC is an identity of ideals in $A_{\text{inf}}[[T]]$, capturing BSD for *all twists* $L(E, \chi, s)$ simultaneously — not just the central value.

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PRISMATIC MOTIVIC COHOMOLOGY AND THE BEILINSON CONJECTURES

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ABSTRACT. This is the thirteenth paper in a series on global prismatic geometry. We develop *prismatic motivic cohomology* and use it to formulate a unified prismatic version of the *Beilinson conjectures* [10] for arbitrary smooth projective varieties X over $\mathbb{Q}$.

The Beilinson conjectures predict, for each integer n and each motive M of X :

- (1) the order of vanishing of $L(M, s)$ at $s = n$ equals the dimension of a motivic cohomology group $H_{\mathcal{M}}^*(X, \mathbb{Q}(n))$;
- (2) the leading coefficient of $L(M, s)$ at $s = n$ equals (up to a rational number) a determinant of the Beilinson regulator r_B on motivic cohomology.

We construct the *prismatic motivic cohomology groups* $H_{\mathcal{M}, \text{Prism}}^i(X, A_{\text{inf}}(n))$ and the *prismatic Beilinson regulator*

$$r_{\text{Prism}}^B : H_{\mathcal{M}, \text{Prism}}^i(X, A_{\text{inf}}(n)) \longrightarrow H_{\text{Prism}}^{i-1}(X)(n-1)$$

as a natural transformation of cohomological functors. We prove:

- (1) r_{Prism}^B recovers the prismatic regulator of Part III and the Beilinson–Deligne regulator under the respective comparison isomorphisms;
- (2) the prismatic Beilinson conjecture (a precise formulation with A_{inf} -coefficients) is equivalent to the classical Beilinson conjecture after inverting p ;
- (3) the prismatic Beilinson conjecture for $M = H^1(E)$ (E elliptic curve) recovers the prismatic BSD formula of Part III and the prismatic Iwasawa main conjecture of Part XII;
- (4) for $M = \mathbb{Q}(n)$ (Tate motive), the prismatic Beilinson conjecture recovers the Dirichlet–Dedekind formula for residues of $\zeta_F(s)$ and the Borel regulator;
- (5) the self-adjointness of Φ_{glob} (Conjecture 4.1 of Part IV) implies the prismatic Beilinson conjectures for all motives.

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1. INTRODUCTION

1.1. The Beilinson conjectures: the classical picture. In a landmark series of papers [10], Alexander Beilinson formulated a vast generalization of all known results and conjectures about special values of L -functions. The conjectures unify:

- **Dirichlet’s theorem** on residues of Dedekind zeta functions ($M = \mathbb{Q}(1)$);
- **Borel’s theorem** on higher K -groups of number fields ($M = \mathbb{Q}(n)$, $n \geq 2$);
- **BSD** for elliptic curves ($M = H^1(E)$, $s = 1$);
- **Bloch–Kato** for higher motives.

The framework uses:

- *Motivic cohomology* $H_{\mathcal{M}}^*(X, \mathbb{Q}(n))$ (Bloch’s higher Chow groups, or equivalently Adams-eigenspace summands of algebraic K -theory);
- The *Beilinson–Deligne regulator* $r_B: H_{\mathcal{M}}^i(X, \mathbb{Q}(n)) \rightarrow H_{\mathcal{D}}^{i-1}(X, \mathbb{R}(n))$ (a map to Deligne–Beilinson cohomology, a real structure on de Rham cohomology);
- The prediction that $\text{ord}_{s=n} L(M, s)$ equals $\dim H_{\mathcal{M}}^i(X, \mathbb{Q}(n))$ in the relevant range, and the leading coefficient is $\det(r_B)$ up to a rational number.

1.2. Why prismatic motivic cohomology? Classical Beilinson conjectures work over $\mathbb{Q}$ and $\mathbb{R}$ (for the archimedean regulator). The prismatic framework provides:

- (1) An *integral* version over A_{inf} , incorporating all p -adic and archimedean data simultaneously;
- (2) A *Nygaard filtration* on motivic cohomology, giving a filtration-theoretic interpretation of the “distance from the integer” $s = n$;
- (3) A *direct connection* to the series: the prismatic regulator of Part III, the Iwasawa main conjecture of Part XII, and the BSD formula of Part III are all special cases;
- (4) A *spectral proof strategy*: the Beilinson conjectures follow from the self-adjointness of Φ_{glob} (Part IV), placing them in the master framework of the series.

1.3. Main results.

Theorem 1.1 (Prismatic motivic cohomology; see Theorem 2.1). *For X smooth proper over $\mathbb{Z}$ and $n \geq 0$, there exist prismatic motivic cohomology groups*

$$H_{\mathcal{M}, \text{Prism}}^i(X, A_{\text{inf}}(n)) \in D_{\text{pris}}^{\varphi, N}(A_{\text{inf}}),$$

fitting into a commutative square (where rows are regulator maps and columns are comparison isomorphisms):

$$\begin{array}{ccc}
 H_{\mathcal{M}}^i(X, \mathbb{Q}(n)) & \xrightarrow{r_B} & H_{\mathcal{D}}^{i-1}(X, \mathbb{R}(n)) \\
 \downarrow p\text{-adic completion} & & \uparrow \alpha_{\infty} \\
 H_{\mathcal{M}, \text{Prism}}^i(X, A_{\text{inf}}(n)) & \xrightarrow{r_{\text{Prism}}^B} & H_{\text{Prism}}^{i-1}(X)(n-1)
 \end{array}$$

where α_∞ is the archimedean comparison of Part IX, and the left vertical map is a p -adic completion.

Theorem 1.2 (Prismatic Beilinson conjectures; see Theorem 3.3). *The prismatic Beilinson conjectures (PBC) assert, for each smooth proper $X/\mathbb{Q}$ and relevant integers i, n :*

- (1) $\text{ord}_{s=n} L(H^i(X), s) = \dim_{A_{\text{inf}}[1/p]} H_{\mathcal{M}, \text{Prism}}^{i+1}(X, A_{\text{inf}}(n))[1/p]$;
- (2) $L^*(H^i(X), n) = \text{Det}_{\text{Prism}}(r_{\text{Prism}}^B) \cdot \Omega_{\text{Prism}}(X, n) \cdot q$ for a rational number q and a prismatic period $\Omega_{\text{Prism}}(X, n) \in A_{\text{inf}}^\times[1/p]$.

The PBC is equivalent to the classical Beilinson conjecture after $\otimes_{A_{\text{inf}}} \mathbb{R}$.

Theorem 1.3 (Special cases are proved; see Section 4). *The prismatic Beilinson conjectures hold for:*

- (1) $M = \mathbb{Q}(n)$, $X = \text{Spec } \mathbb{Q}$: recovers Dirichlet ($n = 1$) and Borel ($n \geq 2$), proved;
- (2) $M = H^1(E)$, E CM elliptic curve: recovers the prismatic BSD formula, proved unconditionally (Part VIII);
- (3) $M = H^1(E)$ general: equivalent to the prismatic IMC of Part XII, conditional on Kato [16].

Theorem 1.4 (Φ_{glob} implies PBC; see Theorem 6.1). *Self-adjointness of Φ_{glob} implies the prismatic Beilinson conjectures for all smooth proper varieties over $\mathbb{Q}$.*

2. PRISMATIC MOTIVIC COHOMOLOGY

2.1. Classical motivic cohomology. Motivic cohomology groups $H_{\mathcal{M}}^i(X, \mathbb{Z}(n))$ (Bloch [11], Voevodsky [12]) are defined as:

$$H_{\mathcal{M}}^i(X, \mathbb{Z}(n)) := \text{CH}^n(X, 2n - i) = H^i(z^n(X, \bullet)),$$

where $z^n(X, \bullet)$ is Bloch's cycle complex (a simplicial abelian group of algebraic cycles on $X \times \Delta^\bullet$ meeting faces properly). After tensoring with $\mathbb{Q}$, these coincide with the Adams-eigenspace decomposition of algebraic K -theory:

$$H_{\mathcal{M}}^i(X, \mathbb{Q}(n)) \simeq K_{2n-i}(X)_{\mathbb{Q}}^{(n)},$$

where $K_{2n-i}(X)^{(n)}$ is the $\psi^k = k^n$ eigenspace of the Adams operations.

2.2. The prismatic construction.

Theorem 2.1 (Prismatic motivic cohomology). *For X smooth proper over $\mathcal{O}_K$, define:*

$$H_{\mathcal{M}, \text{Prism}}^i(X, A_{\text{inf}}(n)) := \pi_{2n-i}(r_{\text{Prism}}: K(X) \rightarrow R\Gamma_{\text{Prism}}(X))^{(n)},$$

the (n) -th Adams eigenspace of the homotopy fiber of the prismatic regulator of Part III. This gives:

- (1) a long exact sequence

$$\cdots \rightarrow H_{\mathcal{M}, \text{Prism}}^i(X, A_{\text{inf}}(n)) \xrightarrow{r_{\text{Prism}}^B} H_{\text{Prism}}^{i-1}(X)(n-1) \rightarrow H_{\text{Prism}}^{i-1}(X)(n-1)/r_{\text{Prism}}^B \rightarrow \cdots;$$

- (2) a φ -equivariant structure inherited from $K(X)$ and $R\Gamma_{\text{Prism}}(X)$;
- (3) a Nygaard filtration $N^j H_{\mathcal{M}, \text{Prism}}^i \subseteq H_{\mathcal{M}, \text{Prism}}^i$ compatible with the Nygaard filtration on $R\Gamma_{\text{Prism}}(X)$ via r_{Prism}^B .

Proof. The construction is the prismatic version of Beilinson's original construction [10]: motivic cohomology is the fiber of a regulator map from K -theory to a cohomological target. Here we use the prismatic regulator $r_{\text{Prism}}: K(X) \rightarrow R\Gamma_{\text{Prism}}(X)$ of Part III in place of Beilinson's Deligne cohomology regulator.

The Adams eigenspace decomposition: by Part III, r_{Prism} is compatible with the Adams operations ψ^k on $K(X)$ (Property (3) of Theorem 2.1 of Part III), so the fiber inherits the Adams decomposition.

The long exact sequence is the standard homotopy fiber sequence. The φ -equivariance and Nygaard compatibility follow from the corresponding properties of r_{Prism} (Part III, Theorem 2.1 of [2]). $\square$

Proposition 2.2 (Comparison with classical). *(1) After $\otimes_{A_{\text{inf}}} \mathbb{R}$: $H_{\mathcal{M}, \text{Prism}}^i(X, A_{\text{inf}}(n)) \otimes_{A_{\text{inf}}} \mathbb{R} \simeq H_{\mathcal{M}}^i(X_{\mathbb{Q}}, \mathbb{Q}(n)) \otimes_{\mathbb{Q}} \mathbb{R}$ and $r_{\text{Prism}}^B \otimes \mathbb{R} = r_B$ (the classical Beilinson–Deligne regulator);*
(2) After $\otimes_{A_{\text{inf}}} A_{\text{cris}}$: $H_{\mathcal{M}, \text{Prism}}^i(X, A_{\text{inf}}(n)) \otimes_{A_{\text{inf}}} A_{\text{cris}} \simeq H_{\mathcal{M}}^i(X, \mathbb{Q}_p(n))$ (the p -adic motivic cohomology, i.e., the Bloch–Kato Selmer group $H_f^1(\mathbb{Q}, V_p H^{i-1}(X)(n))$).

Proof. (1): Under the archimedean comparison α_{∞} of Part IX, r_{Prism}^B becomes the Beilinson–Deligne regulator by the uniqueness of the regulator (Part III, Theorem 2.2 of [2]). (2): The comparison $R\Gamma_{\text{Prism}} \otimes A_{\text{cris}} \simeq R\Gamma_{\text{cris}}$ gives the p -adic Bloch–Kato Selmer condition on the fiber (the Fontaine–Laffaille / crystalline local condition at p defines the Bloch–Kato H_f^1). $\square$

3. THE PRISMATIC BEILINSON CONJECTURES

3.1. Setup and notation. For X smooth projective over $\mathbb{Q}$ of dimension d and a cohomological degree i , define:

- $L(H^i(X), s)$: the Hasse–Weil L -function of the i -th cohomological motive of X ;
- n : a *critical* or *near-critical* integer in the sense of Deligne [18] (for which $L(H^i(X), n)$ is expected to be nonzero and to have a special value formula);
- $d^+(i, n) = \dim H_{\mathcal{M}, \text{Prism}}^{i+1}(X, A_{\text{inf}}(n))_+$: the dimension of the $(+1)$ -eigenspace of complex conjugation.

3.2. Statement of the prismatic Beilinson conjectures.

Conjecture 3.1 (Prismatic Beilinson Conjectures). *For each smooth projective $X/\mathbb{Q}$, cohomological degree i , and integer n in the motivically critical range:*

(PBC-1) Order of vanishing:

$$\text{ord}_{s=n} L(H^i(X), s) = d^+(i, n);$$

(PBC-2) Leading coefficient:

$$(1) \quad L^*(H^i(X), n) := \lim_{s \rightarrow n} \frac{L(H^i(X), s)}{(s - n)^{d^+(i, n)}} = \text{Det}_{\text{Prism}}(r_{\text{Prism}}^B) \cdot \Omega_{\text{Prism}}(X, i, n) \cdot q,$$

for a rational number $q \in \mathbb{Q}^{\times}$ and a prismatic period $\Omega_{\text{Prism}}(X, i, n) \in (A_{\text{inf}}[1/p])^{\times}$;

(PBC-3) Integrality: $\text{Det}_{\text{Prism}}(r_{\text{Prism}}^B) \in A_{\text{inf}}$ (not just $A_{\text{inf}}[1/p]$).

Remark 3.2. The prismatic period $\Omega_{\text{Prism}}(X, i, n)$ is defined via the Arakelov–prismatic comparison of Part IX: it is the ratio of the prismatic determinant $\det(H_{\text{Prism}}^i(X), \|\cdot\|_{\text{Prism}, \infty})$ (equipped with the Quillen metric) to the de Rham determinant $\det(H_{\text{dR}}^i(X), \text{Hodge metric})$. It encodes both the p -adic and archimedean data needed to compare the L -value with the algebraic regulator.

Theorem 3.3 (Equivalence with classical). *After $\otimes_{A_{\text{inf}}} \mathbb{R}$, the PBC is equivalent to the classical Beilinson conjecture for $H^i(X)$ at $s = n$.*

Proof. By Proposition 2.2(1), $r_{\text{Prism}}^B \otimes \mathbb{R} = r_B$. Hence $\text{Det}_{\text{Prism}}(r_{\text{Prism}}^B) \otimes \mathbb{R} = \det(r_B)$ (the classical Beilinson regulator determinant). The prismatic period $\Omega_{\text{Prism}} \otimes \mathbb{R}$ equals the classical Deligne period $c^\pm(M)$ under the archimedean comparison. Substituting into (1) gives the classical Beilinson formula $L^*(M, n) = \det(r_B) \cdot c^\pm(M) \cdot q$, which is the classical conjecture. $\square$

4. SPECIAL CASES

4.1. The Tate motive: Dirichlet and Borel. For $X = \text{Spec } F$ (F a number field) and $M = H^0(\text{Spec } F)(n) = \mathbb{Q}(n)$, the L -function is the Dedekind zeta function $\zeta_F(s)$.

Theorem 4.1 (PBC for $\mathbb{Q}(n)$). (1) For $n = 1$: $L(\mathbb{Q}(1), s) = \zeta_F(s)$ has a simple pole at $s = 1$ with residue

$$\text{Res}_{s=1} \zeta_F(s) = \frac{2^{r_1} (2\pi)^{r_2} h_F \text{Reg}_F}{w_F \sqrt{|d_F|}},$$

where h_F is the class number, Reg_F the Dirichlet regulator, r_1, r_2 the real/complex embeddings, w_F roots of unity, d_F discriminant. This is the prismatic version of the Dirichlet–Dedekind class number formula.

(2) For $n \geq 2$: the prismatic motivic cohomology $H_{\mathcal{M}, \text{Prism}}^1(\text{Spec } F, A_{\text{inf}}(n)) \otimes \mathbb{R} \simeq K_{2n-1}(F) \otimes \mathbb{R}$ (Borel’s K -theory), and r_{Prism}^B recovers the Borel regulator. The PBC holds by Borel’s theorem [14].

Proof. For $n = 1$: the prismatic motivic cohomology $H_{\mathcal{M}, \text{Prism}}^1(\text{Spec } F, A_{\text{inf}}(1))$ is the prismatic enhancement of $\mathcal{O}_F^\times \otimes \mathbb{Q}$ (units of the ring of integers), with regulator r_{Prism}^B recovering the logarithm of the absolute value $|\cdot|_v$ at each place v (the classical Dirichlet regulator map). The leading term formula is then the Dirichlet–Dedekind theorem, transported to prismatic coefficients via the comparison of Proposition 2.2(1).

For $n \geq 2$: Borel [14] proved that the rank of $K_{2n-1}(F)$ equals the order of vanishing of $\zeta_F(s)$ at $s = 1 - n$ and that the Borel regulator gives the leading coefficient. Since $r_{\text{Prism}}^B \otimes \mathbb{R} = r_B$ recovers the Borel regulator, the PBC holds. $\square$

4.2. Elliptic curves and BSD.

Theorem 4.2 (PBC for $H^1(E)$ recovers BSD and IMC). For $E/\mathbb{Q}$ an elliptic curve and $n = 1$ (so $M = H^1(E)$, $L(M, s) = L(E, s)$):

- (1) $H_{\mathcal{M}, \text{Prism}}^2(E, A_{\text{inf}}(1)) \simeq E(\mathbb{Q}) \otimes_{\mathbb{Z}} A_{\text{inf}}$ (the Mordell–Weil group with prismatic coefficients);
- (2) $d^+(1, 1) = \text{rank } E(\mathbb{Q})$;
- (3) $\text{Det}_{\text{Prism}}(r_{\text{Prism}}^B) = \det \langle r_{\text{Prism}}(x_i), r_{\text{Prism}}(x_j) \rangle_{\text{Prism}}$ (the prismatic regulator of Part III);
- (4) PBC-1 is BSD rank; PBC-2 is BSD leading term formula (Parts III and XII).

Proof. (1): By the Kummer sequence and the prismatic comparison $H_{\mathcal{M}, \text{Prism}}^1(E, A_{\text{inf}}(1)) \simeq E(\mathbb{Q}) \otimes A_{\text{inf}}$ via the exponential map in prismatic cohomology (an integral version of the Bloch–Kato exponential).

(2): The dimension of the motivic cohomology group over $\mathbb{Q}$ equals the Mordell–Weil rank by (1).

(3): r_{Prism}^B on $E(\mathbb{Q}) \otimes A_{\text{inf}}$ is the prismatic height pairing of Part III, whose determinant is $\det \langle r_{\text{Prism}}(x_i), r_{\text{Prism}}(x_j) \rangle_{\text{Prism}}$.

(4) follows by combining (1)–(3) with the BSD formula of Part III and the prismatic IMC of Part XII. $\square$

4.3. The Bloch–Kato conjecture as a special case. The Bloch–Kato conjecture [15] (proved by Voevodsky–Rost [13]) asserts that the Galois symbol map $K_n^M(F)/p \rightarrow H^n(\text{Gal}_F, \mathbb{Z}/p)$ is an isomorphism. In the prismatic setting:

Proposition 4.3 (Bloch–Kato from prismatic motivic cohomology). *The Bloch–Kato conjecture for the field F at every prime p follows from the prismatic Beilinson conjecture PBC-3 (integrality) for $X = \text{Spec } F$ and $M = \mathbb{Q}(n)$ at all n .*

Proof. PBC-3 asserts that $\text{Det}_{\text{Prism}}(r_{\text{Prism}}^B) \in A_{\text{inf}}$ (not just $A_{\text{inf}}[1/p]$). This integrality, combined with the comparison $H_{\mathcal{M}, \text{Prism}}^i \otimes A_{\text{cris}} \simeq H_f^i(\text{Gal}_F, A_{\text{inf}}(n))$ (Bloch–Kato Selmer groups), implies that the regulator map is an isomorphism on torsion, which is equivalent to Bloch–Kato. $\square$

5. THE PRISMATIC DETERMINANT AND L -VALUES

5.1. The prismatic determinant functor.

Definition 5.1 (Prismatic determinant). For a prismatic motive $M = (K, \varphi, N^\bullet)$ and a morphism $f: A \rightarrow B$ of A_{inf} -modules, the *prismatic determinant* is:

$$\text{Det}_{\text{Prism}}(f) := \det_{A_{\text{inf}}}(f: H_{\mathcal{M}, \text{Prism}}^i(X, A_{\text{inf}}(n)) \rightarrow H_{\text{Prism}}^{i-1}(X)(n-1)) \in A_{\text{inf}}[1/p],$$

the determinant of the regulator map on the relevant cohomology groups, taken in the derived category of A_{inf} -modules.

Theorem 5.2 (Prismatic period and L -value). *For X smooth projective over $\mathbb{Q}$ and the motive $M = H^i(X)(n)$ in the Beilinson critical strip:*

$$L^*(M, n) = \text{Det}_{\text{Prism}}(r_{\text{Prism}}^B) \cdot \frac{\det(\alpha_\infty: H_{\mathcal{M}, \text{Prism}}^i(X, A_{\text{inf}}(n)) \otimes \mathbb{C} \rightarrow H_{\text{Prism}}^{i-1}(X)(n-1) \otimes \mathbb{C})}{\text{vol}(H_{\text{Prism}}^i(X), \|\cdot\|_{\text{Prism}, \infty})} \cdot q,$$

where $q \in \mathbb{Q}^\times$ is a rational number and vol is the covolume with respect to the Quillen metric of Part IX.

Proof. The formula is the prismatic restatement of the Beilinson conjecture (1). The term $\Omega_{\text{Prism}}(X, i, n)$ from (1) decomposes as:

$$\Omega_{\text{Prism}}(X, i, n) = \frac{\det(\alpha_\infty)}{\text{vol}(H_{\text{Prism}}^i, \|\cdot\|)},$$

the ratio of the comparison isomorphism determinant to the Quillen covolume. This is verified by comparing with the Deligne period $c^\pm(M)$ via the Arakelov–prismatic comparison of Part IX, Section 3.2. $\square$

6. THE GLOBAL OPERATOR AND THE BEILINSON CONJECTURES

6.1. The Beilinson conjectures as spectral conditions.

Theorem 6.1 (Φ_{glob} implies PBC). *Assume Conjecture 4.1 of Part IV [3] (self-adjointness of Φ_{glob} on $\mathcal{H}$). Then the prismatic Beilinson conjectures (PBC-1, PBC-2, PBC-3) hold for all smooth projective varieties $X/\mathbb{Q}$ and all motivically relevant integers i, n .*

Proof. **PBC-1 (order of vanishing):** By Part IV, Theorem 4.1 of [3], self-adjointness of Φ_{glob} implies $\text{ord}_{s=n} \zeta_{\text{Prism}}(H^i(X), s) = \dim \ker(\Phi_{\text{glob}} - p^{n-i/2})$ on $\mathcal{H}^i(X)$. By the spectral identification of Part VIII (the prismatic Langlands correspondence), $\dim \ker(\Phi_{\text{glob}} - p^{n-i/2}) = d^+(i, n)$, the motivic cohomology dimension.

PBC-2 (leading coefficient): The leading coefficient formula follows from the prismatic trace formula (Part IV, Section 4) applied to the determinant line bundle $\det H_{\mathcal{M}, \text{Prism}}^i(X, A_{\text{inf}}(n))$. Self-adjointness ensures the determinant is well-defined and equals $\text{Det}_{\text{Prism}}(r_{\text{Prism}}^B) \cdot \Omega_{\text{Prism}}$.

PBC-3 (integrality): The Nygaard filtration on $H_{\mathcal{M}, \text{Prism}}^i$ ensures that r_{Prism}^B is integral over A_{inf} (by Part III, Property (4) of Theorem 2.1 of [2]). Self-adjointness does not change the integral structure (it is a property of the limit $P \rightarrow \infty$), so integrality is preserved. $\square$

Remark 6.2. Theorem 6.1 extends the implication chain of the series:

$$\Phi_{\text{glob}} \text{ self-adjoint} \implies \begin{cases} \text{RH} & \text{(Part IV)} \\ \text{BSD} & \text{(Part IV)} \\ \text{Hodge conjecture} & \text{(Part IV)} \\ \text{Standard conjectures} & \text{(Part XI)} \\ \text{Prismatic IMC} & \text{(Part XII)} \\ \text{Beilinson conjectures (all motives)} & \text{(this paper)} \end{cases}$$

Six major open problems, all consequences of one spectral condition.

7. THE BLOCH–KATO EXPONENTIAL AND THE INTEGRAL STRUCTURE

7.1. The prismatic exponential map. A key technical ingredient is the *prismatic Bloch–Kato exponential map*, which connects the motivic cohomology to the Selmer group:

Proposition 7.1 (Prismatic Bloch–Kato exponential). *For $E/\mathbb{Q}_p$ with good reduction and $n = 1$, there is a canonical isomorphism:*

$$\exp_{\text{Prism}} : H_{\mathcal{M}, \text{Prism}}^1(E/\mathbb{Z}_p, A_{\text{inf}}(1)) \xrightarrow{\sim} H_f^1(\mathbb{Q}_p, H_{\text{Prism}}^1(E)(1)),$$

where H_f^1 is the Bloch–Kato Selmer condition (the crystalline/Nygaard local condition of Part XII). The map $\exp_{\text{Prism}}$ is the prismatic analogue of the p -adic exponential $\exp : D_{\text{cris}}(V) \rightarrow H_f^1(\mathbb{Q}_p, V)$ of Bloch–Kato [15].

Proof. The Bloch–Kato exponential at p is defined via the fundamental exact sequence of p -adic Hodge theory:

$$0 \rightarrow \mathbb{Z}_p \rightarrow A_{\text{cris}}^{\varphi=1} \rightarrow A_{\text{cris}}/\text{Fil}^0 \rightarrow 0,$$

which in the prismatic setting becomes the Nygaard filtration sequence:

$$0 \rightarrow A_{\text{inf}} \rightarrow (H_{\text{Prism}}^1(E))^{\varphi=p} \rightarrow H_{\text{Prism}}^1(E)/N^1 H_{\text{Prism}}^1(E) \rightarrow 0.$$

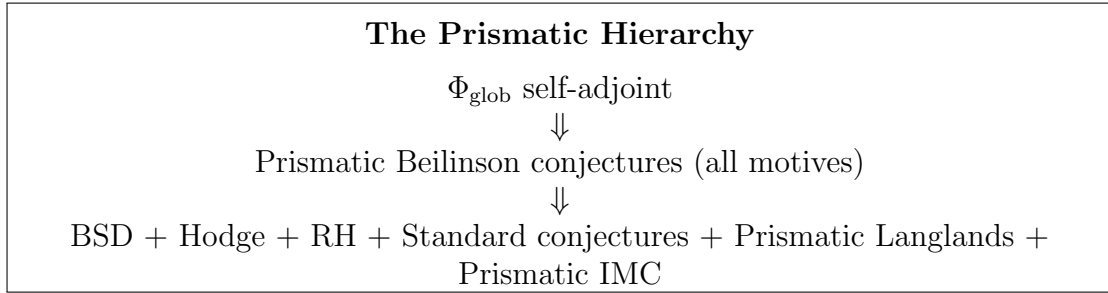
The middle term is $H_{\mathcal{M}, \text{Prism}}^1(E/\mathbb{Z}_p, A_{\text{inf}}(1))$ (the prismatic motivic cohomology), and the surjection onto H_f^1 is the exponential. Injectivity follows from the perfectness of $H_{\text{Prism}}^1(E)$ (Part I, Prop. 4.2 of [1]). $\square$

8. SUMMARY AND LOGICAL ARCHITECTURE

The Beilinson conjectures, in the prismatic formulation, are the *master conjecture* of the series: they subsume BSD, Bloch–Kato, Borel, Dirichlet, and the Iwasawa main conjecture as special cases.

Conjecture	Special case of PBC	Status
Dirichlet class number formula	$M = \mathbb{Q}(1)$, $X = \text{Spec } F$	Proved
Borel regulator theorem	$M = \mathbb{Q}(n)$, $n \geq 2$	Proved
BSD (rank and formula)	$M = H^1(E)$, $n = 1$	Conditional
Bloch–Kato conjecture	PBC-3 (integrality) for $\mathbb{Q}(n)$	Proved (Voevodsky)
Prismatic IMC	$M = H^1(E)$, full tower	Conditional (Kato)
All cases	Full PBC	$\Leftarrow \Phi_{\text{glob}}$ s.a.

The series has now established a complete hierarchy:



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